MD-11
Flight Crew Operations Manual

Volume I - Quick Reference Handbook

Shanghai Airlines Cargo

Revision Number: 59

Revision Date: May 15, 2010

Customer airplane configuration determines the data provided in this manual. The Boeing Company keeps a list of each airplane configuration as it is built and modified through the service bulletin process. The FCOM does not reflect customer originated modificationswithout special contract provisions.

Preface — Chapter 0
Manual Effectivity — Section 1

General

The airplanes listed in the table below are covered in the Flight Crew Operations Manual (FCOM). The table information is used to distinguish data peculiar to one or more, but not all of the airplanes. Where data applies to all airplanes listed, no reference is made to individual airplanes.

Registry number is supplied by the national regulatory agency. Airplane, serial and tabulation numbers are supplied by Boeing.

Aircraft Number	Registry Number	Serial Number	Tabulation Number
	B-2176	48415	1PC701
	B-2177	48544	1FA301
	B-2178	48543	1PC702
	B-2179	48545	1FA302

Intentionally
Blank

Revision Transmittal Letter

This revision reflects the most current information available to the company. The following revision highlights explain changes in this revision. The Revision Record page preceding this page explains the use of revision bars to identify new or revised information.

Revision Record

Revisions 1-50 incorporated

No.	Revision Date	Date Filed	No.	Revision Date	Date Filed
51	April 15, 2006	--	52	Nov 15, 2006	--
53	May 15, 2007	--	54	Nov 15, 2007	--
55	May 15, 2008	--	56	December 15, 2008	--
57	May 15, 2009	--	58	Oct 15, 2009	--
59	May 15, 2010	--			

General

Texts amended by SALC are marked in shaded background with revision bar(s). Deleted paragraph is identified by empty-shaded line with revision bar(s).

The company issues Flight Crew Operations Manual revisions to provide new or revised procedures and information. Formal revisions also incorporate appropriate information from previously issued Temporary Revisions and Interim Operating Procedures.

The revision date is assigned to correspond with the Boeing Airplane Company flight crew operating manual revision date. Due to the need for correlation with the company, the flight crew operating manual will always pre-date the issue date.

Formal revisions include a Transmittal Letter, a new Revision Record, Revision Highlights, and a current List of Effective Pages. Use the information on the new Revision Record and List of Effective Pages to verify the operations manual content.

Pages containing revised material have revision bars associated with the changed text or illustration.

The Revision Record should be completed by the person incorporating the revision into the manual.

Filing Instructions

Keep applicable Temporary Revisions and Interim Operating Procedure unless instructed to remove them by the highlights. The manual is revised by pages. To file a revision package, use the list of effective pages (LEP) to verify the correct content of the manual. On the LEP, pages identified with an asterisk(*) are replacement, new (original) issue or deleted pages. Use the pages provided in the package to add new pages, or replace the corresponding pages in the manual. Remove pages that are marked “Deleted” on the LEP; there will be no replacement pages for deleted pages.

Revision Highlights

Entire sections (but not chapters) are printed, even if there is only one change in a section.

This section (0.2) replaces the existing section 0.2 in your manual.

Be careful when inserting changes not to throw away pages from the manual that are not replaced. Using the List of Effective Pages (0.3) can help determine the correct content of the manual.

Throughout the manual, aircraft effectivity may be updated to reflect coverage as listed on the Preface - Model Identification page. Registry or tabulation numbers are used as available at the time of printing. Highlights are not supplied.

Highlights and revision bars are provided for technical changes. In some sections, text may be rewritten or reformatted for clarity or other editorial purposes; these changes will have revision bars, but may not have highlights.

When changes are minor, or of a grammatical nature, and the meaning remains essentially unchanged there will be no revision bar or highlights.

Pages may also be republished, without revision bars, due to slight changes in the presentation of the document caused by the publishing system.

AP 50 ABNORMAL PROCEDURES - FUEL

AP.50.6 Add the missing “X” to the correct XFEED Switch.

QR 10 ALL WEATHER

QR.10.2 Amend the NO AUTOLAND to APPROACH ONLY to reflect the actual FMA change.

Intentionally
Blank

LIST OF EFFECTIVE PAGES

Volume 1 - QRH

Page	Date
*Title.1-2	May 15, 2010
0.1.1-2	May 15, 2008
*0.2.1-4	May 15, 2010
*0.3.1-2	May 15, 2010
0.4.1-2	March 15, 2003
0.5.1-2	March 15, 2003
0.6.1-2	May 15, 2008
0.7.1-2	May 15, 2008

Normal Checklists

NC.TOC.00.1-2	Deleted
NC.10.1-2	Deleted
NC.10.3-4	Deleted
NC.20.1-4	Deleted
NC.20.5-6	Deleted

Emergency Alert

EP.TOC.10.1-2	March 15,2002
EP.10.1	March 15,2002
EP.10.2-4	September 15, 2003
EP.10.5	May 15, 2009
EP.10.6-7	March 15, 2002
EP.10.8	September 15, 2003
EP.10.9-10	March 15, 2002
EP.10.11-12	September 15, 2003
EP.10.13	March 15, 2002
EP.10.14	November 15, 2006
EP.10.15	November 15, 2005
EP.10.16	May 15, 2009
EP.10.17	November 15, 2004
EP.10.18-19	September 15, 2003
EP.10.20	February 15, 2004
EP.10.21	September 15, 2003
EP.10.22	November 15, 2004
EP.10.23-26	September 15, 2003

Page	Date
Emergency Non-Alert	
EP.TOC.20.1	October 15, 2009
EP.TOC.20.2	March 15, 2002
EP.20.1-2	May 15, 2007
EP.20.3	April 15, 2006
EP.20.4	November 15, 2005
EP.20.5-6	May 15, 2007
EP.20.7	September 15, 2003
EP.20.8	March 15, 2002
EP.20.9-28	October15,2009

Abnormal Procedures

AP.TOC.00.1-2	November 15, 2002

Air

AP.TOC.10.1-2	March 15, 2002
AP.10.1-2	March 15, 2002
AP.10.3-6	February 15, 2004
AP.10.7-9	March 15, 2002
AP.10.10	February 15, 2004
AP.10.11-12	March 15, 2002

Abnormal Procedures
Config

AP.TOC.20.1-2	May 15, 2007
AP.20.1-3	May 15, 2007
AP.20.4-6	May 15, 2005
AP.20.7	April 15, 2006
AP.20.8-28	May 15, 2005

Abnormal Procedures
Elec

AP.TOC.30.1-2	February 15, 2004
AP.30.1-2	March 15, 2002
AP.30.3	September 15, 2003
AP.30.4-7	March 15, 2002
AP.30.8	Dec 15, 2008
AP.30.9-11	February 15, 2004
AP.30.12	November 15, 2004

Page	Date
AP.30.13	February 15, 2004
AP.30.14	November 15, 2004
AP.30.15-19	February 15, 2004
AP.30.20	November 15, 2004
Eng/APU	
AP.TOC.40.1-2	March 15, 2002
AP.40.1	March 15, 2003
AP.40.2	September 15, 2003
AP.40.3-4	March 15, 2002
AP.40.5	May 15, 2007
AP.40.6	April 15, 2006
Fuel	
AP.TOC.50.1-2	May 15, 2009
AP.50.1-4	March 15, 2002
AP.50.5-10	September 15, 2003
*AP.50.6	May 15, 2010
AP.50.7-10	September 15, 2003
AP.50.10A/B-11	May 15, 2009
AP.50.12	September 15, 2003
AP.50.13	March 15, 2002
AP.50.14	September 15, 2003
AP.50.15-20	March 15, 2002
Hyd	
AP.TOC.60.1-2	March 15, 2002
AP.60.1-2	September 15, 2003
AP.60.3-4	November 15, 2004
AP.60.5	March 15, 2002
AP.60.6	September 15, 2003
Misc	
AP.TOC.70.1-2	March 15, 2002
AP.70.1-3	March 15, 2002
AP.70.4	May 15, 2005
Non-Alert	
AP.TOC.80.1	October 15, 2009
AP.TOC.80.2	November 15, 2007
AP.80.1-3	February 15, 2004
AP.80.4	May 15, 2007
AP.80.5-12	March 15, 2002
AP.80.13-15	May 15, 2007
AP.80.16	October 15, 2009
AP.80.17-18	April 15, 2006
AP.80.19-20	May 15, 2007

Page	Date
AP.80.21	April 15, 2006
AP.80.22-23	November 15, 2007
AP.80.24	April 15, 2006
AP.80.25	May 15, 2008
AP.8026	April 15, 2006
AP.80.27	May 15, 2008
AP.80.28-32	April 15, 2006
AP.80.33	May 15, 2009
AP.80.34-52	April 15, 2006
AP.80.53	May 15, 2009
AP.80.54	April 15, 2006
AP.80.55	May 15, 2009
AP.80.56-58	April 15, 2006
Level 1 Alerts	
AP.TOC.90.1-2	November 15, 2006
AP.90.1-4	March 15, 2002
AP.90.5-6	February 15, 2004
AP.90.7	May 15, 2005
AP.90.8	February 15, 2004
AP.90.9-10	March 15, 2002
AP.90.11	Dec 15, 2008
AP.90.12-16	November 15, 2006
Quick Reference	
QR.TOC.00.1	May 15, 2009
QR.TOC.00.2	Nov 15, 2007
QR.10.1	May 15, 2009
*QR.10.2-3	May 15, 2010
QR.10.4	Dec 15, 2008
QR.10.5	May 15, 2007
QR.10.6-10	November 15, 2007
QR.20.1-4	May 15, 2007
QR.30.1	Dec 15, 2008
QR.30.2	May 15, 2007
QR.30.3	May 15, 2009
QR.30.4-5	October 15, 2009
QR.30.6-8	May 15, 2009
QR.40.1-30	May 15, 2007
QR.50.1	May 15, 2008
QR.50.2-3	October 15, 2009
QR.50.4	May 15, 2007
QR.60.1	May 15, 2007
QR.60.2	October 15, 2009

Preface

Temporary Revision Record

TR Number	Date Inserted

TR Number	Date Inserted

Intentionally
Blank

Preface — Chapter 0

Interim Operating Procedure Record — Section 5

IOP Number	Date Inserted	IOP Number	Date Inserted

Intentionally
Blank

Preface — Chapter 0

Temporary Revision Summary Record — Section 6

TR Number	Issue Date	Revision Date Incorp.
1-001~567		Not Applicable Cancelled or Previously Incorporated

All other TRs not listed above are cancelled, not applicable or incorporated.

Intentionally
Blank

Preface — Chapter 0
Interim Operating Procedure Summary — Section 7

IOP Number	LOCATE TO FACE PROCEDURE	Issue Date
1-86A	Abnormal Alert, FUEL section	Mar 16, 2007
1-87D	Level 1/0 Alerts, LVL1 ALERT tab	Mar 16, 2007
1-102	Abnormal Non-Alert, FLIGHT CONTROLS, JAMED OR RESTRICTED	Nov 15, 2004
1-103A	Abnormal Non-Alert, FUEL LEAK	Oct 16, 2007
1-104	AIR CONDITIONING PACK SURGE	Feb 06, 2008

All other IOPs not listed above are cancelled, not applicable or incorporated.

Intentionally
Blank

EMERGENCY LEVEL 3 ALERTS

Intentionally
Blank

AIR MANF__FAIL

When flight conditions permit,

Throttle (Affected Engine) . . . . IDLE

When alert is no longer displayed, operate associated engine at a thrust level that will keep alert from being displayed.

Land at nearest suitable airport.

NOTE: Do not repressurize affected air system.

[END]

APU FIRE

APU FIRE Handle . . . . .PULL

APU FIRE AGENT 3 LOW light . . . . .CHECK

NOTE: If the APU FIRE AGENT 3 LOW light did not illuminate, push the APU FIRE AGENT 3 switch to manually discharge APU FIRE AGENT 3.

APU START/STOP Switch . . . . OFF

AIR SYSTEM MANUAL

NO

BLEED AIR 2 Switch . . . . OFF

1-2 ISOL Switch. . . . . OFF

PACK 2 . . . . OFF

After 30 seconds,

FIRE WARNING CONTINUES

NO

APU FIRE
Handle,AGT LOW Light . . . . .PULL & TURN,CHECK

FIRE WARNING CONTINUES

NO

Remaining Agent . . . . .DISCH,CHECK

NOTE: All APU and engine 2 fire agents have been depleted.

Land at nearest suitable airport.

[END]

BLD AIR__TEMP HI

BLEED AIR Source (Affected System) . . . . OFF

After 30 seconds,

"BLD AIR___TEMP HI" ALERT
DISPLAYED AGAIN

NO

If flight conditions permit, slowly reduce thrust on associated engine until alert is no longer displayed. Operate engine at a thrust level which will keep alert from being displayed for rest of flight.

[END]

PACK Switch (Affected System) . . . . OFF

Associated ISOL Switch . . . . ON

[END]

CABIN ALTITUDE

Oxygen Masks	ON ,100%
Crew, Courier(s) Communication	ESTABLISH
Outflow VALVE	VERIFY CLOSED
AVNCS FAN Switch	VERIFY OVRD

CABIN ALTITUDE CONTROLLABLE

NO

Operate cabin pressure system as required.

[END]

Perform an emergency descent.

Altitude Select Knob	REDUCE,PULL

Initate descent to 10,000 feet or minimum safe altitude, whichever is higher.

SPOILER Handle	SPD BRK FULL

WARNING: ***If structural damage is suspected or turbulence present, do not exceed .82 Mach/305 KIAS.***

IAS/MACH Select Knob	SELECT .85 MACH/320-350 KIAS
Descent	MAX PITCH 10°,MAX BANK 30°
Transponder (Unless Otherwise Required)	7700
NO SMOKE and SEAT BELTS Switches	ON

To re-activate boom mike when O_2 mask is no longer required,

Eros O_2 Mask Doors . . . . CLOSE

PRESS TO TEST AND RESET Lever . . . . PUSH

[END]

CABIN SMOKE

NOTE: In addition to the "CABIN SMOKE" alert displayed, an aural warning will sound.

Oxygen Masks ON,100%

Don smoke goggles as required.

Use emergency oxygen pressure, as required to purge mask and goggles of smoke and/or fumes.

Crew, Courier(s) Communication . . . . ESTABLISH

AIR SYSTEM MANUAL

NO

Reduce to single pack operation.

CABIN AIR Switch . . . . OFF

CABIN PRESS Panel . . . .SYSTEM MANUAL,CLIMB

"CAB AIR NOT OFF" ALERT DISPLAYED

NO

AIR SYSTEM SELECT Switch. . . . . MANUAL

Operating PACK Switch . . . . OFF

Descend as required to maintain maximum cabin altitude of 25,000 feet to avoid oxygen pressure breathing and to starve fire.

When aircraft is depressurized,

Outflow VALVE Indicator . . . . SET 9:00 POSITION

NOTE: With no packs operating, selection of a position greater than 9:00 can cause a negative pressure in the aircraft. This will cause cabin doors to unseat and allow outside air to flow into the cabin.

(CONTINUED)

CABIN SMOKE (Continued)

"CAB AIR NOT OFF" ALERT DISPLAYED

NO

(CONTINUED)

When cockpit is clear of smoke and/or fumes, move oxygen dilution control lever to NORMAL in order to extend usable oxygen time.

Land at nearest suitable airport.

After landing and prior to opening door,

Outflow VALVE Indicator . . . . SET FULL OPEN

[END]

AIRCRAFT AT OR ABOVE FL270

NO

Maintain 25,000 feet cabin altitude until approaching FL250 during descent.

Below 27,000 feet,

Maintain 0.5 psi cabin differential pressure.

When cockpit is clear of smoke and/or fumes, move oxygen dilution control lever to NORMAL in order to extend usable oxygen time.

Just prior to landing,

CABIN PRESS Manual Rate Selector . . . . .CLIMB

When aircraft is depressurized,

Outflow VALVE Indicator. SET 10:30 POSITION

> *NOTE: With a pack operating, selection of a position greater than 10:30 can cause a negative pressure in the aircraft. This will cause cabin doors to unseat and allow outside air to flow into the cabin.*

Land at the nearest suitable airport.

(CONTINUED)

CABIN SMOKE (Continued)

After landing and prior to opening door,

Outflow VALVE Indicator . . . . SET FULL OPEN

[END]

CAC MANF FAIL

PACK Switch(es) . . . . OFF

AVNCS FAN Switch . . . . OVRD

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

All Engine BLEED AIR Switches . . . . . ON (5 SECONDS), THEN OFF

Compare air system pressure decay rates on synoptic.

BLEED AIR Switch (System With Slowest Rate Of Decay) . . . . ON

PACK Switch (System With Slowest Rate Of Decay). . . . . ON

MANF lights on the overhead panel should go off within approximately 5 minutes.

After 5 minutes,

"CAC MANF FAIL" ALERT REMAINS DISPLAYED

NO

Recheck manifold decay rates to verify correct system has been selected.

Restoration of an additional system may be attempted if required.

CAUTION: Do not repressurize the manifold that has the suspected failure.

AVNCS FAN Switch . . . . AS REQD

Avoid icing conditions.

[END]

CRG FIRE LWR__

Flashing CARGO FIRE AGENT DISCH Switch . . . . PUSH

NOTE: CARGO FIRE AGENT DISCH switch will continue to flash until LOW light illuminates.

Associated CARGO FLOW Switch . . . . OFF

Associated CARGO TEMP Selector . . . . OFF

"CRG FLO AFT DISAG" ALERT DISPLAYED

NO

NOTE: "CRG FLO AFT DISAG" alert may be displayed after a cargo fire procedure is completed and airflow through aft cargo compartment is detected.

AIR SYSTEM SELECT Switch. . . . . MANUAL

BLEED AIR 1 Switch . . . . OFF

PACK 1 Switch. . . . . OFF

1-2 and 1-3 ISOL Switches . . . . OFF

Depart icing conditions (if applicable).

After approximately 1 minute elapsed time,

CARGO FIRE AGENT DISCH LOW LIGHT ILLUMINATED

NO

(CONTINUED)

CARGO FIRE LWR __ (Continued)

CARGO FIRE AGENT DISCH LOW
LIGHT ILLUMINATED

NO

(CONTINUED)

Approximately 90 minutes after agent 1 has been discharged, “DISCH CARGO AGENT” alert will be displayed on EAD and CARGO FIRE AGENT 2 DISCH switch will flash. The flashing CARGO FIRE AGENT DISCH switch should be pushed at that time.

Land at nearest suitable airport.

[END]

Associated CARGO FIRE AGENT 2 DISCH Switch
(Located Below Flashing Switch) . . . . PUSH

Land at nearest suitable airport.

[END]

ENG 2 A–ICE DUCT

ENGINE 2 Throttle . . . . IDLE

ENGINE 2 FUEL Switch . . . . OFF

AIR SYSTEM MANUAL

NO

Engine 2 BLEED AIR Switch . . . . OFF

PACK 2 Switch . . . . OFF

1–2 ISOL Switch . . . . ON

Transponder / TCAS Selector . . . . .TA

Land at nearest suitable airport

[END]

ENGINE__FIRE

OR
SEVERE DAMAGE

Throttle (Affected Engine) . . . . .IDLE

FUEL Switch (Affected Engine) . . . . OFF

ENG FIRE Handle,AGT LOW Light (Affected Engine)
. . . . .DOWN & DISCH,CHECK

After 30 seconds,

“ENGINE __ FIRE” ALERT REMAINS DISPLAYED OR “FIRE DET __ FAIL” DISPLAYED

NO

Remaining Agent . . . . .DISCH,CHECK

NOTE: Discharging both fire agents to engine 2 leaves no engine fire agent for APU

AIR SYSTEM MANUAL

NO

BLEED AIR Switch (Affected Engine). . . . . OFF

Associated ISOL Switch . . . . OFF

(CONTINUED)

ENGINE__FIRE (Continued)

CONTINUOUS HIGH AIRFRAME VIBRATION PRESENT

NO

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level.

NOTE: If high vibration returns and further airspeed reduction and descent are not practicable, increasing airspeed may reduce vibration.

Transponder/TCAS Selector . . . . .TA

Land at nearest suitable airport.

[END]

HYD 1 & 2 FAIL

NOTE:

1. Increased fuel consumption, up to approximately 15%, may result due to control surface float.

2. Outboard slats will not retract if they were extended before the loss of pressure occurred. "SLAT DISAG" alert will be displayed when flap/slat handle is in the 0/RET position.

3. The "LSAS ALL FAIL" Level 2 alert will be displayed and LSAS channels cannot be restored.

"HYD 3 ELEV OFF" ALERT DISPLAYED

NO

Elevators are inoperative. Pitch control is available from engine thrust and/or stabilizer trim (one-half rate).

Rudders are inoperative. Directional control is available from ailerons, spoilers, and engine thrust.

"RUDDER BOTH INOP" ALERT DISPLAYED

NO

Directional control is available from ailerons, spoilers and engine thrust.

If a wing engine is shut down, a missed approach should not be attempted.

(CONTINUED)

HYD 1 & 2 FAIL (Continued)

FLAPS EXTENDED

NO

Leave FLAP/SLAT handle in existing position.

GPWS Switch .... FLAP OVRD

Recommended maximum crosswind componet is 12 knots.

V_{MCA} is 160 KIAS

NOTE: IF " HYD 3 ELEV OFF" alert displayed, refer to FCTM HYDRAULIC SYSTEM 1 AND 2 FAILURE WITH "HYD 3 ELEV OFF" ALERT before continue to the next procedure.

Reduce gross weight as required.

When ready for approach,

FLAPS RETRACTED

NO

0/EXT APPROACH SPEEDS
HYDRAULIC SYSTEMS 1 & 2 FAIL

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{APP} (V_{REF} + 15)	167	172	176	181	185	189	193	197

0/EXT ESTIMATED LANDING DISTANCES (FEET)
HYDRAULIC SYSTEM 1 & 2 FAIL

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	DRY	5938	6206	6594	6955	7349	7710	8136	8563
STD=15°C	WET	7415	7776	8169	8563	8989	9583	9810	10269
2000 FT	DRY	6332	6660	7021	7447	7841	8277	8694	9186
STD=11°C	WET	7874	8300	8727	9153	9615	10039	10499	10991
4000 FT	DRY	6758	7119	7513	7972	8432	8858	9350	9875
STD=7°C	WET	8432	8858	9318	9810	10301	10761	11286	11811
6000 FT	DRY	7218	7644	8071	8563	9055	9547	10103	10663
STD=3°C	WET	9022	9479	10006	10531	11056	11581	12139	12730
8000 FT	DRY	7776	8202	6894	9219	9777	10335	10925	11324
STD=–1°C	WET	9678	10203	10761	11319	11909	12500	13123	13747

(CONTINUED)

HYD 1 & 2 FAIL (Continued)

FLAPS RETRACTED

NO

(CONTINUED)

0/EXT ESTIMATED LANDING DISTANCES (FEET) HYDRAULIC SYSTEM 1 & 2 FAIL

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
10000 FT	**DRY**	8366	8858	9383	9974	10597	11319	12106	12959
STD=–5°C	**WET**	10433	10991	11614	12237	12894	13648	14436	15256

NOTE: Standard day, no wind, zero slope, three engines at maximum rverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then three engines at forward idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C

FEET PER °C	Dry	Wet
BELOW standard day	–20	–23
ABOVE standard day	+66	+69

Slope: Valid from –2% downhill +2% uphill

FEET PER 1% SLOPE	Dry	Wet
UPHILL	–116	–194
DOWNHILL	+659	+906

Wind: Valid from –10 knot tailwind +20 knot headwind

FEET PER KNOT	Dry	Wet
HEADWIND	–49	–66
TAILWIND	+213	+226

FLAP/SLAT Handle. 0/EXT

NOTE: Slats may not extend until speed is reduced.

When ready to extend landing gear,

Airspeed . . . . MAX 230 KIAS

Alternate Gear Extension Lever . . . . RAISE,LATCH

HYD 1 & 2 FAIL (Continued)

After three green lights illuminate,

Center Gear Alternate Extension Handle,Lights . . . . . . PULL,4 GREEN

GEAR Handle . . . . DOWN

After 2 minutes,

Alternate Gear Extension Lever . . . . .STOW

AUTO BRAKE Selector. . . . . OFF

Cross threshold at V_{APP}, reduce sink rate slightly, disconnect autothrottles, retard throttles to idle and fly a positive touchdown. Do not hold aircraft off. Excessive flare will result in float and excessive use of runway.

CAUTION: Tail strike may occur at pitch attitudes greater than 10°.

Manually assist spoiler handle as it deploys.

NOTE: If go-around is required, it is recommended that landing gear not be retracted. If gear retraction is necessary, delay until aircraft is clear of obstacles.

[END]

HYD 1 & 3 FAIL

NOTE: Increased fuel consumption, up to approximately 15%, may result due to control surface float.

Reduce gross weight as desired.

When ready for approach,

SLATS RETRACTED

NO

SLAT STOW Switch . . . . .SLAT STOW

FLAP/SLAT Handle . . . . 0/EXT POSITION

V_{MCA} is 140 KIAS

"RUDDER UPR INOP" ALERT DISPLAYED

NO

V_{MCA} is 160 KIAS

Recommended maximum crosswind component is 12 knots.

Stabilizer is inoperative.

Complete Abnormal Non-Alert Procedure- STABILIZER INOPERATIVE then continue with this procedure.

FLAP/SLAT Handle . . . . 28/EXT POSITION

When ready to extend landing gear,

Airspeed. . . . . .MAX 230 KIAS

Alternate Gear Extension Lever . . . . RAISE,LATCH

(CONTINUED)

HYD 1 & 3 FAIL (Continued)

After three green lights illuminate,

Center Gear Alternate Extension Handle,Lights . . . . . . PULL,4 GREEN

GEAR Handle . . . . DOWN

AUTO BRAKE Selector . . . . OFF

SLATS EXTENDED

NO

Plan a 35/EXT approach and landing.

Manually assist spoiler handle as it deploys.

[END]

GPWS Switch . . . . FLAP OVRD

Plan a 28/RET approach and landing.

28/RET APPROACH SPEEDS
HYDRAULIC SYSTEMS 1 & 3 FAIL

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{APP} (V_{REF} + 5)	175	180	185	190	195	200	205	209

(CONTINUED)

HYD 1 & 3 FAIL (Continued)

28/RET ESTIMATED LANDING DISTANCES (FEET) HYDRAULIC SYSTEMS 1 & 3 FAIL

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	7546	7940	8207	8661	8989	9350	9711	10105
STD=15°C	**WET**	9810	10269	10490	10728	11647	12172	12664	13156
2000 FT	**DRY**	8005	8397	8760	9153	9514	9908	10302	10695
STD=11°C	**WET**	10433	10925	11417	11942	12434	12959	13481	14009
4000 FT	**DRY**	8497	8891	9285	9711	10105	10351	10925	11352
STD=7°C	**WET**	11122	11680	12172	12730	13254	13812	14370	14960
6000 FT	**DRY**	9022	9239	9875	10335	10728	11188	11614	12073
STD=3°C	**WET**	11909	12500	13025	13615	14173	14796	15354	15978
8000 FT	**DRY**	9580	10072	10499	10991	11417	11909	12336	12828
STD=–1°C	**WET**	12730	13350	13943	14600	15190	15814	16437	17126
10000 FT	**DRY**	10203	10728	11188	11713	12172	12828	13484	14140
STD=–5°C	**WET**	13648	14337	14960	15649	16273	17159	18012	18865

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–20	–30
ABOVE standard day	+49	+69

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–197	–390
DOWNHILL	+561	+1293

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–56	–85
TAILWIND	+116	+194

(CONTINUED)

HYD 1 & 3 FAIL (Continued)

Cross threshold at V_{APP}, reduce sink rate slightly. Disconnect autothrottles, retard throttles to idle and raise nose of aircraft to at least a level attitude. Do not hold aircraft off. Excessive flare will result in float and excessive use of runway.

CAUTION: Tail strike may occur at pitch attitudes greater than 10°.

Manually assist spoiler handle as it deploys.

[END]

HYD 2 & 3 FAIL

NOTE: I

1. Increased fuel consumption, up to approximately 15%, *may result due to control surface float.*
2. *Outboard slats will not retract if they were extended before the loss of pressure occurred. "SLAT DISAG" alert will be displayed when flap/slat handle is in the 0/RET position.*

Lower rudder is inoperative.

V_{MCA} is 180 KIAS.

CAUTION: Do not attempt a go-around at speeds below V_{MCA}.

Recommended maximum crosswind component is 12 knots.

Reduce gross weight as desired.

When ready for approach,

FLAP/SLAT Handle 28/EXT

When ready to extend landing gear,

Airspeed. . . . . .MAX 230 KIAS

Alternate Gear Extension Lever . . . . RAISE,LATCH

After three green lights illuminate,

Center Gear Alternate Extension Handle,Lights. . . . . . PULL,4 GREEN

GEAR Handle . . . . DOWN

AUTO BRAKE Selector . . . . OFF

GPWS Switch . . . . FLAP OVRD

28/EXT APPROACH SPEEDS
HYDRAULIC SYSTEMS 2 & 3 FAIL

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{APP} (V_{REF} + 8)	148	153	157	161	165	169	173	176

(CONTINUED)

HYD 2 & 3 FAIL (Continued)

28/EXT ESTIMATED LANDING DISTANCES (FEET) HYDRAULIC SYSTEMS 2 & 3 FAIL

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	5184	5446	5709	6004	6299	6594	6955	7316
STD=15°C	**WET**	6496	6824	7152	7480	7808	8136	8464	8858
2000 FT	**DRY**	5512	5646	5935	6398	6726	7054	7415	7808
STD=11°C	**WET**	6992	7249	7617	7972	8333	8661	9055	9449
4000 FT	**DRY**	5840	6168	6463	6824	7185	7546	7940	8180
STD=7°C	**WET**	7382	7743	8104	8497	8891	9285	9678	10138
6000 FT	**DRY**	6234	6594	6922	7283	7710	8071	8497	8990
STD=3°C	**WET**	7874	8268	8694	9088	9547	9941	10400	10892
8000 FT	**DRY**	6993	7054	7448	7841	8268	8694	9154	9711
STD=–1°C	**WET**	8465	8891	9818	9777	10236	10696	11188	11745
10000 FT	**DRY**	7185	7579	8005	8432	8924	9482	10039	10597
STD=–5°C	**WET**	9088	9547	10007	10532	11056	11647	12205	12828

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then three engines at forward idle to stop (includes air run distances)

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–16	–20
ABOVE standard day	+49	+56

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–105	–200
DOWNHILL	+302	+778

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–46	–66
TAILWIND	+161	+190

(CONTINUED)

HYD 2 & 3 FAIL (Continued)

Cross threshold at V_{APP}, reduce sink rate slightly. Disconnect autothrottles, retard throttles to idle and fly to a positive touchdown. Do not hold aircraft off. Excessive flare will result in float and excessive use of runway.

CAUTION:Tail strike may occur at pitch attitudes greater than 10°.

Manually assist spoiler handle as it deploys.

[END]

NO MASKS

NO MASKS Switch . . . . PUSH AND HOLD 3 TO 5 SECONDS

[END]

TNK__FUEL QTY LO

DUMP Switch . . . . OFF

DUMP SWITCH ON LIGHT FAILS TO EXTINGUISH

NO

FUEL DUMP EMER STOP Switch. . . . . STOP

Reschedule fuel distribution as required.

[END]

Intentionally
Blank

Emergency Procedures — Chapter EP

Table Of Contents — Section 20

EMERGENCY NON - ALERT

Intentionally
Blank

AIRSPEED: LOST, SUSPECT OR ERRATIC

NOTE: The following information and displays can be considered reliable: PFD attitude, NAV display, ground speed, engine N_1 and stickshaker.

AFS OVRD Switches . . . . OFF

Aircraft Pitch/Thrust. . . . . STABILIZE

Disregard IAS/Flight Director pitch bar and high speed warnings. Use pitch attitude and thrust as the primary flight reference. Should stick shaker be encountered, lower nose to horizon and increase thrust. Resume pitch/thrust reference using the AIRSPEED : LOST, SUSPECT OR ERRATIC table after the stick shaker ceases.

Flight Director . . . . OFF

Disregard all alerts and warnings, except stick shaker, until after aircraft is stabilized and safe operations achieved. Alerts and aural warnings can produce conflicting and disorienting cues.

CAUTION: Under certain failures FPA and PLI may be unreliable. Check against primary flight references before using FPA or PLI.

If practical fly to VFR conditions at earliest possible opportunity.

After the aircraft is safely stabilized in flight, ensure terrain avoidance.

NOTE: Approximately 10 degrees pitch attitude and MCT thrust will provide a safe initial climb condition if a climb is required.

Pilot and Standby Flight Instruments. . . . . COMPARE

(CONTINUED)

AIRSPEED: LOST, SUSPECT OR ERRATIC (Continued)

ABLE TO IDENTIFY UNRELIABLE AIR DATA SOURCE

NO

CADC (Unreliable Side) . . . . SELECT TO OTHER SIDE

NOTE: The OVERSPEED aural warning may continue since there is no connection between the CAWS and the CADC switch.

Static Air Switch (Unreliable Side) . . . . ALT

AIR DATA RETURNS TO NORMAL

NO

AFS OVRD OFF Switch
(Reliable Side) . . . . NORMAL POSITION

Use autopilot and autothrottles associated with the reliable ADC.

Flight directors may be engaged using the reliable ADC.

If ADC 1 is reliable,

• Select FLT DIR . . . . F/O ON 1

If ADC 2 is reliable,

• Select FLT DIR . . . . CAPT ON 2

NOTE: The following information and displays may or may not be reliable: FMC (unreliable side) data associated with air data and TAS and WIND on ND (unreliable side).

Continue to monitor pitch, thrust, and airspeed to ensure accuracy of selected instruments.

[END]

Attitude and Thrust . . . . ADJUST

(CONTINUED)

AIRSPEED: LOST, SUSPECT OR ERRATIC (Continued)

Maintain nomal pitch attitude and thrust for the phase of flight.

NOTE: The following information and displays can be considered reliable: PFD attitude, NAV display, groundspeed, engine N_1 and stickshaker.

The following may or may not be reliable depending on the cause of lost or suspect airspeed: FPA, PLI, low speed pitch protection, IVSI, altimeter, altitude reporting and TCAS. FMS NAV function may be inoperative.

NOTE: The following will not be reliable: Flight director pitch bar, autothrottle speed protection, high speed pitch protection and overspeed warning.

Use the following AIRSPEED: LOST, SUSPECT OR ERRATIC tables to determine thrust/pitch relation for remainder of flight.

NOTE: IAS and vertical speed (VS) values in the following tables are approximate.

FLIGHT PHASE	CONFIG	PRESSURE ALTITUDE	REF	WEIGHT (1000 KG)			
				200	**240**	**280**	**285**
CLIMB	UP/RET	5000	PITCH	14.0	12.0	10.0	9.5
			IAS	*247*	*270*	*295*	*299*
		FL 100	PITCH	13.0	10.5	8.5	8.0
Use max thrust			*IAS*	*248*	*277*	*307*	*311*
(throttles to		FL150	PITCH	11.0	8.5	7.0	6.5
overboost bar)			*IAS*	*256*	*288*	*317*	*321*
		FL 200	PITCH	8.5	6.5	5.5	5.0
			IAS	*266*	*298*	*327*	*331*
CRUISE	UP/RET	FL 100	PITCH	2.0	2.5	3.0	3.0
			N_1	76.5	78.6	80.8	81.1
			IAS	*330*	*330*	*330*	*330*
		FL 200	PITCH	2.0	2.5	3.0	3.0
			N_1	83.4	85.5	87.9	88.2
			IAS	*330*	*330*	*330*	*330*
Use N_1 for thrust setting		FL 300	PITCH	2.0	2.5	3.0	3.0
			N_1	89.0	91.6	95.0	95.5
			MACH	.827	.827	.827	.827
			IAS	*315*	*315*	*315*	*315*
		FL 350	PITCH	2.5	--	--	--
			N_1	90.9	--	--	--
			MACH	.830	--	--	--
			IAS	*283*	--	--	--

(CONTINUED)

AIRSPEED: LOST, SUSPECT OR ERRATIC (Continued)

FLIGHT PHASE	CONFIG	PRESSURE ALTITUDE	REF	WEIGHT (1000 KG) 160	180	200	220
DESCENT	UP/RET	FL 350	PITCH	1.0	--	--	--
			MACH	.768	--	--	--
			IAS	*260*	--	--	--
			VS	2020	--	--	--
		FL 300	PITCH	1.5	1.5	1.5	--
			MACH	.693	.715	.772	--
Use Idle thrust			*IAS*	*260*	*269*	*292*	--
			VS	1920	2000	2170	--
		FL 200	PITCH	1.5	2.5	2.5	--
			IAS	*260*	*260*	*278*	--
			VS	1770	1760	1880	--
		FL 100	PITCH	2.0	2.5	--	--
			IAS	*250*	*262*	--	--
			VS	1500	1560	--	--
ARRIVAL LVL FLT	UP/RET	5000	PITCH	5.0	5.0	5.0	5.0
			N_1	59.1	62.1	64.9	67.5
			IAS	*222*	*235*	*247*	*259*
	0/EXT	3000	PITCH	8.5	8.5	8.5	8.5
			N_1	61.7	64.9	67.8	70.6
			IAS	*183*	*193*	*203*	*213*
Use N_1 for thrust setting	15/EXT	3000	PITCH	6.0	6.0	6.5	6.5
			N_1	64.9	68.2	71.2	74.0
			IAS	174	*184*	*194*	*203*
	28/EXT	3000	PITCH	4.0	4.0	4.5	4.5
			N_1	70.4	73.8	76.9	79.7
			IAS	*169*	*178*	*187*	*196*
APPROACH *IAS* APPROX V_{REF} + 15 Use N_1 for thrust setting	35/EXT GEAR DOWN	DESCENT	PITCH	2.5	2.5	2.5	3.0
			N_1	62.2	65.4	68.3	70.8
			IAS	*153*	*162*	*170*	*177*
	Maintain pitch and adjust power to maintain glide path.						
GO AROUND	28/EXT GEAR UP	SEA LVL	PITCH	20.0	20.0	19.5	18.0
			IAS	*180*	*172*	*170*	*177*
		5000	PITCH	20.0	18.5	17.0	15.5
			IAS	*159*	*162*	*170*	*177*

When ready for approach and landing,

- Maintain VFR conditions.
- Establish landing configuration early.
- Use IRS ground speed and reported winds to verify airspeed.
- Use radar altimeter.
- Use a runway with electronic or visual glideslope.

[END]

ALL ENGINE FLAMEOUT

NOTE: All engine flameout can be recognized by decrease in EGT, N_2 and fuel flow. This will be followed closely by a decrease in N_1.

ENG IGN OVRD Switch . . . . OVRD ON

ADG . . . . DEPLOY

Air-start envelope is 250 KIAS to V_{MO}, SL to FL300. Control aircraft at an IAS to obtain a minimum N_2 of 15% for air start.

MINIMUM AIRSPEED FOR CONTROLLABILITY (KIAS)

Gross Weight (1000 KG)	160	180	200	230	250
UP/RET	188	200	209	224	234
0/EXT	155	161	170	182	190
28/EXT	155	155	157	168	176

NOTE: If required, and time permits, the CABIN PRESS System may be operated in MANUAL, with the outflow selected CLOSED, to minimize cabin rate of climb, until an engine restart is achieved. When an engine restarts, return the CABIN PRESS system to automatic mode.

Throttles (All) . . . . IDLE

NOTE: Airstarts, especially at high altitudes, may result in the engine(s) accelerating to idle very slowly. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N2 is steadily increasing and EGT remains within limits, the start is progressing normally.

ANY ENGINE RESTARTS

NO

ADG ELEC Switch . . . . ON

Remaining Engine(s) . . . . RESTART

Refer to Abnormal Non-Alert procedure - ENGINE RESTART IN FLIGHT.

(CONTINUED)

ALL ENGINE FLAMEOUT (Continued)

ANY ENGINE RESTARTS

NO (CONTINUED)

Verify all systems are operating as required.

Land at nearest suitable airport.

[END]

FUEL Switches (All). . . . . .OFF, THEN ON

NOTE: If N2 is steadily increasing, and EGT remians within limits, the start is progressing normally. Additional cycling of the fuel switches will result in delayed engine recovery times.

ANY ENGINE RESTARTS

NO

ADG ELEC Switch . . . . ON

Remaining Engines(s) . . . . RESTART

Refer to Abnormal Non-Alert procedure - ENGINE RESTART IN FLIGHT.

Verify all systems are operating as required.

Land at nearest suitable airport.

[END]

Flaps/Slats . . . . .MAINTAIN

DITCHING REQUIRED

NO

Landing Gear . . . . .UP

Refer to Abnormal Non-Alert procedure - DITCHING.

[END]

Main Landing Gear . . . . .ALTN EXT

Center gear may be extended as desired.

Move gear handle down.

Do not stow alternate gear extension lever.

[END]

EMERGENCY DESCENT

Altitude Select Knob	REDUCE,PULL

Initiate descent to 10,000 feet or minimum safe altitude, whichever is higher.

SPOILER Handle	SPD BRK FULL

WARNING: ***If structural damage is suspected or turbulence present, do not exceed .82 Mach/305 KIAS.***

IAS/MACH Select Knob	SELECT .85 MACH/320-350 KIAS
Descent.	MAX PITCH 10°,MAX BANK 30°
Transponder (Unless Otherwise Required).	7700
NO SMOKE and SEAT BELTS Switches	ON

To reactivate boom mike when O_2 mask is no longer required,

Eros O_2 Mask Doors . . . . CLOSE

PRESS TO TEST AND RESET Lever . . . . PUSH

[END]

REVERSER DEPLOYED OR U/L OR REV DISPLAYED IN FLIGHT

AIRCRAFT BUFFET OR TRIM CHANGE

NO

Take immediate corrective action as necessary to maintain aircraft control.

Throttle (Affected Engine) . . . . IDLE

Reverser Levers . . . . FULL DOWN (FWD IDLE)

U/L OR REV REMAINS DISPLAYED OR AIRCRAFT BEHAVIOR STILL ABNORMAL

NO

Fuel Switch (Affected Engine). . . . . OFF

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

Set autopilot and autothrottles as desired.

Gross Weight . . . . REDUCE, AS REQUIRED

Use 35° flaps for landing.

[END]

Continue use of affected engine at PIC's discretion.

Set autopilot and autothrottles as required.

[END]

Continue normal engine operation.

[END]

SMOKE/FIRE/FUMES

DIVERSION MAY BE REQUIRED.

Oxygen Masks (If Required) ON/100%

Don smoke goggles as required.

Use emergency oxygen pressure, as required to purge mask and goggles of smoke and/or fumes.

Crew/Courier(s) Communications . . . . ESTABLISH

EMER LT Switch . . . . ON

CAB BUS Switch . . . . OFF

Lift guard and push CAB BUS switch. Observe OFF light illuminates.

NOTE: When CAB BUS OFF light is illuminated, power is removed from the cabin AC ground service bus and cargo loading bus.

Anytime smoke or fumes becomes the greatest threat, refer to the Abnormal Non-Alert procedure SMOKE/FUMES REMOVAL.

NOTE: If fumes are identified as fuel/oil and an increase in oil quantity is observed, refer to Abnormal Non-Alert procedure ENGINE OIL QUANTITY INCREASE.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SOURCE OBVIOUS AND ACCESSIBLE

NO

Isolate and extinguish the source. If possible remove power from the affected equipment by switch or circuit breaker on the flight deck or in the cabin.

SOURCE CONFIRMED EXTINGUISHED AND SMOKE/FUMES ARE DECREASING

NO

Flight may be continued at Captain's discretion.

At Captain's discretion,

CAB BUS Switch . . . . ON

Lift guard and push CAB BUS switch. Observe OFF light extinguishes.

NOTE: Pushing the CAB BUS switch restores power to all cabin AC ground service bus and cargo loading bus.

At Captain's discretion,

EMER LT Switch . . . . .ARM

[END]

Diversion . . . . INITIATE

Initiate a diversion to the nearest suitable airport while continuing the checklist.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

WARNING: ***Consider an immediate landing if smoke, fire, or fumes become uncontrollable.***

Do not delay landing in order to accomplish Emergency or Abnormal procedures as well as fuel dump.

LANDING IMMINENT

NO

Time permitting and at Captain's discretion, review OPERATIONAL CONSIDERATIONS located at the end of this procedure.

[END]

Do not delay landing in an attempt to complete all of the following steps:

NOTE: Direct courier(s) to fight the fire.

ATC Switch . . . . SELECT 1

Verify ATC switch 1 is selected on transponder panel.

VHF-1 or HF-1. SELECT VHF-1 AND/OR HF-1

Push the VHF-1 or HF-1 volume control knob, as appropriate, to receive and push the appropriate MIC/CALL light to transmit.

Autothrottles . . . . OFF

Throttles . . . . SET 70% N1 OR LESS

FADEC MODE Switches (All Engines) . . . . .SELECT ALTN

Open cover and push each FADEC MODE switch. Observe each ALTN light illuminates.

"ENG(1, 2, 3) FADEC ALTN" Level 1 alerts will be displayed on secondary engine display.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

AP 1 (If Desired) . . . . .SELECT

If autopilot is desired, verify AP 1 is displayed on FMA.

Fuel dump considerations,

CAUTION: If conditions permit, delay fuel dump until smoke switch is in its final position. if fuel dump is started prior to or during smoke switch operation, various valves and pumps may not be controllable which may result in an uncontrollable fuel dump.

SMOKE ELEC/AIR Selector . . . . 3/1 OFF

Push and rotate SMOKE ELEC/AIR selector to 3/1 OFF.

NOTE: All actions are the same whether systems are in automatic or manual.

Evaluate smoke/fumes for up to 2 minutes.

CAUTION: If smoke and/or fumes increase during the waiting period, proceed immediately to the next step.

NOTES: With SMOKE ELEC/AIR selector in 3/1 OFF, generator channel 3 is unpowered and BLEED AIR 1 and PACK 1 are off.

Observe displayed alerts and consequences that are a result of moving SMOKE ELEC/AIR selector out of NORM.

SMOKE/FUMES DECREASE

NO

Leave SMOKE ELEC/AIR selector in 3/1 OFF for remainder of flight.

NOTE: If "FLAP DISAG" alert is displayed, use left inboard flap indication on CONFIG synoptic or PFD to determine flap position.

ENG IGN. . . . . VERIFY A SELECTED

If required, push ENG IGN switch A and observe switch light A illuminates.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SMOKE/FUMES DECREASE

NO

(CONTINUED)

FLAP LIMIT Selector . . . . OVRD 1

Rotate FLAP LIMIT selector to OVRD 1 and observe MANUAL light illuminates.

Manual Thrust Limits . . . . SELECT G/A

Select G/A thrust limit on MCDU THRUST LIMITS page and observe G/A thrust limit is displayed on EAD.

AUTO BRAKE Selector . . . . OFF

NOTE: The following will be inoperative:

- *Ignition system B*
- *GA switch*
- *Auto brakes*
- *Auto slats*
- *BLEED AIR 1*
- *ECON system*
- *F/O pitot heat*
- *PACK 1*
- *Landing gear aural warning*
- *Communications radio panel 2*
- *Audio control panel 2*
- *VHF-2*
- *VHF-3*
- *HF-2*
- *Tail anti-ice*

GO AROUND REQUIRED

NO

Autopilot/Autothrottles . . . . DISCONNECT

Thrust . . . . SET

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SMOKE/FUMES DECREASE

NO

(CONTINUED)

GO AROUND REQUIRED

NO

(CONTINUED)

Speed. . . . . Vref (28) + 5

When positive rate of climb is indicated,

GEAR Handle . . . . UP

Land at nearest suitable airport.

Time permitting and at Captain's discretion, review OPERATIONAL CONSIDERATIONS located at the end of this procedure.

[END]

SMOKE ELEC/AIR Selector . . . . 2/3 OFF

Push and rotate SMOKE ELEC/AIR selector to 2/3 OFF.

Evaluate smoke/fumes for up to 2 minutes.

CAUTION: If smoke and/or fumes increase during the waiting period, proceed immediately to the next step.

NOTES: With SMOKE ELEC/AIR selector in 2/3 OFF, generator channel 2 is unpowered and BLEED AIR 3 and PACK 3 are off.

Generator channel 3, air system 1 and PACK 1 will be restored.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SMOKE/FUMES DECREASE

NO

Leave SMOKE ELEC/AIR selector in 2/3 OFF for remainder of flight.

NOTES: Landing gear position indications may be observed on CONFIG synoptic.

Control wheel trim switches are inoperative. Use LONG TRIM handles when stabilizer trim is desired.

The following will be inoperative:

- *AUX pitot heat*
- *BLEED AIR 3*
- *Engine 2 reverser*
- *PACK 3*
- *Horizontal stabilizer trim switches*
- *Landing gear position indicator lights*

Land at nearest suitable airport.

Time permitting and at Captain's discretion, review OPERATIONAL CONSIDERATIONS located at the end of this procedure.

[END]

AP 2 (If Desired) . . . . SELECT

If autopilot is desired, verify AP 2 is displayed on FMA.

ATC Switch . . . . SELECT 2

Verify ATC switch 2 is selected on transponder panel.

VHF-2 or HF-2. SELECT VHF-2 AND/OR HF-2

Push the VHF-2 or HF-2 volume control knob, as appropriate, to receive and push the appropriate MIC/CALL light to transmit.

SMOKE ELEC/AIR Selector . . . . 1/2 OFF

Push and rotate SMOKE ELEC/AIR selector to 1/2 OFF.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

Evaluate smoke/fumes for up to 2 minutes.

CAUTION: If smoke and/or fumes increase during the waiting period, proceed immediately to the next step.

NOTES: With SMOKE ELEC/AIR selector in 1/2 OFF, generator channel 1 is unpowered and BLEED AIR 2 and PACK 2 are off.

Generator channel 2 and air system 3 will be restored.

SMOKE/FUMES DECREASE

NO

Leave SMOKE ELEC/AIR selector in 1/2 OFF for remainder of flight.

NOTE: If "FLAP DISAG" alert is displayed, use right inboard flap indication on CONFIG synoptic or PFD to determine flap position.

ENG IGN. . . . . VERIFY B SELECTED

If required, push ENG IGN switch B and observe switch light B illuminates.

FLAP LIMIT Selector . . . . OVRD 2

Rotate FLAP LIMIT selector to OVRD 2 and observe MANUAL light illuminates.

Manual Thrust Limits . . . . SELECT G/A

Select G/A thrust limit on MCDU THRUST LIMITS page and observe G/A thrust limits is displayed on EAD.

CAPT FLT DIR OFF Switch . . . . OFF

Push CAPT FLT DIR OFF switch and observe OFF light illuminates. Captain's flight director must be off to allow normal functioning of First Officer's flight director.

AUTO BRAKE Selector. . . . . OFF

NOTES: Autopilot will become inoperative when IRU 1 and IRU AUX become inoperative.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SMOKE/FUMES DECREASE

NO

(CONTINUED)

The following will be inoperative.

- *IGNITION system A*
- *GA switch*
- *Auto slats*
- *Auto GND spoilers*
- *BLEED AIR 2*
- *Captain pitot heat*
- *Engine 1 reverse*
- *Engine 3 reverse*
- *GPWS*
- *GPWS glideslope lights*
- *PACK 2*
- *Communications radio panel 1*
- *Audio control panel 1*
- *VHF-1*
- *HF-1*
- *Wing anti-ice*

GO AROUND REQUIRED

NO

Autopilot/Autothrottles . . . . DISCONNECT

Thrust . . . . SET

Speed . . . . Vref (28) + 5

When positive rate of climb is indicated,

GEAR Handle . . . . UP

Land at nearest suitable airport.

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

SMOKE/FUMES DECREASE

NO

(CONTINUED)

Time permitting and at Captain's discretion, review OPERATIONAL CONSIDERATIONS located at the end of this procedure.

[END]

SMOKE ELEC/AIR Selector . . . . NORM

Push and rotate SMOKE ELEC/AIR selector to NORM.

FADEC MODE Switches (All Engines) . . . . .PUSH

Open cover and push each FADEC MODE switch. Observe each ALTN light extinguishes.

"ENG (1, 2, 3,) FADEC ALTN" level 1 alerts will be removed from secondary engine display.

Land at nearest suitable airport.

WARNING: ***Consider an immediate landing if smoke, fire, or fumes become uncontrollable.***

Do not delay landing in order to accomplish Emergency or Abnormal procedures as well as fuel dump.

OPERATIONAL CONSIDERATIONS

(CONTINUED)

SMOKE/FIRE/FUMES (Continued)

- Declare an emergency
- Accomplish the Abnormal Non-Alert procedure SMOKE/FUMES REMOVAL if required
- Expedite descent/arrival
- Request radar vector direct to the final approach
- Do not delay approach and landing
- Overweight landing may be necessary
- Tailwind landing may be necessary
- Consider use of AUTOLAND
- Use braking appropriate for conditions
- After stopping the aircraft, consider accomplishing the Abnormal Non-Alert procedure - EVACUATION.

[END]

Intentionally
Blank

TWO ENGINES INOPERATIVE

NOTE: During a 2 engine approach, if a second engine fails on final and the gear is down, add power as required. Set flaps to FLAPS 28, maintain speed to reach V_{REF} + 5 for FLAPS 28 and continue approach. Move GPWS switch to FLAP OVRD, conditions permitting.

Throttles . . . . MCT
Flaps (Unless On Final) . . . . UP
Speed (Unless On Final) . . . . DRIFTDOWN OR UP/RET V_{MIN} + 30

NOTE: *Delay gear retraction until flaps are up.*

Gear (Unless Committed) . . . . UP
Slats (Unless On Final) . . . . RET

ENG IGN OVRD Switch . . . . ON
Plan to land at nearest suitable airport.

DRIFTDOWN REQUIRED

NO

Autothrottles . . . . OFF
Thrust . . . . MCT
Driftdown Speed Schedule . . . . CHECK

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

DRIFTDOWN REQUIRED

NO

(CONTINUED)

DRIFTDOWN

ALT (1000 ft)	UNITS	INITIAL GROSS WEIGHT (1000 KG) 140	160	180	200	220	240	260
40	Kts	227	245	249	-Start of Drift Down Speed			
	Ft	25454	22130	19292	-Max 1 Eng Alt (Airfoil & Engine A-ice ON, reduce by 1500ft)			
	NMi	348	376	387	-Distance to Bottom of Driftdown			
	Kg	4265	5117	5802	-Fuel Burned to Bottom of Driftdown			
	Kts	216	230	244	-Bottom of Driftdown Speed 1 Eng Altitude			
	Kg/Hr	4638	5311	6033	-Fuel Flow at 1 Engine Altitude			
35	Kts	224	241	257	272	279		
	Ft	25406	22095	19265	16328	13623		
	NMi	307	342	361	393	407		
	Kg	3974	4878	5603	6660	7410		
	Kts	216	230	244	256	268		
	Kg/Hr	4649	5320	6039	6672	7348		
30	Kts	221	238	253	268	282	296	309
	Ft	25312	22031	19212	16285	13588	10688	7482
	NMi	249	296	323	360	379	427	483
	Kg	3410	4448	5291	6367	7184	8710	10613
	Kts	217	231	244	256	268	279	289
	Kg/Hr	4670	5336	6050	6682	7356	7953	8493
25	Kts		235	250	265	278	292	304
	Ft		21887	19113	16212	13528	10631	7421
	NMi		218	269	316	342	393	453
	Kg		3482	4663	5871	6791	8373	10335
	Kts		231	245	257	269	279	289
	Kg/Hr		5372	6071	6699	7369	7964	8500

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

DRIFTDOWN REQUIRED

NO

(CONTINUED)

ALT (1000 ft)	UNITS	INITIAL GROSS WEIGHT (1000 KG)						
		140	160	180	200	220	240	260
20	Kts			248	262	276	288	301
	Ft			18836	16066	13421	10538	7323
	NMi			156	248	291	351	416
	Kg			2896	4882	6097	7815	9888
	Kts			246	257	269	279	289
	Kg/Hr			6131	6732	7391	7981	8511
15	Kts					273	286	298
	Ft					13142	10344	7141
	NMi					192	284	365
	Kg					4284	6660	9060
	Kts					270	280	290
	Kg/Hr					7451	8017	8531
10	Kts						284	296
	Ft						9786	6733
	NMi						120	277
	Kg						2983	7203
	Kts						282	291
	Kg/Hr						8141	8577

Minimum Safe Altitude/Range Capability . . . . . . DETERMINE

Fuel Dump . . . . CONSIDER

Driftdown Altitude . . . . SELECT

Review the following and then continue checklist.

At bottom of driftdown:

- Level at maximum one engine altitude.
- Maintain altitude and allow aircraft to accelerate to 290 KIAS as gross weight is reduced by fuel burn.

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

DRIFTDOWN REQUIRED

NO

(CONTINUED)

- Maintain 290 KIAS for cruise/climb (autothrottles may be used).

Appropriate Engine Shutdown Procedure . . . . COMPLETE

ENG IGN OVRD Switch . . . . OFF

NOTE: Consider fuel dump to reduce gross weight to minimum practical.

Autothrottles . . . . .AS REQUIRED

Altimeters . . . . SET,CROSS CHECK

GPWS. . . . . .FLAP OVRD

At or below 10,000 feet MSL,

AIR SYS MANUAL

NO

Pack Switches (All) . . . . OFF

HYDRAULIC SYSTEM MANUAL

NO

1-3/2-3 RMPs. . . . . ON

Maintain clean configuration and UP/RET V_{MIN} + 30 until maneuvering has been completed.

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

TWO ENGINES INOPERATIVE SPEEDS

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
UP/RET V_{MIN} + 30	217	223	228	234	240	245	250	255
0/EXT V_{MIN} + 30	182	187	191	196	200	204	208	212
0/EXT V_{MIN} + 15	167	172	176	181	185	189	193	197

0/EXT ESTIMATED LANDING DISTANCES (FEET)
TWO ENGINES INOPERATIVE

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	DRY	4987	5249	5512	5774	6070	6332	6594	6890
STD=15°C	WET	6857	7251	7644	8058	8466	8825	9252	9678
2000 FT	DRY	5282	5545	5840	6135	6430	6693	7021	7316
STD=11°C	WET	7251	7677	8104	8530	8990	9416	9843	10302
4000 FT	DRY	5577	5873	6168	6496	6824	7119	7448	7776
STD=7°C	WET	7710	8169	8629	9088	9580	10039	10532	11024
6000 FT	DRY	5906	6234	6562	6890	7283	7579	7940	8301
STD=3°C	WET	8202	8694	9186	9678	10236	10728	11253	11778
8000 FT	DRY	6266	6627	6988	7349	7743	8104	8497	8891
STD=–1°C	WET	8727	9285	9810	10367	10958	11483	12073	12664
10000 FT	DRY	6660	7054	7448	7841	8301	8727	9252	9810
STD=–5°C	WET	9318	9908	10499	11089	11745	12434	13189	13976

NOTE: Standard day, no wind, zero slope, no reverse thrust (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–16	–23
ABOVE standard day	+43	+66

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–75	–180
DOWNHILL	+322	+787

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–36	–62
TAILWIND	+135	+213

Review the following MISSED APPROACH Caution.

CAUTION: Do not attempt a go-around under any of the following conditions:

- ***Less than 1,000 feet AGL.***
- ***Airspeed below 0/EXT V_{MIN} + 30.***
- ***Gear is extended.***
- ***Hydraulic 1 or 3 failed. Outboard slats will not retract. "SLAT DISAG" alert will be displayed when flap/slat handle is in the 0/RET position.***
- ***Weight, altitude and temperature in excess of those shown in the following chart.***

MAXIMUM WEIGHT FOR TWO ENGINES INOPERATIVE MISSED APPROACH

PRESS ALT	TEMPERATURE MAXIMUM WEIGHT (1000 KG)								
					STD TEMP				
S.L.	-65°C 235	-45°C 237	-25°C 239	-5°C 240	15°C 241	25°C 242	35°C 222	45°C 206	55°C 189
2000 FT	-69°C 227	-49°C 229	-29°C 231	-9°C 232	11°C 233	21°C 233	31°C 214	41°C 198	51°C 182
4000 FT	-73°C 221	-53°C 222	-33°C 223	-13°C 222	7°C 222	17°C 221	27°C 207	37°C 190	47°C 175
6000 FT	-77°C 211	-57°C 211	-37°C 211	-17°C 211	3°C 210	13°C 209	23°C 201	33°C 185	43°C 169
8000 FT	-- --	-61°C 199	-41°C 198	-21°C 197	-1°C 195	9°C 195	19°C 194	29°C 179	39°C 162
10000 FT	-- --	-65°C 185	-45°C 184	-25°C 183	-5°C 182	5°C 181	15°C 180	25°C 172	35°C 159
NOTE: Packs off, engine and airfoil ice protection off.									

(CONTINUED)

TWO ENGINES INOPERATIVE (Continued)

On final,

FLAP/SLAT Handle 0/EXT

Speed . . . . 0/EXT V_{MIN} + 30

> *NOTE: For go-around protection, maintain 0/EXT V_{MIN} + 30 until committed to land (1,000 feet AGL). Achieve V_{MIN} + 15 at or above 50 feet AGL.*

MISSED APPROACH REQUIRED

NO →

> Go-Around Thrust . . . . SET
>
> FLAP/SLAT Handle . . . . UP/RET
>
> Speed. . . . . UP/RET V_{MIN} + 30
>
> Maintain approach descent rate during slat retraction until attaining UP/RET V_{MIN} + 30, then initiate climb.
>
> **[END]**

At 1,000 feet AGL,

GEAR Handle,Lights. . . . . DOWN,4 GREEN

> *NOTE: Do not use autobrakes.*

Spoilers . . . . ARM

Speed (Achieve at or Above 50 Feet). . . . . 0/EXT V_{MIN} + 15

Zero rudder trim before touchdown.

> *NOTE: Do not attempt to achieve a smooth touchdown. At threshold, reduce throttles to idle and use a slight flare. Excessive flare will result in float, excessive use of runway and possible tail strike.*
>
> Auto ground spoilers will not deploy until nose gear touchdown.

Reverse Thrust . . . . AS REQUIRED

[END]

Intentionally
Blank

Abnormal Procedures — Chapter AP TOC

Table Of Contents — Section 00

Intentionally
Blank

Abnormal Procedures — Chapter AP

Table Of Contents — Section 10

Intentionally
Blank

AIR ALERTS

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"AIR ALERTS" REMAINS DISPLAYED

NO

First Officer's EIS SOURCE Selector. . . ORIGINAL POSITION

NOTE: Some subsequent air system faults will not present alerts on EAD or illuminate MASTER WARNING/CAUTION lights; however, annunciator lights on overhead panel are operative.

[END]

Air alerts and system display will operate normally.

[END]

AIR SYS 1-2 OFF

When manifolds 1 and 2 are no longer displayed red on synoptic,

BLEED AIR 2 Switch . . . . ON

“AIR SYS 1-2 OFF” ALERT DISPLAYED AGAIN WITHIN 15 MINUTES

NO

PACK 2 Switch . . . . OFF

Do not repressurize air system 2.

When manifolds 1 and 2 are no longer displayed red on synoptic,

BLEED AIR 1 Switch . . . . ON

“AIR SYS 1-2 OFF” ALERT DISPLAYED AGAIN WITHIN 15 MINUTES

NO

PACK 1 Switch . . . . OFF

Do not repressurize air system 1 or 2.

[END]

No further crew action required.

[END]

Do not repressurize air system 1.

[END]

AIR SYS__PRES LO

Affected BLEED AIR Source. . . . . OFF
Associated PACK Switch . . . . OFF
Associated ISOL Switch . . . . ON
[END]

AVNCS AIR FLO OFF

AIRCRAFT ON GROUND

NO

Call maintenance.
[END]

AVNCS FAN Switch . . . . .VERIFY OVRD
ECON Switch . . . . OFF

NOTE: With ECON switch off, fuel consumption may increase up to 0.6%. If required, increase PERF FACTOR on MCDU A/C STATUS page by 0.6, to revise FMS predictions.

Land at nearest suitable airport.

NEAREST SUITABLE AIRPORT IS WITHIN 90 MINUTES OF PRESENT POSITION

NO

No further crew action required.
[END]

Pull the following circuit breakers, all located on Pilot's overhead circuit breaker panel:

(CONTINUED)

AVNCS AIR FLO OFF (Continued)

- VHF COMM 1
- VHF COMM 2
- VOR 1
- VOR 2
- ATC 2
- ILS 2

CAUTION: The above circuit breakers should remain off (pulled) until aircraft is within 1 hour of landing, then reset.

MCDU 1 Circuit Breaker . . . . .PULL

After 1 hour has elapsed,

MCDU 1 Circuit Breaker . . . . RESET

Verify rested (MCDU 1) indications agree with working (MCDU 2) indications.

After MCDU verification,

MCDU 2 Circuit Breaker . . . . .PULL

After an additional hour has elapsed,

MCDU 2 Circuit Breaker . . . . RESET

Verify rested (MCDU 2) indications agree with working (MCDU 1) indications.

After MCDU verification,

MCDU 1 Circuit Breaker . . . . .PULL

Repeat MCDU cycling procedure each hour until safely landed.

[END]

AVNCS COMPT OVHT

NOTE: This alert will be accompanied by the illumination of the TRIM AIR AVNCS OVHT light on the overhead panel. This light will remain illuminated until reset by maintenance.

AIR SYSTEM SELECT Switch . . . . MANUAL

PACK 1 and PACK 3 Switches . . . . OFF

When "AVNCS COMP OVHT" alert is no longer displayed,

PACK 1 Switch . . . . ON

"AVNCS COMPT OVHT" ALERT DISPLAYED AGAIN

NO

PACK 1 Switch . . . . OFF

When "AVNCS COMPT OVHT" alert is no longer displayed,

PACK 3 Switch . . . . ON

"AVNCS COMPT OVHT" ALERT DISPLAYED AGAIN

NO

PACK 3 Switch . . . . OFF

PACK 1 and PACK 3 must remain off for remainder of flight.

[END]

PACK 1 must remain off for remainder of flight.

[END]

PACK 3 must remain off for the remainder of flight.

[END]

AVNCS EXH FLO OFF

ECON Switch . . . . OFF

AVNCS FAN Switch . . . . VERIFY OVRD

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

No further crew action required.

NOTE: With ECON switch off, fuel consumption may increase up to 0.6%. If required, increase PERF FACTOR on MCDU A/C STATUS page by 0.6, to revise FMS predictions.

AVNCS FAN will remain in OVRD until system status indicates aircraft is on the ground.

[END]

BLEED AIR__FAULT

Throttle for Affected Bleed Air System . . . . SLOWLY ADVANCE

"BLEED AIR__FAULT" REMOVED

NO

If safety of flight permits, operate engine at or above thrust level necessary to keep "BLEED AIR__FAULT" alert from being displayed.

[END]

Return throttles as needed.

Affected BLEED AIR Source. . . . . OFF

Associated PACK . . . . OFF

Associated ISOL Switch . . . . ON

NOTES: When in icing with only one bleed air source, exit icing area and maintain ice protection until clear of icing.

When airfoil anti-ice use has been terminated, the affected bleed air source may be reinstated for normal usage.

[END]

TAIL A-ICE DISAG

CAUTION: Leading edge of horizontal stabilizer may be damaged if tail anti-ice is operated for more than 30 seconds on the ground.

AIRCRAFT ON GROUND

NO

TAIL ANTI-ICE Switch . . . . OFF

AIR APU Switch. . . . . OFF

Notify maintenance to remove external air.

BLEED AIR 2 Switch . . . . VERIFY OFF

1-2 ISOL Switch. . . . . VERIFY OFF

[END]

TAIL ANTI-ICE SWITCH OFF

NO

Assume tail anti-ice is on.

AIR SYSTEM MANUAL

NO

After landing,

BLEED AIR 2 Switch . . . . OFF

1-2 ISOL Switch . . . . OFF

After landing, do not pressurize pneumatic system 2.

[END]

(CONTINUED)

TAIL A-ICE DISAG (Continued)

TAIL ANTI-ICE SWITCH OFF

NO

(CONTINUED)

Depart icing area.

After departing icing area,

TAIL ANTI-ICE Switch . . . . OFF

Land with maximum of 35° flaps.

[END]

TRIM AIR OFF

COCKPIT Temperature Selector . . . . .SET AS REQUIRED

NOTE: Select cockpit zone temperature selector, set temperature at least 2°C less than cabin zone set temperature (cyan).

CARGO Temperature Selector . . . . OFF

[END]

WNG A-ICE__DISAG

CAUTION: Slats may be damaged if wing anti-ice is operated on ground for more than 30 seconds.

AIRCRAFT ON GROUND

NO

WING ANTI-ICE Switch . . . . OFF

Notify maintenance to remove external air.

Associated BLEED AIR 1 or 3 Switch . . . . VERIFY OFF

Associated ISOL Switch(es). . . . . VERIFY OFF

[END]

WING ANTI-ICE SWITCH OFF

NO

Assume wing anti-ice is on.

AIR SYSTEM MANUAL

NO

After landing,

Associated BLEED AIR 1 or 3 Switch. . . . . OFF

Associated ISOL Switch(es) . . . . OFF

After landing, do not pressurize associated pneumatic system.

[END]

WING ANTI-ICE Switch . . . . OFF

Depart icing area.

[END]

Intentionally
Blank

Abnormal Procedures — Chapter AP

Table Of Contents — Section 20

Intentionally
Blank

AIL DEFLECT DISAG

For Aircraft Without AILERON DEFLECT OVRD Switch Installed

AIRCRAFT ON GROUND

NO

Call maintenance.

NOTE: An attempt to clear the alert may be attempted on the ground by placing the FLAP/SLAT handle to 0/RET then back to the planned takeoff position. Should this clear the alert, no further crew action is required.

[END]

Airspeed . . . . MAX 255 KIAS/.51 MACH

FLAP/SLAT Handle 15/EXT

FLAP/SLAT Handle 0/EXT

"AIL DEFLECT DISAG" ALERT EXTINGUISHED

NO

When desired,

FLAP/SLAT Handle . . . . UP/RET

Make log entry.

[END]

When desired,

FLAP/SLAT Handle . . . . UP/RET

(CONTINUED)

AIL DEFLECT DISAG (Continued)

Do not exceed 280 KIAS/.82 Mach.

Do not use FMS fuel computations for flight planning.

NOTE: Fuel consumption may be increased up to 20%.

Diversion to suitable airport may be required.

Plan a normal FLAP/SLAT landing.

Do not use autoland.

Disconnect the autopilot by 200 feet AGL.

Make log entry.

[END]

AIL DEFLECT DISAG

For Aircraft With AILERON DEFLECT OVRD Switch Installed

AIRCRAFT ON GROUND

NO

Call maintenance.

NOTE: An attempt to clear the alert may be attempted on the ground by placing the FLAP/SLAT handle to 0/RET then back to the planned takeoff position. Should this clear the alert, no further crew action is required.

[END]

AILERON DEFLECT OVRD Switch. OVRD ON

Push the AILERON DEFLECT OVRD switch and observe OVRD ON light illuminates.

Airspeed. . . . . .MAX 255 KIAS/.51 MACH

FLAP/SLAT Handle 15/EXT

FLAP/SLAT Handle 0/EXT

(CONTINUED)

AIL DEFLECT DISAG (Continued)

AILERON DEFLECT OVRD Switch . . . . OFF

Push the AILERON DEFLECT OVRD switch and observe OVRD ON light extinguishes.

NOTE: The use of the AILERON DEFLECT OVRD switch is limited to one cycle.

"AIL DEFLECT DISAG" ALERT EXTINGUISHED

NO

When desired,

FLAP/SLAT Handle . . . . UP/RET

Make log entry.

[END]

When desired,

FLAP/SLAT Handle . . . . UP/RET

Do not exceed 280 KIAS/.82 Mach.

Do not use FMS fuel computations for flight planning.

NOTE: Fuel consumption may be increased up to 20%.

Diversion to suitable airport may be required.

Plan a normal FLAP/SLAT landing.

Do not use autoland.

Disconnect the autopilot by 200 feet AGL.

Make log entry.

[END]

ANTI-SKID FAIL

OR

ANTI-SKID__FAIL

Use 50/EXT unless 35/EXT is required. Consider reducing aircraft weight by dumping fuel. Use full length of runway for deceleration. Use full reverse thrust while applying brakes smoothly and gradually. After landing ground roll turn anti-skid off. Compute landing distance from applicable table.

50/EXT ESTIMATED LANDING DISTANCES (FEET)
ANTI-SKID INOPERATIVE

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	4790	4987	5184	5381	5610	5840	6070	6299
STD=15°C	**WET**	6299	6594	6857	7152	7415	7743	8038	8366
2000 FT	**DRY**	5052	5249	5479	5709	5906	6168	6430	6693
STD=11°C	**WET**	6660	6988	7316	7612	7907	8235	8530	8891
4000 FT	**DRY**	5315	5545	5807	6070	6299	6562	6824	7087
STD=7°C	**WET**	7087	7415	7776	8104	8399	8760	9121	9482
6000 FT	**DRY**	5643	5873	6168	6430	6693	6988	7251	7546
STD=3°C	**WET**	7546	7907	8268	8629	8990	9383	9744	10138
8000 FT	**DRY**	5971	6234	6562	6857	7119	7448	7743	8071
STD=–1°C	**WET**	8071	8432	8858	9219	9580	10007	10433	10860
10000 FT	**DRY**	6332	6628	6988	7283	7612	8005	8333	8727
STD=–5°C	**WET**	8596	9022	9482	9875	10302	10827	11286	11778

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–13	–16
ABOVE standard day	+33	+46

(CONTINUED)

ANTI-SKID FAIL or ANTI-SKID __ FAIL (Continued)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–125	–246
DOWNHILL	+394	+935

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–43	–66
TAILWIND	+121	+207

35/EXT ESTIMATED LANDING DISTANCES (FEET) ANTI-SKID INOPERATIVE

WEIGHT 1000KG		**160**	**170**	**180**	**190**	**200**	**210**	**220**	**230**
S.L.	**DRY**	5446	5741	6004	6299	6627	6890	7218	7513
STD=15°C	**WET**	7415	7776	8136	8563	8990	9350	9810	10203
2000 FT	**DRY**	5774	6070	6398	6693	7021	7349	7677	8005
STD=11°C	**WET**	7907	8301	8694	9154	9580	10039	10499	10958
4000 FT	**DRY**	6135	6463	6791	7152	7513	7841	8235	8563
STD=7°C	**WET**	8431	8858	9318	9777	10269	10728	11286	11745
6000 FT	**DRY**	6529	6889	7251	7644	8038	8366	8793	9154
STD=3°C	**WET**	9055	9514	10007	10532	11056	11581	12139	12697
8000 FT	**DRY**	6988	7382	7743	8169	8596	8990	9449	9941
STD=–1°C	**WET**	9711	10236	10761	11352	11942	12467	13123	13812
10000 FT	**DRY**	7480	7907	8301	8760	9288	9810	10335	10892
STD=–5°C	**WET**	10433	11024	11581	12238	12927	13648	14370	15125

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–16	–20
ABOVE standard day	+46	+66

(CONTINUED)

ANTI-SKID FAIL or ANTI-SKID __ FAIL (Continued)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–135	–351
DOWNHILL	+689	+1591

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–52	–75
TAILWIND	+157	+269

[END]

AUTO BRAKE FAIL

IN FLIGHT

NO

AUTO BRAKE Selector . . . . OFF

Anti-Skid operates normally.

After landing, apply anti-skid braking consistent with runway conditions.

[END]

AUTO BRAKE Selector . . . . OFF

“AUTO BRAKE FAIL” ALERT REMAINS DISPLAYED

NO

Takeoff not permitted unless MEL procedure is satisfied.

Call maintenance.

[END]

Takeoff is permitted.

For subsequent landing, AUTO BRAKE LAND mode may be used.

Anti-skid operates normally.

For rejected takeoff, apply anti-skid braking consistent with runway conditions and aircraft speed.

[END]

BRAKE OVERHEAT

AIRCRAFT ON GROUND

NO

Do not take off. Advise ground personnel to remain clear of main gear. Fuse plugs may melt.

Avoid non-essential use of parking brake when BRAKE OVERHEAT alert is displayed.

CAUTION: BRAKE OVERHEAT alert is displayed above 550°C. If temperature exceeds 800°C, stop aircraft and call emergency services.

NOTE: Brakes, wheels and tires will require maintenance before next flight.

Temperatures above 936°C cannot be displayed and will go blank for the affected brake(s). As the temperature decreases below 936°C, the temperature(s) will display again if sensors are not damaged.

[END]

Flight conditions permitting, extend landing gear for cooling for 10 minutes or until alert is not displayed.

Record applicable brake position(s) and maximum temperature reading in aircraft Flight and Maintenance Log.

[END]

FCC__DATA FAULT

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"FCC__ DATA FAULT" ALERT EXTINGUISHED

NO

No further crew action required.

[END]

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

First Officer's EIS SOURCE Selector . . . . ORIGINAL POSITION

Select appropriate procedure below.

NOTE: During the following actions, the associated LSAS and YAW DAMP OFF light(s) will illuminate but "LSAS__OFF" and "YAW DAMP__OFF" alert(s) will not be displayed.

FCC 1A DATA FAULT

LSAS RIGHT OUTBD Switch . . . . OFF

LWR YAW DAMP A Switch . . . . OFF

[END]

FCC 1B DATA FAULT

LSAS LEFT INBD Switch . . . . OFF

LWR YAW DAMP B Switch . . . . OFF

[END]

(CONTINUED)

FCC __ DATA FAULT (Continued)

FCC 2A DATA FAULT

LSAS LEFT OUTBD Switch . . . . OFF

UPR YAW DAMP A Switch . . . . OFF

[END]

FCC 2B DATA FAULT

LSAS RIGHT INBD Switch . . . . OFF

UPR YAW DAMP B Switch . . . . OFF

[END]

FLAP DISAG

Place FLAP/SLAT handle to match flap position. Allow several seconds for system response. If alert remains displayed, select the last symmetrical configuration.

NOTE: *If required to calculate V_{APP} speeds, refer to QRH, NORMAL & ABNORMAL CONFIGURATION REFERENCE SPEEDS (V_{REF}) table located under the REFERENCE DATA tab.*

PROBLEM WAS ASYMMETRIC FLAPS

NO

FLAP POSITION 0 DEGREES

NO

Refer to Abnormal Non-Alert procedure NO FLAP/SLAT EXTENDED LANDING

[END]

Land at nearest suitable airport using existing flap/slat setting.

If final flap setting is less than 35°,

GPWS Switch . . . . FLAP OVRD

Autobrakes . . . . OFF

At 50 feet,

Autothrottles. . . . . OFF

[END]

(CONTINUED)

FLAP DISAG (Continued)

ALERT APPEARED DURING ATTEMPTED FLAP RETRACTION

NO

Land at nearest suitable airport.

If flaps now less than 50°, further extension may be attempted if required.

NOTE: If after selecting a greater flap setting, flaps do not move as slected, place FLAP/SLAT handle to match actual flap position.

If final flap setting is less than 35°,

GPWS Switch. . . . . .FLAP OVRD

Autobrakes . . . . OFF

At 50 feet,

Autothrottles . . . . .OFF

[END]

Airspeed . . . . REDUCE

Reduce speed to minimum maneuver speed displayed on PFD.

NOTE: During the next step, FLAP/SLAT handle forces will be higher than normal.

Place FLAP/SLAT handle to 50/EXT, then return handle to required position. Allow several seconds for system response. (This action may allow flaps to reset to normal operation.)

(CONTINUED)

FLAP DISAG (Continued)

FLAPS EXTEND NORMALLY

NO

No further crew action is required.

[END]

Return FLAP/SLAT handle to match actual flap position.

FLAP POSITION 0 DEGREES

NO

Refer to Abnormal Non-Alert procedure - NO FLAP/SLAT EXTENDED LANDING.

[END]

Land with existing flap/slat setting.

If flaps less than 35°,

GPWS Switch . . . . FLAP OVRD

Autobrakes . . . . OFF

At 50 feet,

Autothrottles . . . . OFF

[END]

LSAS ALL FAIL

Autopilot . . . . DISCONNECT

LSAS Switches . . . . .ALL OFF

Any One LSAS Switch . . . . ON

ASSOCIATED LSAS LIGHT ILLUMINATED

NO

Associated LSAS Switch . . . . OFF

Attempt to restore remaining LSAS channels by pushing any one LSAS switch on and observe its FAIL light. Any LSAS switch that illuminates FAIL should be pushed to OFF prior to pushing next switch.

ANY LSAS CHANNEL OPERATIVE

NO

Attempt to re-engage autopilot.

NOTE: Autopilot requires an operative inboard LSAS channel.

[END]

LSAS is inoperative. Autopilot is not available.

NOTE: Pitch rate damper, pitch protection and positive nose lowering will not be available.

Pitch sensitivity increases with altitude.
Avoid overcontrolling. Also avoid pitch attitudes above 7 degrees during landing.

[END]

SEL ELEV FEEL MAN

"SEL FADEC ALTN" AND/OR "SEL FLAP LIM OVRD" AND/OR "IAS COMPARATOR MONITOR" ALSO DISPLAYED

NO

Refer to Emergency Non-Alert procedure – AIRSPEED: LOST, SUSPECT OR ERRATIC.

[END]

ELEV FEEL Selector . . . . .MANUAL

ELF Speed . . . . SET AS REQUIRED

NOTES: When ELEV FEEL is in MANUAL, ELF speed is displayed on SD CONFIG synoptic page.

Slew ELF reference speed bug to maintain approximate agreement with aircraft indicated airspeed.

[END]

SEL FLAP LIM OVRD

NOTE: FLAP LIMIT MANUAL light will be illuminated

"SEL FADEC ALTN" AND/OR "SEL ELEV FEEL MAN" AND/OR "IAS COMPARATOR MONITOR" ALSO DISPLAYED

NO

Refer to Emergency Non-Alert procedure – AIRSPEED: LOST, SUSPECT OR ERRATIC.

[END]

FLAP LIMIT Selector. . . . . OVRD 1

After 20 seconds,

“FLAP LIMIT DISAG” ALERT DISPLAYED

NO

FLAP LIMIT Selector . . . . OVRD 2

After 20 seconds,

"FLAP LIMIT DISAG" ALERT DISPLAYED

NO

Flap extension may be limited.

[END]

Observe placarded flap limit speed.

[END]

SLAT DISAG

ALERT APPEARED DURING CLIMBOUT, WITH FLAP/SLAT HANDLE IN A SLAT EXTENDED POSITION

NO

Stick shaker may actuate temporarily.

If flaps greater than 15°, retract flaps to 15°.

Accelerate and retract flaps and slats on schedule.

CAUTION: Anticipate "SLAT DISAG" alert to appear during extension. Utilize "ALERT APPEARED DURING ATTEMPTED EXTENSION" decision point in this procedure.

[END]

AIRSPEED ABOVE 280 KIAS/0.55 MACH

NO

FLAP/SLAT Handle . . . . UP/RET

[END]

ALERT APPEARED DURING ATTEMPTED EXTENSION

NO

"SLATS INHIBITED" ALERT DISPLAYED ON SD CONFIG PAGE

NO

(CONTINUED)

SLAT DISAG (Continued)

ALERT APPEARED DURING ATTEMPTED EXTENSION

NO

(CONTINUED)

"SLATS INHIBITED" ALERT DISPLAYED ON SD CONFIG PAGE

NO

(CONTINUED)

FLAP/SLAT Handle. . . . . 10°/EXT OR GREATER

"SLATS DISAG" ALERT REMOVED

NO

Plan a normal flap/slat landing.

[END]

FLAP/SLAT Handle . . . . UP/RET

SLAT STOW Switch . . . . SLAT STOW

FLAP/SLAT Handle . . . . 0/EXT POSITION

GPWS Switch. . . . . FLAP OVRD

Plan a 28/RET landing.

NOTE: Autothrottles will not automatically retard at 50 feet with flaps less than landing range. Autothrottles must be disconnected prior to 50 feet AGL.

(CONTINUED)

SLAT DISAG (Continued)

ALERT APPEARED DURING
ATTEMPTED EXTENSION

NO

(CONTINUED)

15/RET REFERENCE SPEEDS SLATS DISAG

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{REF}	176	181	187	192	197	202	207	212

25/RET REFERENCE SPEEDS SLATS DISAG

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{REF}	171	176	181	186	191	195	200	205

28/RET APPROACH SPEEDS SLATS DISAG

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
V_{APP} (V_{REF} + 5)	175	180	185	190	194	199	204	209

(CONTINUED)

SLAT DISAG (Continued)

ALERT APPEARED DURING ATTEMPTED EXTENSION

NO

(CONTINUED)

28/RET ESTIMATED LANDING DISTANCES (FEET) SLATS DISAGREE

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	6693	7087	7480	7906	8301	8760	9154	9580
STD=15°C	**WET**	8169	8628	9055	9547	10039	10531	10991	11450
2000 FT	**DRY**	7119	7578	8005	8464	8891	9383	9810	10269
STD=11°C	**WET**	8694	9219	9678	10203	10728	11253	11745	12270
4000 FT	**DRY**	7612	8104	8563	9055	9547	10072	10564	11056
STD=7°C	**WET**	9285	9843	10367	10958	11483	12073	12598	13156
6000 FT	**DRY**	8169	8694	9186	9744	10269	10827	11385	11942
STD=3°C	**WET**	9974	10564	11122	11745	12336	12959	13550	14173
8000 FT	**DRY**	8793	9350	9908	10532	11089	11713	12303	12927
STD=–1°C	**WET**	10696	11352	11975	12664	13287	13976	14633	15289
10000 FT	**DRY**	9449	10105	10696	11385	12041	12894	13714	14534
STD=–5°C	**WET**	11516	12238	12927	13648	14370	15289	16175	16995

NOTE: Standard day, no wind, zero slope, two engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop. Tail engine at idle reverse (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–23	–26
ABOVE standard day	+72	+75

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–141	–236
DOWNHILL	+702	+1004

(CONTINUED)

SLAT DISAG (Continued)

ALERT APPEARED DURING ATTEMPTED EXTENSION

NO

(CONTINUED)

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–52	–69
TAILWIND	+203	+226

Cross threshold at V_{APP} and reduce sink rate slightly.

Disconnect autothrottles, retard throttles to idle and raise nose of aircraft to at least a level attitude. Do not hold aircraft off. Excessive flare will result in float and excessive use of runway.

CAUTION: Tail strike may occur at deck angles greater than 10°.

[END]

FLAP/SLAT Handle 0/EXT

Plan a normal flap/slat landing.

[END]

STAB OUT OF TRIM

Observe stabilizer position on SD CONFIG synoptic page for proper autotrim. If no indication of trim response after 5 seconds,

AUTO FLIGHT Switch . . . . PUSH

STABILIZER FAILS TO MOVE

NO

To counteract any abrupt pitching tendencies, firmly grasp control wheel.

Autopilot . . . . DISCONNECT

Trim aircraft using control wheel trim switches or center pedestal LONG TRIM handles.

STABILIZER FAILS TO MOVE

NO

Refer to Abnormal Non-Alert procedure – STABILIZER INOPERATIVE.

[END]

No further crew action required.

[END]

TIRE FAIL

BEFORE TAKEOFF ROLL

NO

Do not take off.

[END]

Reject or continue takeoff, depending on conditions.

LANDING GEAR EXTENDED

NO

Do not retract landing gear.

When ready to land,

Gross Weight . . . . .REDUCE AS REQUIRED

AUTO BRAKE Selector . . . . OFF

[END]

USE MAN SPOILERS

Determine landing distance from the following applicable tables.

At Nose Gear Touchdown. . . . . DEPLOY SPOILERS

NOTE: The pitch rate damper, pitch protection and positive nose lowering may not be available.

50/EXT ESTIMATED LANDING DISTANCES (FEET) USE MAN SPOILERS

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	4255	4439	4623	4797	4957	5148	5318	5489
STD=15°C	**WET**	5085	5328	5571	5794	6020	6270	6503	6736
2000 FT	**DRY**	4449	4649	4849	5030	5203	5407	5591	5774
STD=11°C	**WET**	5371	5630	5889	6129	6368	6640	6893	7146
4000 FT	**DRY**	4662	4875	5089	5282	5469	5686	5883	6079
STD=7°C	**WET**	5679	5958	6237	6496	6755	7044	7313	7582
6000 FT	**DRY**	4902	5125	5348	5558	5758	5991	6201	6411
STD=3°C	**WET**	6027	6322	6617	6902	7182	7490	7782	8074
8000 FT	**DRY**	5144	5387	5630	5853	6070	6319	6549	6778
STD=–1°C	**WET**	6388	6713	7037	7339	7644	7979	8294	8609
10000 FT	**DRY**	5417	5676	5935	6178	6430	6729	7001	7274
STD=–5°C	**WET**	6791	7142	7494	7825	8173	8579	8947	9314

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–13	–13
ABOVE standard day	+26	+36

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–85	–138
DOWNHILL	+230	+443

(CONTINUED)

USE MAN SPOILERS (Continued)

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–33	–46
TAILWIND	+82	+131

35/EXT ESTIMATED LANDING DISTANCES (FEET) USE MAN SPOILERS

WEIGHT 1000KG		**160**	**170**	**180**	**190**	**200**	**210**	**220**	**230**
S.L.	**DRY**	4577	4760	4948	5144	5358	5525	5728	5925
STD=15°C	**WET**	5492	5738	5984	6243	6512	6755	7028	7300
2000 FT	**DRY**	4797	4993	5194	5404	5620	5810	6030	6257
STD=11°C	**WET**	5804	6070	6335	6617	6906	7169	7457	7746
4000 FT	**DRY**	5033	5240	5456	5682	5915	6119	6355	6591
STD=7°C	**WET**	6165	6444	6722	7021	7329	7615	7933	8251
6000 FT	**DRY**	5289	5512	5741	5986	6237	6457	6709	6942
STD=3°C	**WET**	6335	6841	7146	7474	7812	8110	8455	8799
8000 FT	**DRY**	5554	5804	6053	6312	6585	6821	7110	7382
STD=–1°C	**WET**	6929	7270	7612	7963	8323	8652	9039	9426
10000 FT	**DRY**	5853	6122	6391	6703	6978	7310	7625	7940
STD=–5°C	**WET**	7395	7756	8116	8504	8907	9354	9777	10466

NOTE: Standard day, no wind, zero slope, three engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–13	–16
ABOVE standard day	+30	+39

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–95	–154
DOWNHILL	+276	+522

(CONTINUED)

USE MAN SPOILERS (Continued)

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–36	–49
TAILWIND	+95	+144

[END]

YAW DAMP ALL FAIL

YAW DAMP Switches. . . . . .ALL OFF

Any one YAW DAMP Switch . . . . ON

ASSOCIATED YAW DAMP FAIL LIGHT ILLUMINATED

NO

Associated YAW DAMP Switch. . . . . OFF

Attempt to restore remaining yaw damp channels by pushing any one YAW DAMP switch on and observing its FAIL light. Any YAW DAMP switch that illuminates FAIL should be pushed to OFF prior to pushing next switch.

[END]

Intentionally
Blank

Abnormal Procedures — Chapter AP

Table Of Contents — Section 30

Intentionally
Blank

BUS L EMER AC OFF

EMER PWR Selector . . . . OFF

DISPLAY UNITS 1 AND/OR 3 OPERATING

NO

NOTE: The left emergency AC bus sensing circuit has failed.

If subsequent electrical malfunction(s) occurs requiring use of emergency power, rotate EMER PWR selector to ARM or ON and deploy the ADG.

NOTE: The emergency power system is designed to provide power for approximately 15 minutes until the ADG is deployed.

No further crew action required.

[END]

Land at the nearest suitable airport.

Captain’s EIS SOURCE Selector . . . . AUX (OR 2)

Captain’s CADC Switch . . . . CAPT ON 2

Captain’s IRS Switch. . . . . CAPT ON AUX

“ENG IGN NOT ARMED” LEVEL 1 ALERT DISPLAYED

NO

ENG IGN Switch . . . . .SELECT B

(CONTINUED)

BUS L EMER AC OFF (Continued)

"ENG IGN NOT ARMED" LEVEL 1 ALERT DISPLAYED

NO

(CONTINUED)

If subsequent electrical malfunction(s) occurs requiring use of emergency power, rotate EMER PWR selector to ARM or ON and deploy the ADG.

NOTE: The emergency power system is designed to provide power for approximately 15 minutes until the ADG is deployed.

[END]

BUS L EMER DC OFF

EMER PWR Selector . . . . OFF

CAPTAIN'S FLIGHT DIRECTOR AND/OR AUTOPILOT 1 OPERATIONAL

NO

NOTE: The left emergency DC bus sensing circuit has failed.

If subsequent electrical malfunction(s) occurs requiring use of emergency power, rotate EMER PWR selector to ARM or ON and deploy the ADG.

NOTE: The emergency power system is designed to provide power for approximately 15 minutes until the ADG is deployed.

No further crew action is required.

[END]

Land at nearest suitable airport.

If subsequent electrical malfunction(s) occur requiring use of emergency power, rotate EMER PWR selector to ARM or ON and deploy the ADG.

NOTE: The emergency power system is designed to provide power for approximately 15 minutes until the ADG is deployed.

[END]

ELEC ALERTS

NOTE: Under certain conditions ELEC PANEL may be partially inoperative.

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

“BAT BUS OFF” LIGHT ILLUMINATED

NO

Land at nearest suitable airport.

Battery bus is not powered.

Electrical system is in manual mode, but MANUAL light will not be illuminated.

NOTE: The following are inoperative:

- *Standby attitude indicator.*
- *Overspeed aural warning.*
- *MASTER WARNING/MASTER CAUTION lights.*
- *Engine and APU fire detection.*
- *Ignition override.*
- *APU*
- *SD ELEC synoptic page.*

Some subsequent electrical system faults will not present alerts on EAD or illuminate MASTER CAUTION lights.

(CONTINUED)

ELEC ALERTS (Continued)

"BAT BUS OFF" LIGHT ILLUMINATED

NO

(CONTINUED)

If indications of equipment failure in other systems are received, observe overhead ELEC control panel for possible bus failure to determine if new alert is primary or only a result of a faulted electrical bus. If "BUS L EMER AC OFF" and/or "BUS L EMER DC OFF" alerts are displayed, use appropriate Abnormal procedure.

NOTE: Subsequent alerts from other systems may be displayed as a result of not being inhibited.

[END]

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"ELEC ALERTS" REMAINS DISPLAYED

NO

First Officer's EIS SOURCE Selector. . . ORIGINAL POSITION

NOTES: Some subsequent electrical system faults will not present alerts on EAD or illuminate MASTER CAUTION lights.

Electrical system may be in manual mode.

If indications of equipment failure in other systems are received, observe overhead ELEC control panel for possible bus failure to determine if new alert is primary or only a result of a faulted electrical bus. If "BUS L EMER AC OFF" and/or "BUS L EMER DC OFF" alerts are displayed, use appropriate Abnormal procedure.

NOTE: Subsequent alerts from other systems may be displayed as a result of not being inhibited.

[END]

Electrical alerts and system display will operate normally.

[END]

GEN ALL OFF

CAUTION: With all generators off, the HYD, AIR and FUEL panel illumination will not be functioning. Operation of switches on these panels could change system configuration. These changes would not be indicated to the crew.

NOTE: Battery emergency power is limited to 15 minutes until ADG is deployed.

ENG IGN OVRD Switch . . . . .OVRD ON

AIRCRAFT ALTITUDE > 38,000 FEET

NO

While continuing this procedure, begin descent to an altitude of 38,000 feet or less.

NOTE: Engines may not sustain fuel suction feed at altitudes above 38,000 feet.

ELEC SYSTEM MANUAL

NO

GEN Switches With OFF Light Illuminated . . . . PUSH

CAUTION: Only one reset attempt is permitted.

(CONTINUED)

GEN ALL OFF (Continued)

ELEC SYSTEM MANUAL

NO

(CONTINUED)

GENERATOR BUSES 1 AND 3 POWERED

NO

EMER PWR Selector . . . . OFF THEN ARM

If required, start engine 2. Refer to Abnormal Non-Alert procedure – ENGINE RESTART IN FLIGHT.

Land at nearest suitable airport.

[END]

ADG . . . . .DEPLOY

ADG ELEC Switch . . . . ON

NOTE: Horizontal stabilizer trim is available only through the LONG TRIM handles on the pedestal.

Captain’s SISP FMS and APPR Switches . . . . PUSH

NOTE: SISP lights will not illuminate; however, the switches are functional.

If required, restart engine 2. Refer to Abnormal Non-Alert procedure - ENGINE RESTART IN FLIGHT.

NOTE:Landing gear position indications on the instrument panel and SD CONFIG synoptic page are not available when DC bus 2 and DC bus 3 are not powered.

Flap position indications on the PFD and SD CONFIG synoptic page are not available when AC bus 1 and AC bus 3 are not powered.

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

Refer to following table for maximum landing flap setting.

AIRSPEED AT TIME OF FAILURE	MAXIMUM LANDING FLAP SETTING
At or above 211 KIAS	22°
191 KIAS to 210 KIAS	28°
176 KIAS to 190 KIAS	35°
At or below 175 KIAS	50°

Land at neatest suitable airport.
Refer to tables listed below for configuration speeds and estimated landing distances.

**22/EXT APPROACH SPEEDS
GEN ALL OFF**

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
$V_{APP} = V_{REF} + 5$	147	152	156	160	164	168	172	176

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

FLAP 22/EXT ESTIMATED LANDING DISTANCES (FEET) GEN ALL OFF

WEIGHT 1000 KG		160	170	180	190	200	210	220	230
S.L.	DRY	6759	7087	7448	7808	8169	8563	8924	9350
STD=15°C	WET	10433	10925	11417	11942	12467	12992	13550	14140
2000 FT	DRY	7152	7513	7907	8301	8694	9088	9514	9941
STD=11°C	WET	11122	11647	12172	12730	13287	13878	14469	15092
4000 FT	DRY	7612	8005	8399	8825	9252	9678	10138	10597
STD=7°C	WET	11877	12434	12992	13583	14206	14829	15486	16142
6000 FT	DRY	8104	8530	8957	9416	9875	10367	10827	11352
STD=3°C	WET	12697	13287	13911	14534	15190	15879	16568	17290
8000 FT	DRY	8629	9088	9547	10039	10531	11056	11581	12139
STD=–1°C	WET	13550	14206	14862	15551	16273	16995	17749	18504
10000 FT	DRY	9186	9711	10236	10794	11352	11975	12598	13222
STD=–5°C	WET	14501	15190	15945	16732	17520	18373	19259	20144

NOTE: Standard day, no wind, zero slope, no anti-skid, manual spoilers, flight idle and no reverse thrust (Includes Air Run Distances)

CORRECTIONS:		
Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–23	–36
ABOVE standard day	+56	+85

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	Dry	Wet
UPHILL	–240	–669
DOWNHILL	+906	+2654

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	Dry	Wet
HEADWIND	–66	–112
TAILWIND	+217	+417

28/EXT APPROACH SPEEDS GEN ALL OFF

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
$V_{APP} = V_{REF} + 5$	145	150	154	158	162	166	170	174

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

FLAP 28/EXT ESTIMATED LANDING DISTANCES (FEET) GEN ALL OFF

WEIGHT 1000 KG		160	170	180	190	200	210	220	230
S.L.	**DRY**	6594	6955	7316	7677	8038	8399	8760	9121
STD=15℃	**WET**	10138	10630	11122	11614	12106	12631	13156	13681
2000 FT	**DRY**	7021	7382	7776	8136	8530	8924	9318	9711
STD=11℃	**WET**	10794	11319	11844	12402	12927	13484	14042	14600
4000 FT	**DRY**	7448	7841	8268	8661	9088	9514	9941	10367
STD=7℃	**WET**	11516	12073	12631	13222	13812	14403	14993	15584
6000 FT	**DRY**	7907	8366	8793	9252	9678	10138	10597	11056
STD=3℃	**WET**	12303	12894	13517	14140	14764	15387	16043	16699
8000 FT	**DRY**	8432	8891	9383	9843	10335	10827	11319	11844
STD=–1℃	**WET**	13123	13780	14436	15092	15781	16470	17159	17881
10000 FT	**DRY**	8990	9482	10007	10531	11089	11680	12270	12894
STD=–1℃	**WET**	14042	14731	15453	16207	16962	17782	18602	19455

NOTE: Standard day, no wind, zero slope, no anti-skid, manual spoilers, flight idle and no reverse thrust (Includes Air Run Distances)

CORRECTIONS:		
Temperature: Valid from STD –20°C to STD +40℃		
FEET PER ℃	**Dry**	**Wet**
BELOW standard day	–23	–36
ABOVE standard day	+52	+82

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–236	-650
DOWNHILL	+876	+2523

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–66	–108
TAILWIND	+207	+397

35/EXT APPROACH SPEEDS
GEN ALL OFF

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
$V_{APP} = V_{REF} + 5$	143	147	152	156	160	164	167	170

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

FLAP 35/EXT ESTIMATED LANDING DISTANCES (FEET) GEN ALL OFF

WEIGHT 1000 KG		160	170	180	190	200	210	220	230
S.L.	**DRY**	6375	6749	7100	7454	7805	8136	8494	8802
STD=15℃	**WET**	9721	10203	10679	11142	11601	12110	12598	13074
2000 FT	**DRY**	6814	7165	7549	7884	8268	8652	9016	9380
STD=11℃	**WET**	10341	10853	11362	11906	12395	12920	13442	13944
4000 FT	**DRY**	7208	7592	8031	8396	8812	9229	9619	10016
STD=7℃	**WET**	11024	11568	12110	12680	13238	13793	14327	14862
6000 FT	**DRY**	7638	8104	8520	8970	9367	9797	10246	10650
STD=3℃	**WET**	11765	12343	12946	13556	14137	14708	15322	15919
8000 FT	**DRY**	8150	8599	9098	9524	10007	10466	10928	11417
STD=–1℃	**WET**	12546	13182	13819	14442	15085	15735	16368	17044
10000 FT	**DRY**	8698	9154	9665	10157	10702	11257	11814	12421
STD=–5℃	**WET**	13415	14085	14767	15476	16191	16969	17726	18537

NOTE: Standard day, no wind, zero slope, no anti-skid, manual spoilers, flight idle and no reverse thrust (Includes Air Run Distances)

CORRECTIONS:		
Temperature: Valid from STD –20°C to STD +40℃		
FEET PER ℃	**Dry**	**Wet**
BELOW standard day	–23	–36
ABOVE standard day	+30	+79

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	Dry	Wet
UPHILL	–230	-617
DOWNHILL	+827	+2336

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	Dry	Wet
HEADWIND	–66	–105
TAILWIND	+197	+374

50/EXT APPROACH SPEEDS
GEN ALL OFF

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
$V_{APP} = V_{REF} + 5$	140	144	148	151	155	158	161	165

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

FLAP 50/EXT ESTIMATED LANDING DISTANCES (FEET) GEN ALL OFF

WEIGHT 1000 KG		160	170	180	190	200	210	220	230
S.L.	**DRY**	5807	6102	6398	6726	7021	7316	7644	7940
STD=15°C	**WET**	8563	8990	9383	9810	10203	10630	11056	11516
2000 FT	**DRY**	6168	6463	6791	7119	7448	7776	8104	8432
STD=11°C	**WET**	9121	9547	9974	10433	10860	11319	11778	12238
4000 FT	**DRY**	6529	6857	7218	7546	7907	8268	8596	8957
STD=7°C	**WET**	9711	10171	10630	11089	11581	12073	12566	13058
6000 FT	**DRY**	6923	7283	7644	8038	8399	8793	9186	9547
STD=3°C	**WET**	10335	10827	11319	11844	12336	12861	13386	13911
8000 FT	**DRY**	7349	7743	8136	8530	8957	9350	9777	10203
STD=–1°C	**WET**	11024	11549	12073	12631	13156	13714	14304	14862
10000 FT	**DRY**	7841	8235	8661	9121	9580	10072	10564	11089
STD=–5°C	**WET**	11745	12303	12894	13484	14108	14764	15453	16142

NOTE: Standard day, no wind, zero slope, no anti-skid, manual spoilers, flight idle and no reverse thrust (Includes Air Run Distances)

CORRECTIONS:		
Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–16	–30
ABOVE standard day	+43	+66

(CONTINUED)

GEN ALL OFF (Continued)

AIRSPEED KNOWN AT TIME OF FAILURE

NO

(CONTINUED)

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–197	–505
DOWNHILL	+663	+1804

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–59	–95
TAILWIND	+171	+322

[END]

Plan a 22/EXT approach and landing.

Refer to 22/EXT APPROACH SPEED and ESTIMATED LANDING DISTANCE tables in this procedure.

Land at nearest suitable airport.

[END]

GEN BUS__FAULT

ELEC SYSTEM MANUAL

NO

Associated GEN Switch . . . . PUSH TO RESET

CAUTION: Only one reset attempt is permitted.

GEN BUS POWERED

NO

“EMER PWR ON” ALERT DISPLAYED AND/OR EMER PWR ON LIGHT ILLUMINATED

NO

EMER PWR Selector . . . . . . OFF THEN ARM

[END]

No further crew action required.

[END]

“GEN BUS 1 FAULT” AND/OR “GEN BUS 3 FAULT” ALERT DISPLAYED

NO

- Generator bus 1 fault condition may cause activation of right stick shaker. Stick shaker may be deactivated by disconnecting cannon plug on control column or pulling F/O STICK SHAKER circuit breaker on overhead circuit breaker panel.

(CONTINUED)

GEN BUS __ FAULT (Continued)

"GEN BUS 1 FAULT" AND/OR "GEN BUS 3 FAULT" ALERT DISPLAYED

NO

(CONTINUED)

- Generator bus 3 fault condition may cause activation of left stick shaker. Stick shaker may be deactivated by disconnecting cannon plug on the control column or pulling CAPT STICK SHAKER circuit breaker on overhead circuit breaker panel.

ADG . . . . DEPLOY

ADG ELEC Switch . . . . ON

"GEN BUS 2 FAULT" ALERT DISPLAYED

NO

Battery charger is inoperative.

If battery charging is required, consider deploying the ADG and pushing the ADG ELEC switch to ON to allow the battery charger to be powered by the right emergency AC bus.

Continue with affected circuits inoperative.

Land at nearest suitable airport.

Review applicable consequences.

[END]

GEN DRIVE __ FAULT

ANY OTHER ENGINE GENERATOR OPERATING

NO

DRIVE Switch. . . . . .DISC

[END]

Do not disconnect IDG. Continue with “GEN DRIVE __ FAULT” alert displayed.

[END]

GEN LOAD HI

Freighter aircraft:

Reduce all non essential loads.

Monitor generator load on ELEC synoptic.

NOTE: Controller may turn generator off when load limits are exceeded. This will cause the "GEN LOAD HI" alert to extinguish and the "GEN __OFF" alert to be displayed.

[END]

GEN__OFF

Associated GEN Switch . . . . .PUSH TO RESET

CAUTION: Only one reset attempt is permitted.

If reset attempt is not successful, continue flight with generator inoperative.

[END]

Abnormal Procedures — Chapter AP

Table Of Contents — Section 40

Intentionally
Blank

ENG__EGT HI

Throttle (Affected Engine) . . . . RETARD

EGT REMAINS ABOVE REDLINE

NO

Shut down affected engine.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

EGT EXCEEDED 1000°C

NO

Throttle (Affected Engine) . . . . IDLE

Use higher thrust only at PIC's discretion.

NOTE: If any engine indications are abnormal at minimum thrust, a precautionary shutdown should be considered.

[END]

Operate affected engine at a throttle setting necessary to maintain EGT below red line.

[END]

ENG__OIL FILTER

Throttle (Affected Engine) . . . . IDLE

"ENG__OIL FILTER" ALERT REMAINS DISPLAYED

NO

Shut down affected engine.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

Advance throttle to a point below that which causes "ENG__OIL FILTER" alert to be displayed. Operate engine at or below thrust level necessary to keep "ENG__OIL FILTER" alert from being displayed.

[END]

ENG__OIL PRES LO

OR

ENGINE OIL PRESSURE BELOW REDLINE

INDICATOR PRESSURE BELOW REDLINE AND "ENG__OIL PRES LO" ALERT DISPLAYED

NO

Shut down affected engine. Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

Oil Quantity, Temperature and Pressure (Affected Engine)
. . . . MONITOR

[END]

ENG__OIL TEMP HI

Throttle (Affected Engine) . . . . ADJUST

NOTE: Advancing throttle results in increased fuel flow and may decrease oil temperature.

Record maximum temperature reading in aircraft Flight and Maintenance Log.

NOTE: Operation in caution range is permissible for 15 minutes. Operation above redline is not permitted.

OIL TEMPERATURE WITHIN LIMITS

NO

Continue engine operation. Monitor oil temperature.

[END]

Shut down engine.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

ENG__RPM HI

Throttle (Affected Engine). . . . . RETARD

RPM REMAINS ABOVE REDLINE

NO

Shut down engine.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

RPM EXCEEDED 124% N_1 OR 114% N_2

NO

Throttle (Affected Engine) . . . . IDLE

Use higher thrust only at PIC’s discretion.

NOTE: If any engine indications are abnormal at minimum thrust, a precautionary shutdown should be considered.

[END]

Operate engine at a throttle setting necessary to maintain N_1 and N_2 below red lines.

[END]

ENG__RPM LO

Observe engine parameters for flameout or failure indications.

RESTART DESIRED

NO

Associated Throttle . . . . IDLE

ENG IGN OVRD Switch . . . . OVRD ON

Push ENG IGN OVRD switch and observe OVRD ON light illuminates.

NOTE: Air starts, expecially at high altitude, may result in the engine accelerating to idle very slowly. Slow acceleration may be incorrectly interpreted as a hung start or an engine malfunction. If N2 is steadily increasing, and EGT remains wihtin limits, the start is progressing normally.

ABNORMAL START

NO

Fuel Switch . . . . OFF

ENG IGN OVRD Switch . . . . OFF

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

ENGINE RESTARTS

NO

After engine is stabilized at idle,

ENG IGN OVRD Switch . . . . OFF

Unless thrust is required for safety of flight, observe 1 minute engine warm-up at idle before gradually increasing thrust to resume normal operations.

[END]

Refer to Abnormal Non-Alert procedure - ENGINE RESTART IN FLIGHT.

[END]

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

SELECT FADEC ALTN

"SEL FLAP LIM OVRD" AND/OR "SEL ELEV FEEL MAN" AND/OR "IAS COMPARATOR MONITOR" ALSO DISPLAYED

NO

Refer to Emergency Non-Alert procedure – AIRSPEED: LOST, SUSPECT OR ERRATIC.

[END]

Autothrottles . . . . DISENGAGE

Associated Throttle . . . . SET 70% N1 OR LESS

Illuminated FADEC MODE Switch . . . . PUSH

Associated FADEC MODE Switch . . . . PUSH AGAIN

NOTE: First push selects FADEC ALTN mode. Second push attempts to return FADEC to normal mode.

"SELECT FADEC ALTN" ALERT EXTINGUISHES

NO

Autothrottles. . . . . ENGAGE

[END]

Illuminated FADEC MODE Switch . . . . PUSH

Associated Throttle . . . . SET AS REQUIRED

Remaining Engines
(One at a Time). . . . . REDUCE THRUST,SELECT ALTN

Autothrottles . . . . ENGAGE

During landing roll, limit reverse thrust to 90% N_1.

[END]

Interim Operating Procedure 1-86A

SUBJECT: Fuel Pump

Following is a combined Effectivity List and Filing Instructions. Filing instructions are by procedures. Should it be necessary to file or retain this Interim Operating Procedure in a later revision where procedure location may have changed, insert or move Interim Operating Procedure so that it faces subject matter to which it applies.

FILE IN: MD-11 Flight Crew Operating Manual, Volume I Quick Reference Handbook, so that pages 1 and 2 follow the FUEL tab and file the remaining pages so they face the appropriate Abnormal Alert procedures in the FUEL section.

This IOP covers the following level 2 FUEL alerts:

- TAIL PUMPS LO
- TANK __PUMPS LO
- TNK __AFT PMP LO
- TNK __FWD PMP LO
- TNK __XFER PMP LO

REASON FOR IOP:

To provide the flight crew with guidance on how to disable selected fuel pump circuitry following a pump failure. These procedure are not to be used for displayed alerts associated with normal fuel management/transfer.

IOP 1-86A is being issued because the fuel pump electrical connector replacement parts specified in the Service Bulletin MD11-28A113 and referenced in IOP 1-86 are showing failure characteristics similar to the original parts.

INSTRUCTIONS:

IOP 1-86 is superseded by IOP 1-86A. Please remove IOP 1-86 and replace with IOP 1-86A. File and retain this IOP until notified to remove it.

Intentionally
Blank

Abnormal Procedures

Chapter AP

Table Of Contents

Section 50

Intentionally
Blank

BALST FUEL DISAG

AIRCRAFT ON GROUND

NO

Compare FMS ballast fuel entry with actual ballast fuel quantity and distribution.

FMS BALLAST FUEL QUANTITY CORRECT

NO

BALLAST TANK FUEL QUANTITY CORRECT

NO

Call maintenance.

[END]

Advise ground personnel to correct ballast tank fuel quantity.

[END]

Enter correct quantity into FMS.

[END]

FUEL SYSTEM SELECT Switch . . . . MANUAL

Operate fuel pumps as necessary to stop erroneous transfer.

[END]

CG OUT OF LIMIT

NOTE: Alert may appear after fuel dump is completed and DUMP switch is moved to OFF, even if fuel system is operating in automatic mode.

FUEL SYSTEM IN AUTO MODE

NO

Observe SD FUEL synoptic page.

FUEL BEING RESCHEDULED

NO

No further action required.

[END]

FUEL SYSTEM SELECT Switch . . . . .MANUAL

Reschedule fuel distribution and monitor fuel usage as required.

[END]

DUMP VLV__DISAG

DUMP SWITCH ON

NO

Fuel dump time will be increased.
[END]

Do not pressurize the fuel crossfeed manifold for remainder of flight.

Tank TRANS Switches (All) . . . . VERIFY OFF

Tank XFEED Switches (All). . . . . VERIFY OFF

[END]

FSC AUTO FAIL

FUEL SYSTEM SELECT Switch . . . . MANUAL

Continue flight with fuel system in manual.

[END]

FUEL DUMP LEVEL

DUMP Switch . . . . OFF

DUMP SWITCH ON LIGHT FAILS TO EXTINGUISH

NO

FUEL DUMP EMER STOP Switch. . . . . STOP
[END]

No further crew action required.
[END]

FUEL OFF SCHEDULE

FUEL SYSTEM IN AUTO MODE

NO

Observe SD FUEL synoptic page.

FUEL BEING RESCHEDULED

NO

No further crew action required.

[END]

FUEL SYSTEM SELECT Switch . . . . .MANUAL

Reschedule fuel distribution and monitor fuel usage as required.

[END]

FUEL QTY ALERTS

FUEL SYSTEM SELECT Switch . . . . MANUAL

OVERHEAD FUEL QUANTITY READOUTS BLANK

NO

NOTE: Fuel quantity system is inoperative. Use FMS UFOB on WEIGHT INIT PAGE to determine fuel remaining. FMS UFOB and GW are now calculated using fuel flow only.

TANKS 1 AND 3 FILL VALVES ARMED

NO

TANK FILL Valve Switches (All) . . . VERIFY ARM,FILL

When “TAIL PUMPS LO” alert is displayed,

TAIL TANK TRANS Switch . . . . OFF

When "AUX UPR PUMPS LO" alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

TANK 2 TRANS Switch . . . . ON

If "TNK__FUEL QTY LO" alert is displayed,

Tank XFEED Switches (All). . . . . ON

NOTE: FMS UFOB and GW are not accurate during or after fuel dump. If fuel dump is required, calculate dump time by using 2,268 kilograms per minute dump rate. When fuel dump time has elapsed, push dump switch off. Subtract the amount of fuel dumped from FMS UFOB. Enter the result as FMS UFOB.

[END]

(CONTINUED)

FUEL QTY ALERTS (Continued)

OVERHEAD FUEL QUANTITY READOUTS BLANK

NO

(CONTINUED)

TANKS 1 AND 3 FILL VALVES ARMED

NO

(CONTINUED)

TANK 2 FILL Valve Switch . . . . ARM,FILL

When “TAIL PUMPS LO” alert is displayed,

TAIL TANK TRANS Switch . . . . OFF

When "AUX UPR PUMPS LO" alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

If "TNK__FUEL QTY LO" alert is displayed,

TANK XFEED Switches (All) . . . . ON

NOTE: FMS UFOB and GW are not accurate during or after fuel dump. If fuel dump is required, calculate dump time by using 2,268 kilograms per minute dump rate. When fuel dump time has elapsed, push dump switch off. Subtract the amount of fuel dumped from FMS UFOB. Enter the result as FMS UFOB.

[END]

First Officer’s EIS SOURCE Selector . . . . AUX (OR 1)

(CONTINUED)

FUEL QTY ALERTS (Continued)

"FUEL QTY ALERTS" REMAINS DISPLAYED

NO

First Officer's EIS Source Selector. . . . . ORIGINAL POSITION

Monitor fuel quantity readouts on overhead panel.

NOTES: On the SD FUEL synoptic page, an X will cover the area of removed or invalid data. Subsequent alerts for removed or invalid data will not be displayed.

FMS UFOB and GW are now calculated using fuel flow only. If fuel dump is required, calculate dump time using 2,268 kilograms per minute rate. When fuel dump time has elapsed, push dump switch off. After fuel dump add total fuel on overhead fuel panel and enter the amount as FMS UFOB.

[END]

Fuel quantity alerts and SD FUEL synoptic page data are normal.

FUEL SYSTEM SELECT Switch . . . . .AS REQUIRED

[END]

FUEL QTY FAULT

(Amber, “X”ed or Frozen)

Select the appropriate procedure below:

– TANK FUEL QUANTITY ON SD FUEL SYNOPTIC PAGE - AMBER

NOTE: Affected tank fuel quantity is valid, but displayed in amber.

Enter fuel quantity fault in aircraft Flight and Maintenance Log.

[END]

– TANK FUEL QUANTITY ON SD FUEL SYNOPTIC PAGE - “X”ED

TANK 1, 2 OR 3 “X”ED OR ANY TANK CONTAINS BALLAST FUEL

NO

FUEL SYSTEM SELECT Switch . . . . .MANUAL

ANY TANK CONTAINS BALLAST FUEL

NO

Maintain ballast fuel as required.

Calculate fuel quantity in affected tank.

NOTES: Calculate fuel quantity in affected tank by subtracting the sum of the total fuel used and the total fuel of operational gages from the departure fuel.

FMS UFOB and GW are now calculated using fuel flow only.

(CONTINUED)

FUEL QTY FAULT (Continued)

FMS UFOB and GW are not accurate during or after fuel dump. If fuel dump is required, calculate dump time by using 2,268 kilograms per minute dump rate. When fuel dump time has elapsed, push dump switch off and subtract the amount of fuel dumped from FMS UFOB. Enter the result as FMS UFOB.

[END]

– FUEL QUANTITIES DO NOT CHANGE (FROZEN)

FUEL SYSTEM SELECT Switch . . . . MANUAL

NOTE: Fuel quantity system is inoperative. Subtract total fuel used from departure fuel. Enter the result as FMS UFOB on WEIGHT INIT page. FMS UFOB and GW are now calculated using fuel flow only.

TANKS 1 AND 3 FILL VALVES CAN BE LATCHED IN ARM

NO

TANK FILL Valve Switches (All) . . . . VERIFY ARM,FILL

When “TAIL PUMPS LO” alert is displayed,

TAIL TANK TRANS . . . . OFF

When “AUX UPR PUMPS LO” alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

TANK 2 TRANS Switch . . . . ON

If “TNK__FUEL QTY LO” alert is displayed,

Tank XFEED Switches (All) . . . . ON

NOTE: If an extra crew member is available, consider cycling FUEL QTY NORMAL POWER and FUEL QUANTITY ALTN POWER C/B’s (located at E20 and F20 respectively, on the upper main C/B panel) simultaneously to attempt to restore system operation.

(CONTINUED)

FUEL QTY FAULT (Continued)

TANKS 1 AND 3 FILL VALVES CAN BE LATCHED IN ARM

NO

(CONTINUED)

FMS UFOB and GW are not accurate during or after fuel dump. If fuel dump required, calculate dump time by using 2,268 kilograms per minute dump rate. When fuel dump time has elapsed, push dump switch off. Subtract the amount of the fuel dumped from FMS UFOB. Enter the result as FMS UFOB.

[END]

Tank 2 FILL Valve Switch . . . . VERIFY ARM,FILL

When “TAIL PUMPS LO” alert is displayed,

TAIL TANK TRANS Switch . . . . OFF

When "AUX UPR PUMPS LO" alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

If "TNK__FUEL QTY LO" alert is displayed,

Tank XFEED Switches (All) . . . . ON

NOTE: If an extra crew member is available, consider cycling FUEL QTY NORMAL POWER and FUEL QUANTITY ALTN POWER C/B’s (located at E20 and F20 respectively, on the upper main C/B panel) simultaneously to attempt to restore system operation.

FMS UFOB and GW are not accurate during or after fuel dump. If fuel dump required, calculate dump time by using 2,268 kilograms per minute dump rate. When fuel dump time has elapsed, push dump switch off. Subtract the amount of the fuel dumped from FMS UFOB. Enter the result as FMS UFOB.

[END]

Intentionally
Blank

FUEL QTY/USED CHK

Consequences:

POSSIBLE FUEL LEAK

FUEL Quantity . . . . EVALUATE
Check departure fuel minus total used is approximately equal to present fuel on board.

NOTES: Pressing the FUEL USED RESET button after engine start may also cause the "FUEL QTY/USED CHK" to be displayed.

Some inflight maneuvers or turbulence may cause fuel quantity indications to fluctuate enough to cause the "FUEL QTY/USED CHK" alert to be displayed. In such cases, the alert should extinguish after the aircraft is established in stable flight for 2 to 3 minutes.

If a fuel leak is suspected, confirmed, or sufficient reserves cannot be verified, continue with the remainder of the FUEL QTY/USED CHK procedure and consider landing at nearest suitable airport.

Continue fuel comparison checks, check fuel quantity against the flight planned fuel, and if conditions permit, scan outside for possible fuel venting.

(Continued)

FUEL QTY/USED CHK (Continued)

After fuel quantity indications stabilize,

FUEL ON BOARD IS APPROXIMATELY EQUAL TO CALCUALTED FUEL

NO

No further crew action required.

[END]

FUEL SYSTEM Select Switch . . . . MANUAL

All TANK TRANS and FILL Valve Switches . . . . OFF

Push all TANK TRANS and FILL valve switches and observe all ON lights extinguish and all ARM/FILL lights extinguish.

NOTE: The previous action isolates each tank to determine which tank quantity is decreasing abnormally.

ABNORMAL QUANTITY DECREASE FROM TANK 1, 2 OR 3

NO

NOTE: The following actions will isolate leak to either the tank or the engine.

TANK TRANS Switch
(Aux or Nonleaking Main Tank) . . . . ON

TANK XFEED Switch (Leaking Tank). . . . . ON

TANK PUMP Switch (Leaking Tank) . . . . OFF

ABNORMAL QUANTITY DECREASE FROM AUX OR NONLEAKING MAIN TANK

NO

Throttle (Leaking Engine) . . . . IDLE

FUEL Switch (Leaking Engine). . . . . OFF

(CONTINUED)

FUEL QTY/USED CHK (Continued)

ABNORMAL QUANTITY DECREASE FROM TANK 1, 2 OR 3

NO

(CONTINUED)

ABNORMAL QUANTITY DECREASE FROM AUX OR NONLEAKING MAIN TANK

NO

(CONTINUED)

TANK PUMP Switch (Leaking Tank) . . . . ON

TANK XFEED Switch (Leaking Tank) . . . . OFF

TANK TRANS Switch
(Aux or Nonleaking Main Tank) . . . . OFF

Use fuel from leaking tank until nearly empty, then supply engine from another source.

NOTE: Do not transfer fuel into the leaking tank. Consider transferring fuel into another tank.

[END]

AUX TANK L and R TRANS Switches . . . . ON

TANK FILL Valve Switches (All) . . . . ARM,FILL

ABNORMAL QUANTITY DECREASE FROM AUX TANK

NO

TANK FILL Valve Switches (All). . . . . OFF

Refer to Abnormal Alert Procedure - TNK__OVERFILL, in this section.

[END]

Transfer fuel out of tail tank as required.

[END]

FUEL SYS ALERTS

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"FUEL SYS ALERTS" REMAIN DISPLAYED

NO

First Officer's EIS SOURCE Selector. . . ORIGINAL POSITION

NOTE: Some subsequent fuel system faults will not present alerts on EAD or illuminate MASTER WARNING/CAUTION lights; however, annunciator lights on overhead panel are operative.

[END]

Fuel alerts and SD FUEL synoptic page will operate normally.

[END]

LAT FUEL UNBAL

NOTE: If a fuel leak is suspected, accomplish the "FUEL LEAK" procedure in Abnormal Non-Alert section of this manual. The "FUEL QTY/USED CHK" alert is inhibited with the fuel system in MANUAL mode. If alert appeared during fuel dump, complete or terminate fuel dump prior to accomplishing this procedure.

FUEL SYSTEM IN AUTO MODE

NO

FUEL BEING RESCHEDULED

NO

No further action required.

[END]

FUEL SYSTEM SELECT Switch . . . . .MANUAL

FUEL CROSSFEED REQUIRED

NO

Tank TRANS Switch (Supplying Tank). . . . . ON

Tank XFEED Switche(s) (Receiving Engine(s)). . . . . ON

Tank PUMPS Switch(es) (Receiving Engine(s)) . . . . OFF

Fuel Quantities. . . . . .MONITOR

When tank quantities are approximately balanced,

Tank PUMPS Switch(es) (Receiving Engine(s)) . . . . ON

Tank XFEED Switch(es) (Receiving Engine(s)). . . . . OFF

Tank TRANS Switch (Supplying Tank). . . . . OFF

[END]

Tank TRANS Switch (Supplying Tank) . . . . ON

Tank FILL Switch(es) (Receiving Tank(s)) . . . . PUSH AND HOLD

(CONTINUED)

LAT FUEL UNBAL (Continued)

Fuel Quantities . . . . MONITOR

When tank quantities are approximately balanced,

Tank FILL Switch(es) (Receiving Tank(s)). . . . . .RELEASE

Tank TRANS Switch (Supplying Tank) . . . . OFF

[END]

TAIL FUEL QTY LO

Tank 2 PUMPS Switch . . . . ON

TAIL TANK ALT PUMP Switch . . . . OFF

CAUTION: Immediate action is required to prevent flameout of engine 2 due to fuel starvation.

[END]

TAIL PUMPS LO

TAIL TANK QUANTITY LESS THAN
450 KILOGRAMS

NO

TAIL TANK TRANS Switch . . . . OFF

[END]

TAIL TANK ALT PUMP Switch . . . . ON

CAUTION: Observe SD FUEL synoptic page to verify fuel supply from tail tank alternate pump to engine 2. The next action will cause engine 2 to flame out if tail tank alternate pump is not supplying fuel to engine 2.

Tank 2 PUMPS Switch . . . . OFF

CAUTION: Do not allow fuel quantity in tail tank to decrease below 450 kilograms as flameout of engine 2 could occur.

When tail tank quantity has decreased to 450 kilograms or “TAIL FUEL QTY LO” alert is displayed,

Tank 2 PUMPS Switch . . . . ON

TAIL TANK ALT PUMP Switch . . . . OFF

[END]

TAIL PUMPS LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

TAIL TANK TRANS Switch . . . . .OFF

TAIL TANK QUANTITY LESS THAN 450 KGS

NO

No further crew action required.

[END]

TAIL TANK ALTN PUMP Switch. . . . . ON

CAUTION: Observe synoptic to verify fuel supply from tail tank alternate pump to engine 2. The next action will cause engine to flame out if tail tank alternate pump is not supplying fuel to engine 2.

Tank 2 PUMPS Switch . . . . .OFF

CAUTION: Do not allow fuel quantity in tail tank to decrease below 450 kilograms as flameout of engine 2 could occur.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Nomenclature	Location
TAIL L XFR	UPPER MAIN Row E
TAIL R XFR	UPPER MAIN Row F

(CONTINUED)

TAIL PUMPS LO (Continued)

When tail tank quantity has decreased to 450 kilograms of “TAIL FUEL QTY LO” alert is displayed,

Tank 2 PUMP Switch . . . . ON

TAIL TANK ALT PUMP Switch. . . . . OFF

[END]

TANK __PUMPS LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

Associated Tank TRANS Switch . . . . ON

Associated XFEED Switch . . . . ON

Associated Tank PUMPS Switch . . . . OFF

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TANK 1 PUMPS LO	TANK 1 AFT TANK 1 FWD	UPPER MAIN Row G UPPER MAIN Row E
TANK 2 PUMPS LO	TANK 2L AFT TANK 2R AFT TANK 2 FWD	OVERHEAD Row G UPPER MAIN Row F UPPER MAIN Row E
TANK 3 PUMPS LO	TANK 3 AFT TANK 3 FWD	UPPER MAIN Row F UPPER MAIN Row G

[END]

TNK__ AFT PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

Associated Tank TRANS Switch . . . . ON

Associated XFEED Switch . . . . ON

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 AFT PMP LO	TANK 1 AFT	UPPER MAIN Row G
TNK 2L AFT PMP LO	TANK 2L AFT	OVERHEAD Row G
TNK 2R AFT PMP LO	TANK 2R AFT	UPPER MAIN Row F
TNK 3 AFT PMP LO	TANK 3 AFT	UPPER MAIN Row F

[END]

TNK__ FWD PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 FWD PMP LO	TANK 1 FWD	UPPER MAIN Row E
TNK 2 FWD PMP LO	TANK 2 FWD	UPPER MAIN Row E
TNK 3 FWD PMP LO	TANK 3 FWD	UPPER MAIN Row G

At top of descent,

TANK FUEL QUANTITY LESS THAN 5,200 KGS

NO

Associated XFEED Switch . . . . ON

Another XFEED Switch . . . . ON

[END]

No further crew action required

[END]

Intentionally
Blank

TANK__PUMPS LO

TANK TRANS Switch (Affected Tank). . . . . ON
XFEED Switch (Affected Tank) . . . . ON
Tank PUMPS Switch (Affected Tank) . . . . OFF
[END]

TNK__AFT PMP LO

Tank TRANS Switch (Affected Tank). . . . . ON
XFEED Switch (Affected Tank) . . . . ON
[END]

TNK__FWD PMP LO

At top of descent,

TANK FUEL QUANTITY LESS THAN 5,200 KILOGRAMS

NO

XFEED Switch (Affected Tank) . . . . ON
Another XFEED Switch . . . . ON
[END]

No further crew action required.
[END]

TNK__OVERFILL

FUEL SYSTEM AUTO

NO

Observe SD FUEL synoptic page.

FSC TAKING CORRECTIVE ACTION

NO

No further crew action required.

[END]

FUEL SYSTEM SELECT Switch . . . . .MANUAL

Select appropriate procedure below:

TNK WING OVERFILL

Depressurize crossfeed manifold.

Pressurize manifold only to maintain fuel quantity to upper aux and main tanks 900 to 1,400 kilograms below full.

[END]

TNK TAIL OVERFILL

Transfer fuel out of tail tank using TAIL TANK TRANS pumps.

[END]

TNK__ XFER PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

Associated XFEED Switch . . . . ON

Associated Tank TRANS Switch. . . . . .OFF

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 XFER PMP LO	TANK 1 XFR	UPPER MAIN Row F
TNK 2 XFER PMP LO	TANK 2 XFR	UPPER MAIN Row G
TNK 3 XFER PMP LO	TANK 3 XFR	UPPER MAN Row E

[END]

Intentionally
Blank

TNK__TIP FUEL LO

NOTES:

- *M_{MO} above 30,704 feet is limited to .85 MACH.*
- *V_{MO} below 30,704 feet is limited to 320 KIAS.*

Tank TRANS Switch (Affected Tank). . . . . ON

Tank FILL Valve Switch (Affected Tank) . . . . PUSH AND HOLD

Two minutes after alert is no longer displayed,

Tank FILL Valve Switch (Affected Tank) . . . . .RELEASE

Tank TRANS Switch (Affected Tank). . . . . OFF

[END]

TNK__XFER PMP LO

XFEED Switch (Affected Tank) . . . . ON

Tank TRANS Switch (Affected Tank). . . . . OFF

[END]

Intentionally
Blank

Abnormal Procedures — Chapter AP

Table Of Contents — Section 60

Intentionally
Blank

HSC AUTO FAIL

HYD SYSTEM SELECT Switch . . . . MANUAL
HYD SYSTEM SELECT Switch . . . . AUTO

"HSC AUTO FAIL" ALERT DISPLAYED

NO

HYD SYSTEM SELECT Switch . . . . MANUAL
Continued flight with system in manual.

[END]

Enter details of malfunction in aircraft Flight and Maintenance Log.

[END]

HYD ALERTS

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"HYD ALERTS" REMAINS DISPLAYED

NO

First Officer's EIS SOURCE Selector . . ORIGINAL POSITION

NOTE: Some subsequent hydraulic system faults will not present alerts on EAD or illuminate MASTER WARNING/CAUTION lights; however, annunciator lights on overhead panel are operative.

[END]

Hydraulic alerts and SD HYD synoptic page will operate normally.

[END]

HYD 1 FAIL

NOTE: Outboard slats will not retract if they were extended before the loss of pressure occurred. "SLAT DISAG" alert will be displayed when flap/slat handle is in the 0/RET position.

"HYD 3 ELEV OFF" ALERT DISPLAYED

NO

Land at nearest suitable airport.

AUTO BRAKE Selector . . . . OFF

Plan a 35/EXT landing.

[END]

HYD 2 FAIL

"HYD 3 ELEV OFF" ALERT DISPLAYED

NO

Land at nearest suitable airport

Lower rudder is inoperative.

V_{MCA} is 180 KIAS.

CAUTION: Do not attempt a go around at speeds below V_{MCA}.

Recommended maximum crosswind component is 12 knots.

Plan a 35/EXT landing.

[END]

HYD 3 FAIL

NOTE: Autoland is not approved. Autopilot must be disconnected by 100 feet AGL.

Autopilot GA not recommended.

Outboard slats will not retract if they were extended before the loss of pressure occurred. “SLAT DISAG” alert will be displayed when flap/slat handle is in the 0/RET position.

Plan a 35/EXT landing.

When gear extension is required,

Airspeed . . . . MAX 230 KIAS

Main Gear Alternate Extension Lever. . . . . RAISE,LATCH

After three green lights illuminate,

Center Gear Alternate Extension Handle/Lights PULL,4GREEN

GEAR Handle . . . . DOWN

AUTO BRAKE Selector . . . . OFF

Autopilot . . . . .DISCONNECT BY 100 FEET AGL

[END]

HYD__PRES HI

HYDRAULIC SYSTEM IN MANUAL MODE

NO

HYD SYS L PUMP Switch (Affected System) . . . . OFF

"HYD___PRES HI" ALERT REMAINS DISPLAYED

NO

After 10 seconds,

HYD SYS L PUMP Switch (Affected System). . . . . ON

Make entry in aircraft Flight and Maintenance Log.

NOTE: If alert remains displayed, system pressure sensor has failed.

[END]

[END]

Make entry in aircraft Flight and Maintenance Log.

NOTE: If alert remains displayed, system pressure sensor has failed.

[END]

HYD__PRES LO

Affected HYD PUMP Switch(es) (Affected System) . . . . OFF

Repressurize system for approach and landing.

If system does not repressurize,

Refer to Abnormal Alert procedure - HYD 1 FAIL, HYD 2 FAIL, or HYD 3 FAIL, as appropriate.

[END]

HYD__QTY LO

HYD SYS PUMP Switches(Affected System). . . . . OFF

Associated RMP Switches . . . . OFF

NOTE: Hydraulic system may be used for approach and landing.

[END]

HYD__TEMP HI

AIRCRAFT IN FLIGHT

NO

If phase of flight permits,

HYD SYS PUMP Switches(Affected System) . . . . OFF

Associated RMP Switches . . . . OFF

NOTE: *Hydraulic system should be on for approach and landing.*

[END]

Flight Controls. . . . . CYCLE

NOTE: Cycling flight controls, with affected engine hydraulic pumps operating, will circulate and cool hydraulic fluid and may cause “HYD__TEMP HI” alert to extinguish. If alert remains, call maintenance.

[END]

Abnormal Procedures — Chapter AP

Table Of Contents — Section 70

Intentionally
Blank

CARGO DOOR__

SINGLE ALERT DISPLAYED

NO

Cargo Door Test . . . . PERFORM

OTHER SENSOR ON FAULTED DOOR TESTS CORRECTLY ON MISC SYSTEM DISPLAY

NO

No further crew action required. Enter fault in aircraft Flight and Maintenance Log.

[END]

CABIN PRESS SYSTEM SELECT Switch . . . . MANUAL

Rotate CABIN PRESS manual rate selector towards CLIMB and observe indicator moves toward OP.

Descend aircraft to 15,000 feet or minimum safe altitude, whichever is higher.

Reduce cabin differential pressure to 2 psi or less.

NOTE: An aircraft altitude of 15,000 feet provides 9,500 feet cabin altitude at 2 psi differential pressure.

Land at nearest suitable airport.

[END]

IRU__FAIL

IRS Mode Selector (Affected Unit) . . . . OFF

IRU 1 OFF

NO

IRS Switch (Captain's SISP Panel) . . . . CAPT ON AUX

IRU 2 OFF

NO

IRS Switch (F/O's SISP Panel) . . . . F/O ON AUX

[END]

MISC ALERTS

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

First Officer's EIS SOURCE Selector . . . . AUX (OR 1)

"MISC ALERTS" REMAINS DISPLAYED

NO

First Officer's EIS SOURCE Selector. . . ORIGINAL POSITION

[END]

Miscellaneous alerts and systems display will operate normally.

[END]

MSC AUTO FAIL

Operate engine ignition manually. Refer to Supplemental Procedure under ENG/APU tab, ENGINE IGNITION MANUAL OPERATION.

The ENG START switch must be held out. Release the switch when the engine reaches 45% N_2.

NOTE: If a cargo fire condition exists, manual timing of agent discharge is required.

If a "CRG FIRE LWR__" alert is received, both AGENT DISCH lights will remain illuminated as long as the fire is detected. Only agent 1 should be discharged, followed in 90 minutes by agent 2.

[END]

TAIL CONE UNLOCK

AIRCRAFT ON GROUND

NO

Call maintenance.

NOTE: Aircraft may be dispatched with this alert provided maintenance verifies what tailcone and number 2 engine service platform (patio) doors are locked in accordance with the MEL. The Level 1 alert “DOOR OPEN” will remain displayed until after engine start.

[END]

AIRCRAFT DISPLAYS BUFFETING OR CONTROL COLUMN PULSES

NO

Reduce airspeed to low cruise.

Enter details of malfunction in aircraft Flight and Maintenance Log.

[END]

Continue normal operations and speed.

Enter details of malfunction in aircraft Flight and Maintenance Log.

[END]

Abnormal Procedures — Chapter AP

Table Of Contents — Section 80

Interim Operating Procedure 1-104

IOP2-215A.(SectNum) Interim Operating Procedure 2-215A-(SectName)

SUBJECT: AIR CONDITIONING PACK SURGE

This Interim Operating Procedure (IOP) is applicable for the operator codes listed in the EFFECTIVITY field. Filing instructions are provided below. IOPs are numbered by Volume Number-IOP Number-Letter Change. IOP numbers may not be consecutive because all IOPs are not applicable to all operators. Letter changes are applied to IOP updates. IOPs are intended to precede the relevant subject matter and should be moved if subsequent revisions change the pagination of the Flight Crew Operations Manual.

EFFECTIVITY:	AZ, CIX, DAC, FM, GMN, KE, LH, SV, SVP2, TBC, TML, UPS, WLD, WLDP2, Y4, YM, YMP2, YS, Z4, Z5, Z8, ZMF
FILE IN:	MD-11 Flight Crew Operations Manual, Volume I, Quick Reference Handbook, Abnormal Non-Alerts, so that this IOP follows the Table of Contents.

REASON FOR IOP:

To provide crews with a procedure to eliminate air conditioning pack surge. This procedure is only effective for airplanes in the freighter configuration.

INSTRUCTIONS:

Retain IOP 1-104 until notified to remove it.

AIR CONDITIONING PACK SURGE

AIRCRAFT ON GROUND

NO

If takeoff is to be accomplished,

Consider a PACKS OFF takeoff.

AFTER TAKEOFF

NO

NOTE: Following a PACKS OFF takeoff, when the aircraft passes through 1,000 feet above airport elevation and climb power is set, but no later than 4,500 feet above airport elevation, the ESC will restore the air conditioning. Allow sufficient time for the ESC to stabilize the air conditioning system before evaluating pack surge.

PACK SURGE STOPS

NO

No further crew action required.

[END]

Crew workload permitting,

Zone Temperature Selector(s) . . . SELECT COLDER TEMPERATURE

One at a time and in the following order (cargo, courier, cockpit), select a temperature at least 5 degrees below the actual temperature of the associated zone. Pause 2 minutes, if surge continues, select zone temperature to full cold and evaluate. If surge persists after 1 minute, proceed to the next zone, leaving previous zone selection at full cold.

(CONTINUED)

AIR CONDITIONING PACK SURGE (Continued)

PACK SURGE STOPS

NO

No further crew action required.

[END]

With all packs operating,

AIR SYSTEM SELECT Switch . . . . MANUAL

Push AIR SYSTEM SELECT switch and observe MANUAL light illuminates.

PACK 2 Switch . . . . OFF

Push PACK 2 switch and observe OFF light illuminates.

PACK SURGE STOPS

NO

No further crew action required.

[END]

PACK 2 Switch . . . . ON

Push PACK 2 switch and observe OFF light extinguishes.

PACK 3 Switch . . . . OFF

Push PACK 3 switch and observe OFF light illuminates.

PACK SURGE STOPS

NO

No further crew action required.

[END]

PACK 3 Switch . . . . ON

Push PACK 3 switch and observe OFF light extinguishes.

(CONTINUED)

AIR CONDITIONING PACK SURGE (Continued)

PACK 1 Switch. . . . . OFF

Push PACK 1 switch and observe OFF light illuminates.

PACK SURGE STOPS

NO

No further crew action required.

[END]

Select any one pack to OFF and make log book entry.

Prior to approach,

Pack Selected to OFF . . . . SELECT ON

AIR SYSTEM SELECT Switch . . . . .AUTO/AS REQUIRED

[END]

DEU FAIL

EIS SOURCE Selector (On Failed Side) . . . . AUX

DISPLAY UNITS RECOVERED

NO

No further crew action required.

[END]

EIS SOURCE
Selector (On Failed Side) . . . . RETURN TO ORIGINAL POSITION

CAUTION: Do not rotate the EIS SOURCE selector on the failed side to the opposite (cross-cockpit) position. Such action may cause the operating displays to replicate the failure pattern of the failed DEU.

DISPLAY ELEC UNIT
Circuit Breaker (On Failed Side) . . . PULL (WAIT 5 SECONDS)/RESET

NOTE: DISPLAY ELEC UNIT 1 circuit breaker is located at the left overhead C/B panel F-3. DISPLAY ELEC UNIT 2 circuit breaker is located at right overhead C/B panel F-28.

DISPLAY UNITS RECOVERED

NO

No further crew action required.

[END]

DISPLAY ELEC UNIT
Circuit Breaker (On Failed Side) . . . . PULL

Complete flight using the operating displays.

[END]

DITCHING

Crew and Courier(s) . . . . ALERT
Transponder . . . . SET 7700
ATC. . . . . ADVISE
Fuel Quantity . . . . REDUCE
V_{APP} . . . . .CHECK
APU . . . . VERIFY OFF
First Aid And Survival Equipment,Loose Equipment . . . . .STOW
Left Observer's Seat . . . . FACING FORWARD
Right Observer's Seat (If Occupied) . . . . FACING FORWARD
NO SMOKE/SEAT BELTS . . . . ON

Freighter aircraft:
Courier(s) Ditching Preparation . . . . COMPLETE
Crew Vests, Belts, Harnesses . . . . .ON,FASTEN
DITCHING Switch. . . . . ON

AIR SYSTEM MANUAL

NO

AVNCS FAN Switch. . . . . .OVRD
At or below 10,000 feet,
PACK,BLEED Switches. . . . . OFF

(CONTINUED)

DITCHING (Continued)

CABIN PRESS CONTROLLER
MANUAL

NO

At or below 10,000 feet,

CABIN PRESS Manual Rate Selector . . . . CLIMB

When aircraft is depressurized (cabin differential pressure less than 0.5 psi),

CABIN PRESS Manual Rate Selector . . . . FULL DESC

Freighter aircraft:

Crew,Courier(s) Briefing . . . . COMPLETE

NOTE: When beginning final approach, advise crew and courier(s) to brace for impact (30 seconds prior to touchdown) and not to release harness until aircraft has come to a complete stop.

GEAR Handle,Lights . . . . UP/OFF

FLAP/SLAT Handle 50/EXT

(CONTINUED)

DITCHING (Continued)

NOTE: Continuous aural warnings will sound.

If time permits, prior to leaving aircraft, move all ENG FIRE handles to full forward.

If cockpit door is jammed, exit via smoke panel. If debris jams exit to cabin, use windows.

After aircraft has stopped,

EVACUATION Command . . . . INITIATE

NOTE: If time permits, prior to leaving aircraft, move all ENG FIRE handles to full forward.

If cockpit door is jammed, exit via smoke panel. If debris jams exit to cabin, use windows.

[END]

ENGINE ABNORMAL START

(Hot, Hung, or No Start)

FUEL Switch (Affected Engine) . . . . OFF

STARTER DISENGAGED

NO

Allow N_2 RPM to decrease to 15% or less.

Verify IGN A or B is selected.

ENG START Switch. . . . . PULL

Motor engine with starter for 30 seconds.

ENG START Switch . . . . .PUSH

Determine type of abnormal start:

HOT START

Record maximum EGT and elapsed time EGT was above 750°C.

Call maintenance.

[END]

HUNG START/NO START - ENGINE 1 OR 3

ENG IGN A IS SELECTED

NO

Call maintenance

[END]

Select ENG IGN A and attempt another start.

Observe starter air pressure, starting fuel flow, maximum N_2 achieved, and maximum EGT.

If unsuccessful, call maintenance.

[END]

ENGINE ABNORMAL START (Continued)

HUNG START/NO START - ENGINE 2

Select other ENG IGN (A or B) and attempt another start.

Observe starter air pressure, starting fuel flow, maximum N_2 achieved, and maximum EGT.

If unsuccessful, call maintenance.

[END]

ENGINE COMPRESSOR STALL(S)

ENGINE OPERATION NORMAL

NO

Continue engine operation.
[END]

Autothrottles . . . . DISENGAGE

Throttle (Affected Engine) . . . . RETARD TO IDLE

ENG IGN OVRD Switch . . . . .OVRD ON

Associated ENG and WING or TAIL ANTI-ICE Switches . . . . ON

ECON Switch . . . . OFF

ENGINE OPERATION NORMAL

NO

CAUTION: Continued operation of an engine that exhibits stall tendencies must be done with extreme caution. If a high EGT becomes evident or a rapid EGT rise occurs during slow throttle advance, or if an increase in vibration level is noted, shut down engine.

Throttle (Affected Engine) . . . . (SLOWLY) ADVANCE

NOTE: If compressor stall recurs, at PIC's discretion, operate engine at a reduced thrust level at which compressor stall is not experienced.

If compressor stall does not recur, continue engine operation. Monitor engine indications.

ENG IGN OVRD Switch. . . . . OFF

ENG, WING and TAIL ANTI-ICE Switches . . . .AS REQUIRED

ECON Switch. . . . . ON

[END]

(CONTINUED)

ENGINE COMPRESSOR STALL (Continued)

ENGINE OPERATION NORMAL

NO

(CONTINUED)

Shut down affected engine. Refer to Abnormal Non-Alert Procedure - ENGINE SHUTDOWN IN FLIGHT, in this section.

[END]

ENGINE OIL QUANTITY INCREASE

NOTE: Slight increase/decrease in oil quantity may be normal. Use this procedure when continuous oil quantity increase is observed, oil quantity increase is accompanied by secondary indications, or oil quantity exceeds 21 units or fuel/oil fumes are detected.

Maintain normal thrust settings.

FUEL/OIL FUMES DETECTED

NO

AIR SYSTEM SELECT Switch. . . . . MANUAL

BLEED AIR Switch (Affected System) . . . . OFF

PACK Switch (Affected System) . . . . OFF

Associated ISOL Switch. . . . . ON

NOTE: When in icing with only one bleed air source, exit icing area and maintain ice protection until clear of icing.

Record malfunction in aircraft Flight and Maintenance Log.

[END]

ENG OIL QUANTITY LO/DECREASING

Continue engine operation.

If oil pressure/temperature become abnormal, shut down affected engine. Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

ENGINE PRIMARY INSTRUMENT LOSS

NOTE: If engine parameters are lost (indicated by amber X's over N_1, N_2, EGT, and fuel flow), use this procedure to determine the status of the engine.

AIRCRAFT ON GROUND

NO

Call maintenance.

[END]

Observe EAD.

"OFF" DISPLAYED ABOVE FUEL FLOW ON AFFECTED ENGINE

NO

Shut down affected engine.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT.

NOTE: If " OFF" alert is displayed, it indicates a FADEC power failure has occurred and the engine has been shut down.

Engine shutdown can be verified by checking the electrical system status.

[END]

Operate the affected engine throttle by continually aligning it with other throttles. Do not allow affected throttle to lead others.

NOTE: If "OFF" alert is not displayed, it indicates a FADEC data failure has occurred. The engine will continue to operate and respond to throttle movement.

Continue flight and enter details of malfunction in aircraft Flight and Maintenance Log.

[END]

ENGINE RESTART IN FLIGHT

NOTE: Do not attempt to restart an engine if it has been shut down because of engine fire or if there are indications of engine damage.

Throttle (Affected Engine) . . . . VERIFY IDLE

FUEL Switch (Affected Engine) . . . . VERIFY OFF

HYD SYSTEM IN MANUAL MODE

NO

HYD SYS L and R PUMP Switches (Affected Engine) . . . OFF

FUEL SYSTEM IN MANUAL MODE

NO

TANK PUMPS Switch (Affected Engine). . . . . ON

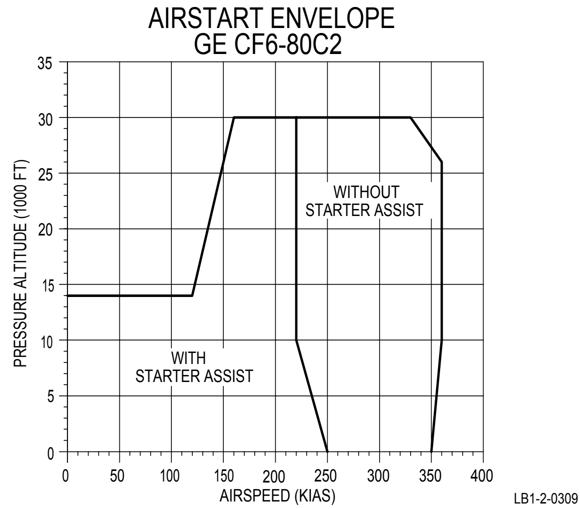
Air starts may be attempted at any altitude and airspeed. Recommended altitude and airspeed for air starts are as follows:

- Above 10,000 feet, greater than 220 KIAS
- Below 10,000 feet, greater than 250 KIAS

(CONTINUED)

ENGINE RESTART IN FLIGHT (Continued)

AIRSTART ENVELOPE



ENG IGN OVRD Switch . . . . .OVRD ON

NOTE: If cyan lightning stroke does not appear on EGT display (ENG IGN OVRD switch inoperative), select engine ignition A and B. Additionally, place engine anti-ice switch to ON for the appropriate engine, as this action also activates the aircraft ignition system. If a restart is not completed within 60 seconds, the engine anti-ice can be selected again.

(CONTINUED)

ENGINE RESTART IN FLIGHT (Continued)

N_2 RPM LESS THAN 15%

NO

AIR SYSTEM IN MANUAL MODE

NO

BLEED AIR Switch (Affected Engine). . . VERIFY OFF

PACK Switch (Affected Engine) . . . . VERIFY OFF

BLEED AIR Switch
(Engine Supplying Bleed For Start). . . . . . VERIFY ON

Associated ISOL Switch . . . . VERIFY ON

ENG START Switch (Affected Engine) . . . . PULL

When N_2 rpm is 15% or greater,

FUEL Switch (Affected Engine) . . . . ON

NOTE:Airstarts,especially at high altitudes, may result in engine(s) accelerating to idle very slowly. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N2 is steadily increasing, and EGT remains within limits, the start is progressing normally.

(CONTINUED)

ENGINE RESTART IN FLIGHT (Continued)

ABNORMAL START

NO

FUEL Switch (Affected Engine) . . . . OFF

ENG IGN OVRD Switch. . . . . OFF

Refer to Abnormal Non-Alert Procedure - ENGINE SHUTDOWN IN FLIGHT.

[END]

Unless thrust is required for safety of flight, observe 1 minute warmup at idle and gradually resume normal operation.

ENG IGN OVRD Switch . . . . OFF

Verify fuel, hydraulic, air and electrical systems are operating in the desired mode.

[END]

ENGINE SHUTDOWN IN FLIGHT

Throttle (Affected Engine) . . . . IDLE

FUEL Switch (Affected Engine) . . . . OFF

AIR SYSTEM MANUAL

NO

BLEED AIR Switch (Affected Engine) . . . . VERIFY OFF

PACK Switch (Affected Engine). . . . . OFF

Associated ISOL Switch . . . . ON

Transponder/TCAS Selector. . . . . .TA

Consider landing at nearest suitable airport.

[END]

Intentionally
Blank

SUBJECT: UNRESPONSIVE RUDDER PEDAL INPUT

This Interim Operating Procedure (IOP) is applicable for the operator code listed in the EFFECTIVITY field. Filing instructions are provided below. IOPs are numbered by Volume Number-IOP Number-Letter Change. IOP numbers may not be consecutive because all IOPs are not applicable to all operators. Letter changes are applied to IOP updates. IOPs are intended to precede the relevant subject matter and should be moved if subsequent revisions change the pagination of the Flight Crew Operating Manual.

EFFECTIVITY:	ALL
FILE IN:	MD-11 Flight Crew Operations Manual, Volume I, Quick Reference Handbook, so that this page faces Abnormal Non-Alert Procedures, FLIGHT CONTROLS JAMMED OR RESTRICTED.

If rudder pedals are movable, but directional control is greatly reduced or unresponsive to rudder pedal input:

- Directional control is available from ailerons, spoilers and asymmetric engine thrust.
- If a wing engine is inoperative, a missed approach should not be attempted.
- Plan landing on runway with the least cross-wind component, conditions permitting.

REASON FOR IOP:

There was a recent occurrence in which a crew experienced a failure of a rudder torque-tube during a control check. A characteristic of this failure is that although the rudder pedals may move, the pilot's control input to the rudders will have no effect or a reduced effect. This IOP is intended to provide guidance to flight crews in such a circumstance.

INSTRUCTIONS:

Retain IOP 1-102 until notified to remove it.

Intentionally
Blank

FLIGHT CONTROL JAMMED OR RESTRICTED

NOTE: If jam occurs during manually controlled flight, attempt to engage autopilot.
If freezing water is the cause, control may be regained by descending into warmer air.

Seat Belts Switch . . . . ON

AILERON CONTROLS JAMMED OR RESTRICTED

NO

If jam was detected during cruise, and if aircraft control is adequate, continue to normal descent point.

NOTE: If Roll Control Wheel Steering is installed, some lateral control may be available through the control wheel position sensors.

If controls remain jammed or restricted,

Autopilot/Autothrottles . . . . DISCONNECT

Maximum Airspeed . . . . 290 KIAS/.80 M

Controls Columns/Wheels

(Lateral Axis) . . . . SEPARATE/DISCONNECT

Use force as required to separate/disconnect the two halves of the control system (approximately 90 pounds).

Autothrottles will be available.

Auto pilot may be available, but should be disconnected during descent and approach and landing.

Select a runway with minimum crosswind.

If roll authority is reduced, fly a wider pattern than normal to allow a stabilized approach.

[END]

(CONTINUED)

FLIGHT CONTROLS JAMMED OR RESTRICTED
(Continued)

ELEVATOR CONTROLS JAMMED OR RESTRICTED

NO

If jam was detected during cruise, and if aircraft control is adequate through use of the autopilot, consider continuing to normal descent point.

If controls remain jammed or restricted, and pitch is not controllable,

Autopilot/Autothrottles. DISCONNECT

Maximum Airspeed. . . . . 290 KIAS/.80 M

Apply force as necessary to free jam.

Autothrottles will be available.

Autopilot may be available.

Elevator authority may be reduced.

[END]

RUDDER JAMMED OR RESTRICTED

NO

Use rudder trim (if available) and ailerons for directional control.

If rudder input is not possible, use operative flight controls, trim, and thrust as required to maintain aircraft control.

Select a runway with minimum crosswind.

The possibility of limited directional control with a wing engine inoperative should be considered prior to executing a missed approach.

AUTOBRAKE Selector. OFF

Rudder pedal steering is inoperative.

If directional control is a concern, use differential braking at high speed.

Nose gear steering wheel may be used as speed decreases.

[END]

[END]

FMC DUAL FAILURE

Both MCDUs . . . . SELECT STANDBY

No FMS database is available in standby mode.

Next Waypoint. . . . . VERIFY

Original routing will be retained in both MCDUs.

Verify active waypoint is correct or enter required waypoint with NAME/LAT/LONG format before continuing this procedure.

SISP Switches . . . . .AS REQUIRED

- FLT DIR - If NAV 1 or NAV 2 is selected, offside roll bar will bias out of view. If required, onside FLT DIR may be selected by pushing CAPT ON 2 (for NAV 2) or F/O ON 1 (for NAV 1).
- FMS - FMS switching should be in normal. If both MCDUs are in standby operation, NDs will display the information depicted in the following table.

FMS SISP SW	CAPT ND DISPLAY	F/O ND DISPLAY
NORMAL	MCDU 1	MCDU 2
CAPT ON 2	MCDU 2	MCDU 2
F/O ON 1	MCDU 1	MCDU 1

NOTE: AP 1 will always follow MCDU 1. AP 2 will always follow MCDU 2 regardless of SISP switching.

- CADC, IRS, VOR and APPR should be left in normal unless anomalies in those systems require switching.

Navigation Routing . . . . VERIFY

Continue to destination at PIC's discretion.

If changes to routing are required, both MCDUs should be updated, time and workload permitting. If over water or if few navigational aids are available, both MCDU routings should be maintained. If time does not permit, one MCDU may be updated and utilized for navigational guidance. Waypoints must be entered in the NAME/LAT/LONG format. Direct-to is available only for waypoints entered in the route or to a LAT/LONG entered in data field 1(L).

ILS, LOC, VOR and NDB navaids must be tuned by **frequency** on respective MCDUs.

(CONTINUED)

FMC DUAL FAILURE (Continued)

Monitor IRU positions.

IRU/MCDU with most accurate position (i.e. MCDU 1, AP 1 and NAV 1, if number 1 is most accurate) should be used for lateral guidance.

Compare fuel used/remaining to flight plan at each waypoint.

Vertical navigation is not available. Plan cruise, climb, and descent points accordingly.

Autothrottles are available in speed mode; however, autothrottles will disconnect in level change and G/A modes.

In the event of a go-around, pushing the thrust levers to overboost bar below 14,000 feet MSL and 300 KIAS will provide G/A thrust.

> *NOTE: Do not exceed G/A thrust time limit. Below 14,000 feet MSL, five minutes of go-around power is permissible, if required.*

Above 14,000 feet or 300 KIAS throttle advance to overboost bar will provide MCT thrust. During climb use following table to set thrust manually.

APPROXIMATE CLIMB THRUST N_1 (%) SETTINGS

TAT (°C)	PRESSURE ALTITUDE - (FT)									
	0	5,000	10,000	15,000	20,000	25,000	30,000	35,000	40,000	45,000
-30	89.7	93.6	97.9	97.5	98.7	99.5	102.2	102.3	102.2	102.8
-25	90.6	94.5	98.8	98.4	99.6	100.4	103.1	103.2	103.1	103.7
-20	91.5	95.4	99.7	99.3	100.5	101.3	104.0	104.1	104.0	104.5
-15	92.4	96.3	100.6	100.2	101.4	102.2	104.9	105.0	104.7	105.1
-10	93.2	97.2	101.5	101.1	102.3	103.1	105.8	105.6	104.9	104.3
-5	94.1	98.1	102.3	101.9	103.1	103.9	106.6	105.1	103.7	103.1
0	95.0	99.0	103.2	102.8	104.0	104.8	106.6	103.9	102.4	101.9
5	95.8	99.8	104.1	103.7	104.9	105.7	105.6	102.8		
10	96.7	100.7	105.0	104.5	105.8	104.9	104.8			
15	97.5	101.6	105.9	105.4	105.5	104.1	103.9			
20	98.4	102.5	105.7	105.6	104.7	103.5				
25	99.2	103.4	105.1	104.9	104.0	102.9				
30	100.1	102.7	104.5	104.3	103.4					
35	100.4	102.0	103.9	103.6	102.9					
40	99.7	101.3	103.2	103.0						
45	98.9	100.6	102.7							
50	98.1									

(CONTINUED)

FMC DUAL FAILURE (Continued)

Below 10,000 feet, climb at 250 KIAS or UP/RET V_{REF} + 35, whichever is higher. Refer to QRH - NORMAL & ABNORMAL CONFIGURATION REFERENCE SPEEDS (V_{REF}) table, under Reference Data tab. Above 10,000 feet, climb at 330 KIAS to FL270 and .82 Mach to level off. Descend at .82 Mach to FL270, then 330 KIAS to 10,000 feet and 250 KIAS below 10,000 feet.

To compute approach speeds, use QRH - NORMAL & ABNORMAL CONFIGURATION REFERENCE SPEEDS (V_{REF}) table, under Reference Data tab.

Low speed protection is not available.

FMS databases STARS and APPROACHES are not available.

DUAL LAND, SINGLE LAND and APPR ONLY ILS modes are available.

[END]

FMC SINGLE FAILURE

Immediately following any disconnect or switching of the autopilot, verify the desired FMS lateral mode is displayed. Desired mode may be reselected if necessary.

FMS 1 FAIL

NO

CAPT FMS SISP . . . . .CAPT ON 2

Push FMS on CAPT SISP and observe CAPT ON 2 illuminates.

MCDU 1 (MENU Page, LSK 1L) .FMC 2

Push LSK 1L to select FMC 2 and observe FMC 2 <ACT> displays.

The Captain’s ND must display VOR or APPR mode whenever the F/O’s ND displays the VOR or APPR mode; the selected modes need not be identical.

The Captain’s ND display may be in VOR or APPR when the F/O’s display is in MAP or PLAN.

When the Captain’s and F/O’s ND displays are in MAP or PLAN the RANGE must be the same.

[END]

FMS 2 failed,

F/O FMS SISP . . . . F/O ON 1

Push FMS on F/O SISP and observe F/O ON 1 illuminates.

MCDU 2 (MENU Page, LSK 1L) . . . . .FMC 1

Push LSK 1L to select FMC 1 and observe FMC 1 <ACT> displays.

(CONTINUED)

The F/O’s ND must display VOR or APPR mode whenever the Captain’s ND displays the VOR or APPR mode; the selected modes need not be identical.

The F/O’s ND display may be in VOR or APPR when the Captain’s display is in MAP or PLAN.

When the F/O’s and Captain’s ND displays are in MAP or PLAN the RANGE must be the same.

[END]

FUEL DUMP

FMC FUEL INIT Page. . . . . SELECT

Edit DUMP TO GW to required weight.

Verify calculated dump time.

FUEL Cue Switch . . . . PUSH

DUMP Switch . . . . ON

NOTES: When fuel dump is terminated at the FMS DUMP TO GW value, "DUMP VLV L DISAG" and "DUMP VLV R DISAG" alerts will be displayed.

If fuel dump does not terminate at the FMS DUMP TO GW value, "FMS DUMP DISABLED" alert will be displayed.

If fuel dumps below the low level dump shutoff (approximately 5,216 kilograms per tank), "FUEL DUMP LEVEL" alert will be displayed.

When required fuel level is reached or calculated dump time is reached,

DUMP Switch . . . . OFF

FURTHER FUEL DUMP DESIRED

NO

DUMP Switch. . . . . ON

When required fuel level is reached or calculated dump time is reached,

DUMP Switch. . . . . OFF

DUMP SWITCH ON LIGHT FAILS TO EXTINGUISH

NO

FUEL DUMP EMER STOP Switch . . . . STOP

[END]

No further crew action required.

[END]

FUEL LEAK

Check departure fuel minus total used is approximately equal to present fuel on board. Land at nearest suitable airport if a fuel leak is suspected , confirmed, or if sufficient fuel reserves cannot be verified.

FUEL SYSTEM SELECT Switch . . . . MANUAL

All TANK TRANS and FILL Valve Switches . . . . OFF

NOTE : The previous action isolates each tank to determine which tank quantity is decreasing abnormally.

ABNORMAL QUANTITY DECREASE FROM TANK 1, 2, OR 3

NO

NOTE: The following actions will isolate leak to either the tank or the engine.

TANK TRANS Switch (Aux or Nonleaking Main Tank) . . . . ON

TANK XFEED Switch (Leaking Tank). . . . . ON

TANK PUMP Switch (Leaking Tank) . . . . OFF

ABNORMAL QUANTITY DECREASE FROM AUX OR NONLEAKING MAIN TANK

NO

Throttle (Leaking Engine) . . . . IDLE

FUEL Switch (Leaking Engine). . . . . OFF

ENG FIRE Handle (Leaking Engine). . . . . DOWN

FUEL SYSTEM SELECT Switch. . . . . .AS REQUIRED

(CONTINUED)

FUEL LEAK (Continued)

ABNORMAL QUANTITY DECREASE FROM TANK 1, 2, OR 3

NO

(CONTINUED)

ABNORMAL QUANTITY DECREASE FROM AUX OR NONLEAKING MAIN TANK

NO

(CONTINUED)

Consider effects on any system operating in manual mode.

Refer to Abnormal Non-Alert procedure - ENGINE SHUTDOWN IN FLIGHT, as required.

[END]

TANK PUMP Switch (Leaking Tank) . . . . ON

TANK XFEED Switch (Leaking Tank) . . . . OFF

TANK TRANS Switch
(Aux or Nonleaking Main Tank) . . . . OFF

Use fuel from leaking tank until nearly empty, then supply engine from another source.

NOTE: Do not transfer fuel into the leaking tank. Consider transferring fuel into another tank.

[END]

AUX TANK L and R TRANS Switches . . . . ON

(CONTINUED)

FUEL LEAK (Continued)

TANK FILL Valve Switch (All) . . . . ARM,FILL

ABNORMAL QUANTITY DECREASE FROM AUX TANK

NO

TANK FILL Valve Switches (All) . . . . OFF

Refer to TNK__OVERFILL procedure, in the Abnormal Alert - Fuel section of this manual.

[END]

Transfer fuel out of tail tank as required.

[END]

GEAR HANDLE WILL NOT MOVE TO DOWN POSITION

Indicated Airspeed . . . . MAX 230 KIAS

Alternate Gear Extension Lever . . . . RAISE,LATCH

GEAR Lights . . . . 3 GREEN

CAUTION: After alternate extension of main gear, wait until three green lights illuminate before next action.

Center Gear Alternate Extension Handle .PULL

GEAR Lights . . . . 4 GREEN

CAUTION: Main gear alternate extension lever must be left in the raised position to prevent inadvertent gear retraction.

NOTE: Nosewheel steering will be 25° to the right and unrestricted to the left.

[END]

INTERIM OPERATING PROCEDURE 1-103A

SUBJECT: FUEL QUANTITY ABNORMAL INDICATION (ALL MAIN TANKS)

This Interim Operating Procedure (IOP) is applicable for the operator codes listed in the EFFECTIVITY field. Filing instructions are provided below. IOPs are numbered by Volume Number-IOP Number-Letter Change. IOP numbers may not be consecutive because all IOPs are not applicable to all operators. Letter changes are applied to IOP updates. IOPs are intended to precede the relevant subject matter and should be moved if subsequent revisions change the pagination of the Flight Crew Operations Manual

EFFECTIVITY: All

FILE IN: MD-11 Flight Crew Operations Manual, Volume I, Quick Reference Handbook, Abnormal Procedures, Non-Alerts, so that this IOP follows the last page of the FUEL LEAK procedure.

REASON FOR IOP:

To provide crews with a procedure to follow should they encounter a fuel system anomaly that causes indications of a Fuel Quantity Abnormal Decrease in All Main Tanks.

IOP 1-103A adda a metric equivalent fuel quantity weight previously lomitted in the new procedure.

INSTRUCTIONS:

Remove IOP 1-103 and replace with IOP 1-103A. Retain IOP 1-103A until notified to remove it.

FUEL QUANTITY ABNORMAL INDICATION (ALL MAIN TANKS)

ABNORMAL DECREASE IN ALL MAIN TANK QUANTITY INDICATIONS

NO

LOW PRESSURE (INTERMITTENT OR STEADY) FROM FORWARD PUMP IN ANY MAIN TANK

NO

NOTE: At constant speed and normal cruise attitudes, Tank 2 FWD Pump low pressure should occur at approximately 1,500 pounds (700 kilograms) and Tank 1 and 3 FWD Pumps at approximately 700 pounds (300 kilograms).

Land at nearest suitable airport.

FUEL SYSTEM SELECT Switch . . . . .MANUAL

Tank 2 FILL Valve Switch . . . . VERIFY ARM/FILL

When "TAIL PUMPS LO" alert is displayed,

TAIL TANK TRANS Switch . . . . OFF

When "AUX UPR PUMPS LO" alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

If "TNK 1 or 3 FUEL QTY LO" alert is displayed and Tank 2 fuel quantity is >2000 pounds (900 kilograms),

TANK XFEED Switches (All). . . . . ON

[END]

FUEL QTY NORMAL POWER and
FUEL QUANTITY ALTN POWER
Circuit Breakers (Upper Main C/B Panel) . . . . CYCLE

NOTE: Circuit breakers should be opened simultaneously and then closed again after 30 seconds.

(CONTINUED)

FUEL QUANTITY ABNORMAL INDICATION (ALL MAIN TANKS) (Continued)

ABNORMAL DECREASE IN ALL MAIN TANK QUANTITY INDICATIONS

NO

(CONTINUED)

While quantity system is unpowered, FUEL QTY ALERT and SEL FUEL SYS MAN alerts may be displayed.

NORMAL QUANTITY OPERATION RESTORED

NO

FUEL SYSTEM SELECT Switch . . . . . . AS REQUIRED

Check departure fuel minus total fuel used is approximately equal to present fuel on board.

[END]

FUEL SYSTEM SELECT Switch . . . . VERIFY MANUAL

NOTE: Fuel quantity system is inoperative. Subtract total fuel used from departure fuel. Enter the result as FMS UFOB on WEIGHT INIT page. FMS UFOB and GW are now calculated using fuel flow only.

Tank 2 FILL Valve Switch . . . . VERIFY ARM/FILL

When "TAIL PUMPS LO" alert is displayed,

TAIL TANK TRANS Switch . . . . OFF

When "AUX UPR PUMPS LO" alert is displayed,

AUX TANK L and R TRANS Switches . . . . OFF

If "TNK 1 or 3 FUEL QTY LO" alert is displayed,

TANK XFEED Switches (All). . . . . .ON

Land at nearest suitable airport.

[END]

Refer to Abnormal Non-Alert Procedure - FUEL LEAK.

[END]

Intentionally
Blank

GEAR HANDLE WILL NOT MOVE TO UP POSITION

Indicated Airspeed . . . . MAX 230 KIAS

GEAR HANDLE REL Button. . . . . PUSH,HOLD

Gear Handle . . . . UP

GEAR HANDLE REMAINS DOWN

NO

Landing gear trim system has failed. Failure should have no adverse effect on landing.

GEAR Handle,Lights . . . . DOWN,4 GREEN

Indicated Airspeed . . . . MAX 300 KIAS/.70 MACH

[END]

GEAR Lights . . . . EXTINGUISHED

If ground sensor is not in flight mode, as indicated by the following systems not being in normal mode, refer to Abnormal Non-Alert procedure - GROUND SENSOR FAILURE.

- Autopilot cannot be engaged.
- FCP functions cannot be engaged.
- Takeoff warning will sound when flaps or slats are retracted.
- Auto cabin pressurization is inoperative.

[END]

GEAR PRIMARY OR SECONDARY LT(S) ILLUMINATE WITH GEAR HANDLE UP

CONFIG Cue Switch . . . . PUSH

NOTES: The gear primary lights are adjacent to the gear handle; the gear secondary lights are on the SD CONFIG synoptic page.

If either the gear primary or secondary lights are extinguished, the gear is up and locked.

FOR EACH GEAR, THE PRIMARY OR SECONDARY DISPLAY INDICATES GEAR UP

NO

Consider the gear up and locked.

[END]

Airspeed . . . . MAX 230 KIAS

Gear Handle . . . . DOWN

FOR EACH GEAR, A PRIMARY OR SECONDARY GREEN LIGHT IS ILLUMINATED

NO

Gear Handle . . . . .UP

FOR EACH GEAR THE PRIMARY OR SECONDARY DISPLAY INDICATES GEAR UP

NO

Consider the gear up and locked.

[END]

(CONTINUED)

GEAR PRIMARY OR SECONDARY LT(S) ILLUMINATE WITH GEAR HANDLE UP (Continued)

FOR EACH GEAR, A PRIMARY OR SECONDARY GREEN LIGHT IS ILLUMINATED

NO

(CONTINUED)

FLIGHT CONDITIONS AND TIME PERMIT A LANDING GEAR VISUAL UPLOCK CHECK

NO

Landing Gear Visual Uplock Markings . . . . . . .CHECK

VISUAL UPLOCK MARKINGS INDICATE GEAR IS UP AND LOCKED

NO

Consider the gear up and locked.

[END]

Do not exceed gear extension speed (260 KIAS/.70 Mach). Diversion to a suitable airport may be required.

[END]

Do not exceed gear extension speed (260 KIAS/.70 Mach). Diversion to a suitable airport may be required.

When ready for landing,

Refer to Abnormal Non-Alert procedure – GEAR UNSAFE INDICATION WITH GEAR HANDLE DOWN.

[END]

GEAR UNSAFE INDICATION WITH GEAR HANDLE DOWN

Gear Handle . . . . VERIFY DOWN AND FORWARD

CONFIG Cue Switch. . . . . PUSH

NOTE: The gear primary lights are adjacent to the gear handle. The gear secondary lights are on the SD CONFIG page.

Green indication of either the gear primary or secondary lights indicates the gear is down and locked.

If any gear primary light indicates unsafe, the landing gear aural warning will sound until touchdown.

FOR EACH GEAR, A PRIMARY OR SECONDARY GREEN LIGHT IS ILLUMINATED

NO

Consider the gear down and locked.

[END]

Airspeed . . . . MAX 230 KIAS

Alternate Gear Extension Lever . . . . RAISE,LATCH

(CONTINUED)

GEAR UNSAFE INDICATION WITH GEAR HANDLE DOWN (Continued)

LEFT, RIGHT AND NOSE GEAR PRIMARY OR SECONDARY LIGHTS ARE GREEN

NO

CENTER GEAR PRIMARY OR SECONDARY LIGHT IS GREEN

NO

Nosewheel steering is restricted 25° to the right and is unrestricted to the left.

[END]

CAUTION: After alternate extension of the main gear, wait until gear is extended before next action.

Center Gear Alternate Extension Handle. . . . . PULL

CENTER GEAR PRIMARY OR SECONDARY LIGHT IS GREEN

NO

Nosewheel steering is restricted 25° to the right and is unrestricted to the left.

[END]

Consider maximum landing weight for center gear retracted.

[END]

Alternate Gear Extension Lever . . . . .UNLATCH,STOW

Gear Handle . . . . UP

When gear is retracted,

CTR GEAR Isolation Switch . . . . UP

Gear Handle. . . . . DOWN

(CONTINUED)

GEAR UNSAFE INDICATION WITH GEAR HANDLE DOWN (Continued)

LEFT, RIGHT AND NOSE GEAR PRIMARY OR SECONDARY LIGHTS ARE GREEN

NO

Consider maximum landing weight for center gear retracted.

[END]

Alternate Gear Extension Lever . . . . RAISE,LATCH

LEFT, RIGHT AND NOSE GEAR PRIMARY OR SECONDARY LIGHTS ARE GREEN

NO

Consider maximum landing weight for center gear retracted.

Nosewheel steering is restricted 25° to the right and is unrestricted to the left.

[END]

FLIGHT CONDITIONS AND TIME PERMITS A LANDING GEAR VISUAL DOWNLOCK CHECK

NO

Landing Gear Visual Downlock Indicators . . . . .CHECK

(CONTINUED)

GEAR UNSAFE INDICATION WITH GEAR HANDLE DOWN (Continued)

FLIGHT CONDITIONS AND TIME PERMITS A LANDING GEAR VISUAL DOWNLOCK CHECK

NO

(CONTINUED)

VISUAL DOWNLOCK(S) INDICATE GEAR IS DOWN AND LOCKED

NO

Consider maximum landing weight for center gear retracted.

Nosewheel steering is restricted 25° to the right and is unrestricted to the left.

[END]

Do not exceed gear extension speed (260 KIAS/.70 Mach)

If nose gear is down and locked, nosewheel steering is restricted 25° to the right and is unrestricted to the left.

Refer to Abnormal Non-Alert procedure – LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION.

[END]

GROUND SENSOR FAILURE

NOTE: When ground shift mechanism does not shift to flight mode, some systems will not be in normal mode.

- *Autopilot cannot be engaged.*
- *FCP functions cannot be engaged.*
- *Takeoff warning will sound when flaps or slats are retracted.*
- *Auto cabin pressurization is inoperative.*

For a list of other affected systems, refer to expanded procedures in Vol. II-2 Systems Description.

GROUND SENSING Circuit Breakers (Left And Right) . . . . .PULL

(Located at C3 and E3 respectively on left avionics C/B panel)

During approach and just prior to final approach fix,

WING and TAIL ANTI-ICE Switches. . . . . VERIFY OFF

CABIN PRESS SYSTEM SELECT Switch . . . . .MANUAL

Outflow VALVE Indicator . . . . SET 10:30 POSITION

NOTE: Normal engine 2 reversing and automatic cabin depressurization are not available until ground sensing circuit breakers are reset.

Upon touchdown,

Manually assist spoiler handle as it deploys.

After landing,

GROUND SENSING Circuit Breakers (Left and Right) . . . . RESET

CABIN PRESS SYSTEM SELECT Switch . . . . AUTO

Outflow VALVE Indicator . . . . .VERIFY OPEN

[END]

LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION

CAUTION: Land with center gear retracted and as many other gear extended as possible.

NOTES: This procedure is based on the assumption that all methods to extend the landing gear have been unsuccessful.

If a main gear is partially or fully retracted, consideration should be given to selecting the widest runway available for landing due to the possibility of direction control difficulties.

Fly a power on approach on computed approach speed, carrying power to touchdown, if required, so as to contact the runway as gently as possible.

Complete normal approach and landing checklist and prepare evacuation at the appropriate time.

All slide rafts will be usable with aircraft stopped in any failed landing gear configuration.

Freighter aircraft:

Courier(s) . . . . ALERT

Crew . . . . .BRIEF

Fuel Quantity . . . . REDUCE

APU START/STOP Switch . . . . VERIFY OFF

Loose Equipment . . . . .STOW

Cabin Ready Report . . . . COMPLETE

CAWS TO/LDG/CABIN ALTITUDE C/B
(Right Avionics C/B Panel E33) . . . . PULL

GND PROXIMITY WARN C/B
(Right Avionics C/B Panel B31) . . . . PULL

(CONTINUED)

LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION (Continued)

AUTO BRAKE Selector . . . . OFF

Plan a 50/EXT landing, normal approach and flare.

NOSE GEAR PARTIALLY OR FULLY RETRACTED

NO

Spoiler Handle . . . . ARM

Make a normal approach and touchdown.

At touchdown,

Stabilizer Trim . . . . TRIM TO NOSE UP

Gently lower nose before losing elevator effectiveness.

After nose contact, maintain directional control with rudder and differential braking.

Do not use reverse thrust.

FUEL Switches 1 and 3 . . . . OFF

[END]

NOSE GEAR EXTENDED AND BOTH MAIN GEAR PARTIALLY OR FULLY RETRACTED

NO

Do not arm spoilers.

Make normal approach and flare.

After touchdown,

Maintain wings level attitude and maintain directional control as long as possible with rudder, nosewheel steering and thrust (if available).

Do not use reverse thrust.

FUEL Switches 1and 3 . . . . OFF

[END]

(CONTINUED)

LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION (Continued)

ONE MAIN GEAR PARTIALLY OR FULLY RETRACTED

NO

Do not arm spoilers

Make a normal approach and flare.

After touchdown,

Maintain pitch attitude of 6° to 8° while holding wings level until nearly all lateral control is used.

> *NOTE: This will be the minimum practical speed to lower wing to the ground (while maintaining pitch attitude).*

Promptly lower nosewheels to the runway.

After wing is on ground (pod contacts runway), extend ground spoilers to aid in deceleration.

Do not use reverse thrust on engine 2, or on the side with the abnormal gear.

Associated FUEL Switch . . . . OFF

Maintain directional control by using opposite engine reverse thrust, brakes, and nosewheel steering.

[END]

(CONTINUED)

LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION (Continued)

ONE MAIN GEAR AND NOSE GEAR PARTIALLY OR FULLY RETRACTED

NO

Do not arm spoilers.

Make a normal approach and flare.

At touchdown,

Stabilizer Trim . . . . TRIM TO NOSE UP

After touchdown,

Maintain pitch attitude of 6° to 8° while holding the wings level until nearly all lateral control is used.

NOTE: This will be the minimum practical speed to lower the wing to the ground (while maintaining pitch attitude).

Gently lower wing and nose to runway before control is lost.

Do not use reverse thrust.

As the engine pod contacts runway,

Associated FUEL Switch . . . . OFF

Maintain directional control using brakes and rudder.

[END]

(CONTINUED)

LANDING WITH ABNORMAL LANDING GEAR CONFIGURATION (Continued)

ALL GEAR RETRACTED OR PARTIALLY EXTENDED

NO

Do not arm spoilers.

NOTE: Aircraft will touchdown on aft fuselage and engine pods.

Anticipate a lower than normal cockpit elevation with respect to runway at touchdown because of lack of landing gear.

Do not use reverse thrust.

As soon as aircraft contacts runway,

FUEL Switches. . . . . OFF

[END]

[END]

NO FLAP/NO SLAT LANDING

Reduce gross weight to lowest practicable.

UP/RET ESTIMATED LANDING DISTANCES (FEET)
NO FLAP/NO SLAT LANDING

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	7382	7841	8301	8760	9252	9678	10171	10630
STD=15°C	**WET**	9022	9547	10072	10597	11122	11614	12139	12664
2000 FT	**DRY**	7874	8399	8891	9416	9941	10400	10958	11450
STD=11°C	**WET**	9613	10203	10761	11352	11909	12467	13025	13583
4000 FT	**DRY**	8432	9022	9547	10138	10696	11220	11844	12402
STD=7°C	**WET**	10302	10925	11549	12172	12795	13386	14009	14633
6000 FT	**DRY**	9088	9711	10302	10958	11581	12172	12828	13451
STD=3°C	**WET**	11056	11778	12434	13123	13812	14436	15125	15781
8000 FT	**DRY**	9810	10499	11155	11877	12598	13255	13976	14698
STD=–1°C	**WET**	11909	12697	13419	14140	14928	15584	16371	17093
10000 FT	**DRY**	10630	11417	12139	12927	13747	14469	15354	16371
STD=–5°C	**WET**	12894	13714	14501	15354	16175	16929	17815	18865

NOTE: Standard day, no wind, zero slope, two engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop. Tail engine at idle reverse (includes air run distances).

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C

FEET PER °C	Dry	Wet
BELOW standard day	–26	–30
ABOVE standard day	+85	+92

Slope: Valid from –2% downhill +2% uphill

FEET PER 1% SLOPE	Dry	Wet
UPHILL	–138	–230
DOWNHILL	+751	+1037

Wind: Valid from –10 knot tailwind +20 knot headwind

FEET PER KNOT	Dry	Wet
HEADWIND	–56	–72
TAILWIND	+230	+253

(CONTINUED)

NO FLAP/NO SLAT LANDING (Continued)

Plan a wide pattern for speed stabilization on final.

GPWS Switch . . . . .FLAP OVRD

AUTO BRAKE Selector. . . . . OFF

Perform appropriate normal checklist.

After stabilization on approach, reduce speed in level flight until approach speed is achieved.

UP/RET APPROACH SPEEDS
NO FLAP/NO SLAT LANDING

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
VAPP (V_{REF} + 5)	192	198	203	209	215	220	225	230

NOTE: In the NO FLAP/NO SLAT configuration, the go-around switch provides go-around thrust limits only without autopilot, autothrottles, or flight director guidance. Autoflight and autothrottles must be controlled by the FCP during the approach. Autothrottles will maintain speed, but must be disconnected and retarded manually below 50 feet AGL.

Fly approach using ATS, if available, at V_{REF} + 5.

Fly a normal glideslope.

Disconnect autothrottles prior to 50 feet or approaching the threshold.

Retard throttles to idle and fly no slower than V_{REF} to touchdown with a slight flare.

Do not attempt to achieve a smooth touchdown.

On touchdown, positively lower nose gear to runway and immediately apply full reverse thrust and brakes as required.

NOTE: Spoilers will not deploy until nose gear touchdown.

Full reverse thrust may be used to a complete stop.

[END]

NO FLAP/SLAT EXTENDED LANDING

Reduce gross weight as desired.

0/EXT APPROACH SPEEDS NO FLAP/SLAT EXTENDED LANDING

WEIGHT (1000 KG)	160	170	180	190	200	210	220	230
VAPP (V_{REF} + 15)	167	172	176	181	185	189	193	197

0/EXT ESTIMATED LANDING DISTANCES (FEET) NO FLAP/SLAT EXTENDED LANDING

WEIGHT (1000 KG)		160	170	180	190	200	210	220	230
S.L.	**DRY**	5577	5873	6168	6495	6824	7119	7480	7841
STD=15°C	**WET**	6955	7316	7677	8071	8465	8825	9219	9646
2000 FT	**DRY**	5906	6266	6594	6923	7316	7644	8005	8366
STD=11°C	**WET**	7415	7808	8202	8596	9055	9449	9875	10302
4000 FT	**DRY**	6299	6660	7021	7415	7808	8169	8596	8990
STD=7°C	**WET**	7874	8333	8760	9186	9678	10105	10564	11056
6000 FT	**DRY**	6726	7152	7546	7940	8399	8793	9252	9711
STD=3°C	**WET**	8432	8891	9383	9875	10400	10860	11352	11877
8000 FT	**DRY**	7218	7644	8104	8563	9055	9482	10007	10499
STD=–1°C	**WET**	9022	9547	10072	10597	11155	11680	12238	12828
10000 FT	**DRY**	7743	8235	8727	9219	9810	10367	11024	11745
STD=–5°C	**WET**	9711	10269	10827	11417	12041	12697	13419	14206

NOTE: Standard day, no wind, zero slope, two engines at maximum reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then forward idle to stop. Tail engine at idle reverse (includes air run distances)

CORRECTIONS:

Temperature: Valid from STD –20°C to STD +40°C		
FEET PER °C	**Dry**	**Wet**
BELOW standard day	–16	–23
ABOVE standard day	+62	+66

Slope: Valid from –2% downhill +2% uphill		
FEET PER 1% SLOPE	**Dry**	**Wet**
UPHILL	–98	–184
DOWNHILL	+535	+781

(CONTINUED)

NO FLAP/SLAT EXTENDED LANDING (Continued)

Wind: Valid from –10 knot tailwind +20 knot headwind		
FEET PER KNOT	**Dry**	**Wet**
HEADWIND	–46	–62
TAILWIND	+197	+213

NOTE: Minimum speeds are V_{REF} + 15 to preclude tail strike.

GPWS Switch . . . . .FLAP OVRD

AUTO BRAKE Selector. . . . . OFF

Perform appropriate normal checklists.

Cross threshold at V_{APP}, reduce sink rate slightly. Disconnect autothrottles, retard throttles to idle and fly to a positive touchdown. Do not attempt a smooth touchdown. Excessive flare will result in float and excessive use of runway.

CAUTION: Tail strike may occur at pitch attitudes greater than 10°.

On touchdown, positively lower nose gear to runway and immediately apply full reverse thrust and brakes as required.

NOTE: Auto ground spoilers will not deploy until nose gear touchdown.

Full reverse thrust may be used to a complete stop.

[END]

SEVERE DAMAGE TO COCKPIT OVERHEAD STRUCTURE

This procedure is to be used any time the cockpit overhead structure is struck by a foreign object.

NOTE: A severe blow to the fuselage structure in the area of the cockpit overhead could result in a loss of all primary electrical power, and all hydraulic power, and all hydraulic system status indications on the EAD and/or SD HYD synoptic page and/or HYD control panel (forward overhead panel).

HYD CUE Switch . . . . PUSH

SYNOPTIC LINES OR SYSTEM PRESSURE DISPLAYED AND INDICATE NORMAL OPERATION OF HYDRAULIC PUMPS

NO

No further action required.

[END]

ADG . . . . .DEPLOY

ENG HYD PUMPS Control Left and Right
Circuit Breakers (A2 and A3, Left Overhead C/B Panel) . . . . PULL

NOTE: Removing electrical power from ENG HYD PUMPS control will cause the associated engine driven hydraulic pumps to come on.

ADG ELEC Switch . . . . ON

NOTE: If electric power is not available, engine 2 may flame out.

CAUTION: Do not dump fuel. If necessary, perform an overweight landing.

[END]

SEVERE TURBULENCE AND/OR HEAVY RAIN INGESTION

NOTE: In areas of turbulence, fly the FMS optimum altitude when possible. Buffet margin and economy will be enhanced. Descend if necessary to improve buffet margin.

Turbulence,
Penetration Speed . . . . 290 TO 305 KIAS OR .80 TO .82 MACH WHICHEVER RANGE IS LOWER

NOTE: Below 10,000 feet, the greater of 250 KIAS or CLIMB SPEED may be used.

ENG IGN OVRD Switch . . . . OVRD ON

Auto Throttles . . . . OFF

NOTE: Adjust throttles only if necessary to correct excessive airspeed variation or to avoid exceeding red line limits. Do not chase airspeed.

Autopilot . . . . MONITOR

NOTE: Because the autopilot cannot respond correctly when inputs are made to the control wheel or column, it is designed to disconnect automatically if there are sustained pilot inputs. However, the pilot should never make control inputs when the autopilot is engaged, because at disconnect there will be a sudden and abrupt movement of some flight control surfaces with an associated but unpredictable aircraft response.

(CONTINUED)

SEVERE TURBULENCE AND/OR HEAVY RAIN INGESTION (Continued)

WARNING: ***Applying force to the control wheel or column while the autopilot is still engaged has resulted in autopilot disconnects and subsequent abrupt aircraft maneuvers. Pilots have over controlled the aircraft while trying to return to stabilized level flight. The pilot should never apply force to the control wheel or column while the autopilot is engaged. If the pilot is not satisfied with the autopilot performance, or is unsure that it is operating correctly, It should be immediately disconnected by using one of the autopilot disconnect switches. If the autopilot disengages while a force is applied to the control wheel or column, there will be a rapid, commanded change in some of the control surface positions. This will result in an abrupt and unpredictable aircraft response. Additionally, the pilot should not attempt to disconnect the autopilot while applying a control force. If an inadvertent autopilot disconnect occurs, the pilot must smoothly stabilize the aircraft attitude, releasing the flight controls, if necessary, until the aircraft motion dampens out.***

ENG, WING and TAIL ANTI-ICE Switches . . . . ON

ECON Switch . . . . OFF

When severe turbulence and/or heavy rain no longer exist,

ECON Switch . . . . ON

ENG, WING and TAIL ANTI-ICE Switches . . . . OFF

Auto Throttles . . . . ON

ENG IGN OVRD Switch . . . . OVRD OFF

[END]

SMOKE/FUMES REMOVAL

Oxygen Masks ON,100%

Don smoke goggles as required.

Use emergency oxygen pressure as required to purge mask and goggles of smoke and/or fumes.

Crew/Courier(s) Communication . . . . .ESTABLISH

ECON Switch . . . . OFF

Cockpit Air Outlets . . . . .OPEN

Descend to 10,000 feet (terrain permitting).

CABIN PRESS Control Panel. . . . . .SYSTEM MANUAL,CLIMB

Establish cabin rate at 1,000 to 2,000 fpm up.

When depressurized,

Set outflow VALVE indicator to 10:30 position.

COCKPIT SMOKE SEVERE

NO

Configure aircraft as required to maintain 205 KIAS or minimum maneuvering speed whichever is lower.

Headsets . . . . ON

PM Clearview window . . . . OPEN 7.5CM

Land at nearest suitable airport.

[END]

STABILIZER INOPERATIVE

AIRCRAFT IN TRIM WITH LANDING FLAPS SET

NO

Land with existing flap/slat setting.

[END]

ELEV FEEL Selector . . . . PULL, ROTATE AS NECESSARY

Slew elevator feel reference speed on SD CONFIG synoptic page to desired speed. Update as required.

SLATS AVAILABLE

NO

Plan a 35/EXT approach and landing.

Determine stabilizer inoperative speed additive from following table and add to V_{APP}, not to exceed V_{REF} + 25.

STABILIZER INOPERATIVE SPEED ADDITIVE 35/EXT

STABILIZER SETTING (DEGREES)	CENTER OF GRAVITY – % MAC / SPEED ADDITIVE (KIAS) 12	14	16	18	20	22	23-34
1 AND	20	20	17	12	8	5	0
0	20	16	12	7	5	0	0
1 ANU	16	11	7	5	0	0	0
2 ANU	11	7	5	0	0	0	0
3 ANU	7	5	0	0	0	0	0
4 ANU	5	0	0	0	0	0	0
5 - 16 ANU	0	0	0	0	0	0	0

NOTES: If stabilizer and/or CG is unknown, add 20 knots to normal V_{APP}, not to exceed V_{REF} + 25.

Landing distance will increase approximately 1% for each 1 knot added to V_{REF}.

(CONTINUED)

STABILIZER INOPERATIVE (Continued)

SLATS AVAILABLE

NO (CONTINUED)

Maintain adjusted V_{APP} until touchdown.

NOTE: If go-around is required, use normal procedure, but do not allow speed to go below adjusted V_{APP}.

[END]

Plan a 28/RET approach and landing.

GPWS Switch . . . . .FLAP OVRD

[END]

START VALVE LIGHT ILLUMINATED AFTER START OR IN FLIGHT

ASSOCIATED ENG START SWITCH EXTENDED

NO

ENG START Switch (Affected Engine) . . . . PUSH IN

START VALVE LIGHT EXTINGUISHED

NO

No further crew action required.

[END]

AIR SYSTEM SELECT Switch . . . . .MANUAL

BLEED AIR Switch (Affected Engine). . . . . OFF

Associated ISOL Switch(es) . . . . OFF

AIRCRAFT ON GROUND

NO

AIR APU Switch/External Air . . . . OFF/AS REQD

Verify all air sources to associated starter are off.

FUEL Switch (Affected Engine) . . . . OFF

Call maintenance.

[END]

Do not repressurize affected pneumatic system unless necessary.

Avoid icing conditions.

[END]

TAILPIPE FIRE

CAUTION: Tailpipe fires are not displayed in the cockpit. The first notification of a tailpipe fire may be from an external source. Discharging the engine fire extinguishing agent will not extinguish the fire. If the fire cannot be extinguished by motoring or if motoring is not possible, the use of ground fire fighting equipment may be required.

FUEL Switch (Affected Engine) . . . . OFF

STARTER DISENGAGED

NO

Allow N_2 RPM to decrease to 30% or less.

Verify IGN A or B is selected.

ENG START Switch (Affected Engine) . . . . PULL

Pull ENG START switch and observe light illuminates.

After fire is extinguished,

ENG START Switch (Affected Engine) . . . . PUSH

Call maintenance.

[END]

VOLCANIC ASH

If a volcanic ash cloud is entered inadvertently during flight, depart the area by the shortest route possible. Consider a descending 180 degree turn.

Do not use windshield wipers.

CAUTION: If airspeed indications are abnormal, refer to Emergency Non-Alert procedure - AIRSPEED: LOST SUSPECT OR ERRATIC for pitch guidance information.

Perform the following actions as rapidly as possible:

Autothrottles . . . . DISENGAGE

Throttles (All) . . . . IDLE

ENG IGN OVRD Switch . . . . .OVRD ON

ECON Switch . . . . OFF

ENG ANTI-ICE Switches . . . . ON

WING and TAIL ANTI-ICE Switches. . . . . ON

ENGINE(S) FLAMED OUT OR STALLED, OR EGT BEYOND LIMITS OR INCREASING RAPIDLY TOWARD LIMITS

NO

FUEL Switch (Affected Engine(s)). . . . . OFF THEN ON

N_2 RPM LESS THAN 15%

NO

ENGINE START Switch (Affected Engine(s)). . . . PULL

Monitor start.

(CONTINUED)

VOLCANIC ASH (Continued)

ENGINE(S) FLAMED OUT OR STALLED, OR EGT BEYOND LIMITS OR INCREASING RAPIDLY TOWARD LIMITS

NO

(CONTINUED)

NOTE: Engines accelerate to idle very slowly at high altitudes.This may be incorrectly interpreted as a hung start or an engine malfunction not caused by volcanic ash.

Land at nearest suitable airport.

[END]

WEATHER RADAR DUAL DISPLAY FAILURE

Weather Radar SYS selector . . . . 1 OR 2

Switch SYS selector to other system (from 2 to 1, or 1 to 2).

NOTE : Ensure SYS selector has been changed before selecting SISP to AUX.

WEATHER RADAR RETURNS
DISPLAYED ON EITHER ND

NO

No further crew action required.

[END]

EIS SOURCE
Selector (On PF's side) . . . . AUX

Rotate EIS SOURCE selector (on PF's side) to AUX.

WEATHER RADAR RETURNS
DISPLAYED ON PF'S ND

NO

No further crew action required.

[END]

EIS SOURCE

Selector (On PF's side) . . . . RETURN TO ORIGINAL POSITION

Rotate EIS SOURCE selector (on PF's side) to original position.

EIS SOURCE
Selector (On PM's side) . . . . AUX

Rotate EIS SOURCE selector (on PM's side) to AUX.

(CONTINUED)

WEATHER RADAR DUAL DISPLAY FAILURE (Continued)

WEATHER RADAR RETURNS DISPLAYED

NO

No further crew action required.
[END]

EIS SOURCE

Selector (On PM's side) . . . . RETURN TO ORIGINAL POSITION

Evaluate enroute weather conditions.
Consider diverting to a suitable airport.

[END]

WINDSHIELD/CLEARVIEW/AFT WINDOW CRACKED OR ARCING

WINDSHIELD CRACKED AND/OR ARCING

NO

WINDSHIELD ANTI-ICE Switch (Affected Window) . . . . . . OFF

ARCING CONTINUES

NO

WINDSHIELD DEFOG Switch (Affected Window). OFF

WINDSHIELD DEFOG C/B(Affected Windshield) (L WINDSHIELD B4, R WINDSHIELD D2 - Left Avionics C/B Panel) . . . . PULL

WINDSHIELD DEFOG Switch . . . . .ON

CLEARVIEW OR AFT WINDOW CRACKED AND/OR ARCING

NO

WINDSHIELD DEFOG Switch . . . . OFF

Associated WINDSHLD DEFOG OR AFT WINDOW DEFOG C/B . . . . PULL
(CLEARVIEW DEFOG L-B3, R-D2; AFT WINDOW DEFOG L-B6, R-D3 - Left Avionics C/B Panel))
After associated CLEARVIEW or AFT WINDOW DEFOG circuit breaker has been pulled,WINDSHLD DEFOG Switch . . . . .ON

Monitor damaged unit.

[END]

Interim Operating Procedure 1-87D

SUBJECT: Fuel Pump

Following is a combined Effectivity List and Filing Instructions. Filing instructions are by procedure. Should it be necessary to file or retain this Interim Operating Procedure in a later revision where procedure location may have changed, insert or move Interim Operating Procedure so that it faces subject matter to which it applies.

EFFECTIVITY: AZ, DAC, FMGE, FMPW, GMN, KE, KL, LH, SR, SV, SVP2, TG, VP, Y4, YS, Z5, Z8, ZG

FILE IN: MD-11 Flight Crew Operating Manual, Volume I, Quick Reference Handbook, so that these pages follow the LVL1 ALERT tab.

REASON FOR IOP:

To provide the flight crew with guidance on how to disable selected fuel pump circuitry following a pump failure. These procedures are not to be used for displayed alerts associated with normal fuel management/transfer.

IOP 1-87D is being issued because the fuel pump electrical connector replacement parts specified in the Service Bulletin MD11-28A113 and referenced in IOP 1-87C are showing failure characteristics similar to the original parts.

INSTRUCTIONS:

IOP 1-87C is superseded by IOP 1-87D. Please remove IOP 1-87C and replace with IOP 1-87D. File and retain this IOP until notified to remove it.

Intentionally
Blank

ALERT INDEX

ALERT	Tab
AUX__ PUMP LO AUX LWR PUMPS LO AUX UPP PUMPS LO AUX__ PMP OFF	1
TAIL__ PUMP LO TAIL__ PUMP OFF TANK__ PUMPS LO TANK__ PUMPS OFF	2
TNK__ AFT PMP LO TNK__ FWD PMP LO TNK__ PMP OFF TNK__ XFER PMP LO	3
TNK__ XFER PMP OFF	4
NO TAKEOFF LEVEL 1 ALERTS	4&5

Intentionally
Blank

AUX___PUMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s). 1

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
AUX LWR L PUMP LO	AUX TANK L LWR	UPPER MAIN Row F
AUX LWR R PUMP LO	AUX TANK R LWR	UPPER MAIN Row G
AUX UPR L PUMP LO	AUX TANK L UPR	UPPER MAIN Row F
AUX UPR R PUMP LO	AUX TANK R UPR	UPPER MAIN Row E

[END]

AUX LWR PUMPS LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
AUX LWR PUMPS LO	AUX TANK L LWR AUX TANK R LWR	UPPER MAIN Row F UPPER MAIN Row G

[END]

AUX UPR PUMPS LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

1 FUEL SYSTEM SELECT Switch . . . . MANUAL

AUX TANK L TRANS & R TRANS Switches . . . . OFF

AUX TANK QUANTITY LESS THAN 1,000 LBS/450 KGS

NO

No further crew action required.

[END]

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Nomenclature	Location
AUX TANK L UPR	UPPER MAIN Row F
AUX TANK R UPR	UPPER MAIN Row E

[END]

AUX___PMP OFF

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
AUX LWR L PMP OFF	AUX TANK L LWR	UPPER MAIN Row F
AUX LWR R PMP OFF	AUX TANK R LWR	UPPER MAIN Row G
AUX UPR L PMP OFF	AUX TANK L UPR	UPPER MAIN Row F
AUX UPR R PMP OFF	AUX TANK R UPR	UPPER MAIN Row E

[END]

TAIL___PUMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight. **2**

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TAIL ALT PUMP LO	TAIL TANK ALTN	OVERHEAD Row G
TAIL L PUMP LO	AUX TANK TAIL L XFR	UPPER MAIN Row E
TAIL R PUMP LO	AUX TANK TAIL R XFR	UPPER MAIN Row F

[END]

TAIL___PUMP OFF

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TAIL ALT PUMP OFF	TAIL TANK ALTN	OVERHEAD Row G
TAIL L PUMP OFF	AUX TANK TAIL L XFR	UPPER MAIN Row E
TAIL R PUMP OFF	AUX TANK TAIL R XFR	UPPER MAIN Row F

[END]

TANK___PUMPS LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

AIRCRAFT ON GROUND

NO

2

Do not takeoff. Open (pull) associated circuit breaker(s) and contact maintenance.

[END]

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TANK 1 PUMPS LO	TANK 1 AFT	UPPER MAIN Row G
	TANK 1 FWD	UPPER MAIN Row E
TANK 2 PUMPS LO	TANK 2L AFT	OVERHEAD Row G
	TANK 2R AFT	UPPER MAIN Row F
	TANK 2 FWD	UPPER MAIN Row E
TANK 3 PUMPS LO	TANK 3 AFT	UPPER MAIN Row F
	TANK 3 FWD	UPPER MAIN Row G

[END]

TANK__PUMPS OFF

CAUTION: Do not reset any tripped fuel pump circuit breakers.

AIRCRAFT ON GROUND

NO

Do not takeoff. Contact maintenance.

[END]

No crew action required.

[END]

TNK___AFT PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s). 3

Alert	Nomenclature	Location
TNK 1 AFT PMP LO	TANK 1 AFT	UPPER MAIN Row G
TNK 2L AFT PMP LO	TANK 2L AFT	OVERHEAD Row G
TNK 2R AFT PMP LO	TANK 2R AFT	UPPER MAIN Row F
TNK 3 AFT PMP LO	TANK 3 AFT	UPPER MAIN Row F

[END]

TNK___FWD PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

AIRCRAFT IN CLIMB OR GO-AROUND AND ASSOCIATED TANK QUANTITY LESS THAN 18,000 LBS/8,150 KGS

NO

No further crew action required.

[END]

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TANK 1 FWD PMP LO	TANK 1 FWD	UPPER MAIN Row E
TANK 2 FWD PMP LO	TANK 2 FWD	UPPER MAIN Row E
TANK 3 FWD PMP LO	TANK 3 FWD	UPPER MAIN Row G

[END]

TNK___PMP OFF

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

3 Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 AFT PMP OFF	TANK 1 AFT	UPPER MAIN Row G
TNK 1 FWD PMP OFF	TANK 1 FWD	UPPER MAIN Row E
TNK 2L AFT PMP OFF	TANK 2L AFT	OVERHEAD Row G
TNK 2R AFT PMP OFF	TANK 2R AFT	UPPER MAIN Row F
TNK 2 FWD PMP OFF	TANK 2 FWD	UPPER MAIN Row E
TNK 3 AFT PMP OFF	TANK 3 AFT	UPPER MAIN Row F
TNK 3 FWD PMP OFF	TANK 3 FWD	UPPER MAIN Row G

[END]

TNK___XFER PMP LO

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 XFER PMP LO	TANK 1 XFR	UPPER MAIN Row F
TNK 2 XFER PMP LO	TANK 2 XFR	UPPER MAIN Row G
TNK 3 XFER PMP LO	TANK 3 XFR	UPPER MAIN Row E

[END]

TNK___XFER PMP OFF

CAUTION: Do not reset any tripped fuel pump circuit breakers.

As time permits, open (pull) the **FUEL PUMP CONTROL** circuit breaker(s) for the applicable alert(s).

Do not reset circuit breaker(s).

Do not use pump for remainder of flight.

If on the ground prior to takeoff, open (pull) associated circuit breaker(s) and contact maintenance.

Use the table below to locate the applicable circuit breaker(s).

Alert	Nomenclature	Location
TNK 1 XFER PMP OFF	TANK 1 XFR	UPPER MAIN Row F
TNK 2 XFER PMP OFF	TANK 2 XFR	UPPER MAIN Row G
TNK 3 XFER PMP OFF	TANK 3 XFR	UPPER MAIN Row E

[END]

4

NO TAKEOFF LEVEL 1 ALERTS

Do not takeoff with any of the following level 1 alerts displayed unless MEL relief for the related system discrepancy is documented in the aircraft's maintenance log.

AC TIE FAULT

AC TIE___OFF

AIR MANF TST FAIL

AIR SYS TEST FAIL

APU FSO NOT CLSD

AUTO SLAT FAIL

AUTO TRIM FAIL

BAT CHARGER INOP

4 BAT CHARGING

BAT DISCHARGING

BAT LOW

BAT SWITCH OFF

BUS___OFF

DC TIE___OFF (1,3)

DEU OP DISAG

DEU___OP DISAG

DEU P/N DISAG

DEU___P/N DISAG

EMER LT BAT___LO (Takeoff Permitted For Freighters Per MEL)

EMER PWR ON

EMER PWR TST FAIL

ENG 2 A-ICE OFF (FUS 575 and Subs)

ENG DUCT TST FAIL (Optional)

ENG___FADEC MAINT

ENG FIRE AGENT LO

ENG___FSO CLOSED

ENG___FSO NOT CLOSED

ENG___FUEL FILTER

ENG IGN NOT ARMED

ENG___OIL BYPASS (Optional) (PW ONLY)

ENG___SUCT FEED

FADEC___B/U (Optional)

FIRE DET___FAIL

FSC CONFIG (Optional)

FSC MODE FAULT

FUEL CONTAMINATED

FUEL MANF DRAIN

FUEL SYS TST FAIL

FUEL TEMP LO

(CONTINUED)

NO TAKEOFF LEVEL 1 ALERTS (Continued)

HYD 3 ELEV OFF

HYD___OFF

HYD___PRES LO

HYD PRES TST FAIL

HYD PUMP TST FAIL

HYD___QTY LO

HYD___RMP DISAG (1-3, 2-3)

HYD SYS 3 ISOL

HYD___TEMP HI

PITOT HEAT AUX

PITOT HEAT___

REV___PRESS FAULT

RUDDER BOTH INOP

RUDDER LWR INOP

RUDDER UPR INOP 5

RUD STBY LWR OFF

RUD STBY UPR OFF

SEL HYD PMP___OFF

SLATS INHIBITED

SLAT STOW

STALL WARN FAIL

TIRE DIFF PRESS

TIRE PRES LO

TNK___FUEL QTY LO

Intentionally
Blank

Abnormal Procedures — Chapter AP

Table Of Contents — Section 90

Intentionally
Blank

LEVEL 1 ALERTS

If any level 1 or 0 alerts are displayed, refer to the following tables to determine the appropriate action to be taken. Refer MEL Chapter 1 for additional details.

If in flight, if not restricted by the consequence message, continue to destination.

Make appropriate entry in aircraft Flight and Maintenance Log.

Level 1 alerts are accompanied by an action code which is one of the following:

NO T/O — Do not take off unles MEL relief for the related system discrepancy is documented in the aircraft's maintenance log.

If in flight, review Consequence message(s). Continue to an appropriate destination considering Consequence message(s) and maintenance/MEL relief requirements for subsequent departures. Make an appropriate Flight and Maintenance Log entry.

MAINT — Consult maintenance prior to takeoff for appropriate disposition. MEL procedures and limitations may apply. In flight, if not restricted by the consequence message, continue to destination and make appropriate log book entry.

N/A — No specific flight crew action is required. These alerts generally appear to inform the crew of an automatic system controller normal action, a result of a maintenance action taken to comply with the MEL, or alerts that appear only in flight as a result of an associated problem.

SW — This alert is the result of flight crew inaction or a deliberate flight crew action and reflects an abnormal switch or control position. The flight crew should confirm the required configuration.

Intentionally
Blank

ALERT	CODE
AC TIE FAULT	NO T/O
AC TIE__OFF (1,2,3)	NO T/O/SW
ADG ELEC SW ON	SW
AFSC FAULT	MAINT
A-ICE SENSOR FAIL	MAINT
AIL DEFLECT INOP	MAINT
AIR COND DOOR	MAINT
AIR DATA HTR ON	MAINT
AIR__ISOL DISAG (1-2, 1-3)	MAINT
AIR LRU INOP	N/A
AIR MANF TST FAIL	NO T/O
AIR SYS 1 OFF	MAINT/SW
AIR SYS 2 OFF	MAINT/SW
AIR SYS 3 OFF	MAINT/SW
AIR SYS MANUAL	SW
AIR SYS TEST FAIL	NO T/O
ANTI-SKID FAULT	MAINT
ANTI-SKID OFF	SW
AOA HEAT__FAIL (L, R)	MAINT
APU AUTO SHUTDOWN	MAINT
APU DOOR DISAG	MAINT
APU FAIL	MAINT
APU FAULT	MAINT
APU FIRE AGENT LO	MAINT
APU FSO NOT CLSD	NO T/O
APU FUEL PRES LO	MAINT/SW
APU MAINT DOOR	MAINT
APU STARTER FAULT	MAINT
ATC XPDR__FAIL (1, 2)	MAINT
AUTO BRAKE OFF	SW
AUTOPILOT SINGLE	MAINT
AUTO SLAT FAIL	NO T/O
AUTO TRIM FAIL	NO T/O
AUX LWR PUMPS LO	MAINT
AUX LWR__PMP LO (L, R)	MAINT
AUX LWR__PMP OFF (L, R)	MAINT

ALERT	CODE
AUX UPR PUMPS LO	MAINT
AUX UPR__PUMP LO (L, R)	MAINT
AUX UPR__PUMP OFF (L, R)	MAINT
AVNCS EXT ACC DR	MAINT
AVNCS FAN OVRD	MAINT/SW
AVNCS NOSE WHL DR	MAINT
BALST SW/FMS XCHK	MAINT/SW
BAT CHARGER INOP	NO T/O
BAT CHARGING	NO T/O
BAT DISCHARGING	NO T/O
BAT LOW	NO T/O
BAT SWITCH OFF	NO T/O SW
BLEED AIR__OFF (1, 2, 3)	MAINT/SW
BLEEDS NOT OFF	SW
BRAKE DIFF TEMP	MAINT
BUS AC 1 OFF	NO T/O
BUS AC 2 OFF	NO T/O
BUS AC 3 OFF	NO T/O
BUS DC 1 OFF	NO T/O
BUS DC 2 OFF	NO T/O
BUS DC 3 OFF	NO T/O
BUS DC CABIN OFF	NO T/O
BUS DC GND OFF	NO T/O
BUS R EMER AC OFF	NO T/O
BUS R EMER DC OFF	NO T/O
CAB AIR NOT OFF (Freighter)	MAINT
CAB DOOR OVWING__ (L, R)	MAINT/SW
CAB PRES SYS MAN	SW
CABIN AIR OFF (Freighter)	SW
CABIN BUS SW OFF	SW
CABIN DOOR__ (AFT L, AFT R, FWD L, FWD R, MID L, MID R)	MAINT/SW
CABIN INFLO LO	N/A
CABIN PRES RELIEF	MAINT
CABIN RATE	N/A
CAC AIR FLO OFF	MAINT

ALERT	CODE
CAC DOOR	MAINT
CAC MANF DECAY CK	N/A
CARGO FIRE AGT LO	MAINT
CARGO FLO AFT OFF	SW
CARGO FLO FWD OFF	SW
CAWS FAULT	MAINT
CG DISAG	SW
COLD FUEL RECIRC	N/A
CPC FAULT	MAINT
CRG DOOR TST FAIL	MAINT
CRG DR__DISAG (FWD, AFT, CRT, UPR)	MAINT
CRG FIRE TST FAIL	MAINT
CRG FLO AFT DISAG	MAINT
CRG FLO FWD DISAG	MAINT
CRG TEMP CTL OFF	SW
DC TIE__OFF (1, 3)	NO T/O/SW
DEU OP DISAG	NO T/O
DEU__OP DISAG (1, 2, AUX)	NO T/O
DEU P/N DISAG	NO T/O
DEU__P/N DISAG (1, 2, AUX)	NO T/O
DFDAU FAULT	MAINT
DFDR OFF	MAINT/N/A
DISARM SPOILERS	SW
DISCH CARGO AGENT	MAINT
DOOR OPEN	N/A
ECON OFF	SW
ELEC SYS MANUAL	SW
ELEV FEEL MANUAL	SW
EMER LT BAT__LO (1, 2, 3, 4, 5, 6)	MAINT (Freighter)
EMER LTS DISARM	SW
EMER PWR ON	NO T/O/SW
EMER PWR SW OFF	SW
EMER PWR TST FAIL	NO T/O
ENG__A-ICE DISAG (1, 2, 3)	MAINT

ALERT	CODE
ENG 2 A-ICE DUCT (Effective for registration numbers B-16101/2/3/6/7 and N105EV, burst anti-ice duct detection not installed)	MAINT
ENG 2 A-ICE OFF (Effective for aircraft registration numbers B-16108 and higher, burst anti-ice duct detection installed)	NO T/O
ENG DUCT TST FAIL (Effective for aircraft registration numbers B-16108 and higher, burst anti-ice duct detection installed)	NO T/O
ENG__FADEC ALTN (1, 2, 3)	SW
ENG__FADEC FAULT (1, 2, 3)	MAINT
ENG__FADEC MAINT (1, 2, 3)	NO T/O
ENG FIRE AGENT LO	NO T/O
ENG__FSO CLOSED (1, 3)	NO T/O
ENG__FSO NOT CLSD (1, 3)	NO T/O
ENG__FUEL FILTER (1, 2, 3)	NO T/O
ENGINE IGN MANUAL	MAINT
ENG IGN NOT ARMED	NO T/O/SW
ENG__NAC TEMP HI (1, 2, 3)	MAINT
ENG__SUCT FEED (1, 3)	NO T/O
ENGINE__VIB HI (1, 2, 3)	MAINT
EPGS FAULT	MAINT
FADEC__B/U PWR (1, 2, 3)	NO T/O
FADEC GND PWR ON	SW
FD G/A ONLY	N/A
FIRE DET__FAIL (1, 2, 3)	NO T/O
FIRE DET APU FAIL	MAINT
FIRE DET APU FAULT	MAINT
FIRE DET__FAULT (1, 2, 3)	MAINT
FLAP LIMIT DISAG	MAINT
FLAP LIMIT OVRD	SW
FMS DUMP DISABLED	N/A
FSC CONFIG	NO T/O
FSC FAULT	MAINT
FSC MODE FAULT	NO T/O
FUEL CONTAMINATED	NO T/O
FUEL DUMP ON	SW

ALERT	CODE
FUEL LRU INOP	N/A
FUEL MANF DRAIN	NO T/O/SW
FUEL QTY 2 DISAG	MAINT
FUEL QTY TST FAIL	MAINT
FUEL SYS MANUAL	SW
FUEL SYS TST FAIL	NO T/O
FUEL TEMP FAIL	MAINT
FUEL TEMP LO	NO T/O
FUEL VALVE FAULT	MAINT
FUEL XFEED__DISAG (1, 2, 3)	MAINT
FUEL AUX__PMP LO (1L, 1R, 2L, 2R)	MAINT
GEN APU OFF	MAINT
GEN DRIVE DISC	MAINT/SW
GEN__OFF (1, 2, 3)	MAINT/SW
GPWS FAIL	MAINT
GPWS FAULT	MAINT
HYD 3 ELEV OFF	NO T/O
HYD LRU INOP	N/A
HYD__OFF (1, 2, 3)	NO T/O
HYD__PRES LO (1, 2, 3)	NO T/O
HYD PRES TST FAIL	NO T/O
HYD PUMP__<2800 (1L, 1R, 2L, 2R, 3L, 3R)	MAINT
HYD PUMP__FAULT (1L, 1R, 2L, 2R, 3L, 3R)	MAINT/SW
HYD PUMP__OFF (1L, 1R, 2L, 2R, 3L, 3R)	MAINT/SW
HYD PUMP TST FAIL	NO T/O
HYD__QTY LO (1, 2, 3)	NO T/O
HYD__RMP DISAG (1-3, 2-3)	NO T/O
HYD SYS MANUAL	SW
HYD SYS 3 ISOL	NO T/O
HYD__ TEMP HI (1, 2, 3)	NO T/O
ICE DETECTED	SW
ICE DETECTOR FAIL	MAINT
ICE DET SINGLE	MAINT

ALERT	CODE
IRU BAT LO	MAINT
IRU__NAV FAIL (1, 2, AUX)	MAINT
IRU__ NO ALIGN (1, 2, AUX)	MAINT/SW
IRU OFF	SW
IRU__ON BAT (1, 2, AUX)	MAINT/N/A
LDG ALTITUDE MAN	SW
LSAS ALL OFF	SW
LSAS__OFF (L INBD, LOUTBD, R INBD, R OUTBD)	SW
LWR CARGO TEMP LO	N/A
MANUAL G/A ONLY	MAINT
NO AUTOLAND	MAINT
NO ICE DETECT	SW
OPEN OUTFLO VALVE	MAINT
PACK_ _FLO DISAG (1, 2, 3)	MAINT
PACKS NOT OFF	SW
PACK__OFF (1, 2, 3)	MAINT SW
PITOT HEAT AUX	NO T/O
PITOT HEAT__(CAPT, FO)	NO T/O
PITOT HEAT OFF	MAINT
RETRACT SPD BRK	SW
REV__FAULT (1, 2, 3)	MAINT
REV__PRESS FAULT (1,2,3)	NO T/O
RUDDER BOTH INOP	NO T/O
RUDDER LWR INOP	NO T/O
RUDDER UPR INOP	NO T/O
RUD STBY LWR OFF	NO T/O
RUD STBY UPR OFF	NO T/O
SEL AIR SYS MAN	MAINT
SEL APU AIR OFF	SW
SEL CAB PRES MAN	MAINT

ALERT	CODE
SEL ELEC SYS MAN	MAINT
SEL ELEV FEEL LO	SW
SEL FUEL SYS MAN	MAINT
SEL FWD AUX OFF	SW
SEL HYD PMP__OFF (1L, 1R, 2L, 2R, 3L, 3R)	NO T/O
SEL HYD SYS MAN	MAINT
SEL LSAS__OFF (LOB,ROB,LIB,RIB)	MAINT
SEL PACK__OFF (1, 2, 3)	MAINT
SEL__TEMP OFF (AFT, FWD)	MAINT
SEL YAW___OFF (UPR A, UPR B, LWR A, LWR B)	MAINT
SET LDG ALTITUDE	MAINT
SINGLE LAND	MAINT
SLAT STOW	NO T/O/SW
SLATS INHIBITED	NO T/O
SMOKE SW IN USE	SW
STALL WARN FAIL	NO T/O
START AIR PRES LO	N/A
TAIL ALT PUMP LO	MAINT
TAIL ALT PUMP OFF	MAINT
TAIL FUEL FWD	MAINT
TAIL__PUMP LO (L, R)	MAINT
TAIL__PUMP OFF (L, R)	MAINT
TANK__PUMPS LO (1, 2, 3)	NO T/O
TANK__PUMPS OFF (1, 2, 3)	NO T/O/SW
TAT PROBE HEAT	MAINT
TIRE DIFF PRESS	NO T/O
TIRE PRES LO	NO T/O
TNK__AFT PMP LO (1, 2L, 2R, 3)	MAINT
TNK__AFT PMP OFF (1, 2L, 2R, 3)	MAINT
TNK__FUEL QTY LO (1, 2, 3)	NO T/O
TNK___FWD PMP LO (1, 2, 3)	MAINT
TNK__FWD PMP OFF (1, 2, 3)	MAINT
TNK__TIP FUEL LO (1, 3)	MAINT

ALERT	CODE
TNK__TIP TRAPPED (1, 3)	MAINT N/A
TNK__XFER PMP LO (1, 2, 3)	MAINT
TNK__XFER PMP OFF (1, 2, 3)	MAINT
TR__FAIL (1, 2A, 2B 3)	MAINT
UNABLE RNP	MAINT/SW
WSHEAR DET FAIL	MAINT
WSHLD DEFOG OFF	SW
WSHLD HEAT__FAIL (L, R)	MAINT
YAW DMP___OFF (LWR A, LWR B, UPR A, UPR B)	SW
YAW DAMP ALL OFF	SW
ZONE TEMP SEL MAN	SW

TAIL FUEL FWD

When “TAIL FUEL FWD” alert is displayed, push FUEL SYSTEM SELECT switch from AUTO to MANUAL then back to AUTO.

“TAIL FUEL FWD” ALERT EXTINGUISHED

NO

No further crew action required.

[END]

Cruise performance may be affected.

NOTE: The fuel burn penalty for Tail Fuel Management system inoperative (tail tank empty) is dependent on the zero fuel weight center of gravity as follows:

ZFWCG (%MAC)	FUEL BURN PENALTY (%)
BELOW 22	1.7
22 TO 28	1.0
ABOVE 28	0

The cruise penalty can be entered as a PERF FACTOR (+) on the FMS A/C STATUS page to correct FMS enroute predictions.

TAIL FUEL FWD (Continued)

LESS THAN 4 HOURS REMAIN IN FLIGHT LEG, OR “FUEL TEMP FAIL” ALERT DISPLAYED, CG DISPLAY INOPERATIVE, OR TAIL TANK EMPTY

NO

Continue flight with fuel system in AUTO.

[END]

FUEL SYSTEM SELECT Switch .... MANUAL

Push FUEL SYSTEM SELECT switch and observe MANUAL light illuminates.

NOTE: When switching the fuel system from AUTO to MANUAL, the fuel system controller (FSC) will turn on tank pumps 1, 2 and 3, L and R aux tank transfer and tail tank transfer pumps. Tank 2 transfer pump and 1, 2 and 3 fill valves will remain in previously selected positions. Wait 4 seconds for the FSC to coordinate pump sequence before selecting any pump.

Select SD FUEL synoptic page. Confirm the transfer path from the tail tank to the aux tanks. Confirm the tail tank transfer pumps are on and green flow lines are illuminated, and the upper aux tank fill spigot is displayed. Confirm upper and lower aux pumps are functioning by observing the green flow and pressure indications on the SD FUEL synoptic page.

TAIL TANK TRANS Switch .... OFF

Push TAIL TANK TRANS switch and observe ON light extinguishes.

Tank 1, 2 and 3 FILL Switches...As Required

Monitor aircraft center of gravity and tail tank fuel temperature.

Maintain center of gravity forward of 32% MAC and tail tank fuel temperature warmer than -35°C.

(Continued)

TAIL FUEL FWD (Continued)

CG ≥ 32% MAC

NO

TAIL TANK TRANS Switch ....ON

Push TAIL TANK TRANS switch and observe ON light illuminates.

When CG decreases to 30% MAC,

TAIL TANK TRANS Switch .... Off

Push TAIL TANK TRANS switch and observe ON light extinguishes.

Tank 1, 2 & 3 FILL Switches ....as Required

Monitor aircraft center of gravity and tail tank fuel temperature.

Maintain center of gravity forward of 32% MAC and tail tank fuel temperature warmer than -35°C.

[END]

TAIL FUEL TEMP -35°C OR COLDER,
OR TAIL FUEL QUANTITY ≤ 2268 KG,
OR UPPER AUX TANK EMPTY

NO

FUEL SYSTEM SELECT Switch ....Auto

Push FUEL SYSTEM SELECT switch and observe MANUAL light extinguishes.

NOTE: "TAIL FUEL FWD" alert may be displayed, and fuel will transfer forward or normal tail fuel management will occur.

[END]

Monitor center of gravity.

[END]

UNABLE RNP

NOTE: The level 1 message UNABLE RNP indicates the Navigation performance does not meet required accuracy.

Enroute Phase

Observe IRS/FMC update status on the Navigation display.

A failure of equipment required for RNP operation enroute requires re-evaluation of compliance with the respective RNP requirement.

	IRS/FMC update status required		
RNP 10	DD	VD	IRS (<12 HRS)*
RNP 5	DD	VD	IRS (< 2 HRS)*
RNP 4	DD	VD	IRS (< 2 HRS)*
RNP 1	DD	--	IRS (<30 MIN)

(*) The IRS time limit starts when the system is placed in NAV mode or following the last radio position update, whichever occur later.

In Radio Coverage Airspace

1. If accuracy check is satisfactory (ANP < than required RNP); NAV mode may be used.
2. If accuracy check is not satisactory (ANP > than required RNP); Raw data reference must be used. Consider switching off the terrain function of the EGPWS.

(CONTINUED)

UNABLE RNP (Continued)

In Remote Airspace

Verify the ANP versus RNP value on both FMCs and select NAV mode on the lower ANP side.

If FMC position is based on IRS navigation only, disregard this message. With RNP 10 the “UNABLE RNP” alert may appear 5 hours after the last radio update; this alert is consequence of possible degraded navigation and requires no flight crew response. Check difference between IRS and FMC position.

1. DD update status can be achieved by a single operating DME.
2. VD update status can be achieved by a single operating DME or VOR on the same side.
3. IRS update can be achieved by a single operating IRS.

Approach Phase (Non RNAV Approach):

1. Maintain raw data reference to the navigation aid on which the procedure is based.
2. Consider switching off the terrain function of the EGPWS.

[END]

Intentionally
Blank

Quick Reference — Chapter QR

Table Of Contents — Section 00

Low Visibility Operations

TAKEOFF

Visibility Requirements

Use Higher of:

- Jeppesen Airway Manual Take-off requirements; or
- Flight Operations Supplementary Manual- Low Visibility Operations

When RVR is:

- < 350 meters (1,200 feet), full thrust takeoff only
- < 700 meters (2,400 feet), CM1 takeoff only

Departure Alternate Requirements

A departure alternate shall be selected if the weather conditions at the departure airport are below the lowest published operational landing minimums (CATII/III excluded) or if it is not possible to return to the departure airport for performance reasons. Refer to FOM Ch.6 for detailed information.

Low Visibility Operations (Continued)

APPROACH AND LANDING

Visibility Requirements

Use Higher of:
-Jeppesen Airway Manual if applicable; or
-Flight Operations Supplementary Manual - Low Visibility Operations.

Limitations

Wind components:

- Headwind 25 knots
- Tailwind 10 knots
- Crosswind 15 knots

No windshear conditions reported or suspected

Turbulence < moderate

Max. 200KIAS in DUAL LAND or SINGLE LAND modes

AFS FMA Requirements

AFS FMA	If AFS FMA changes to	Revert to
DUAL LAND (Fail Operational)		
	APPROACH ONLY	CAT I
SINGLE LAND (Fail Passive)	APPROACH ONLY	CAT I

Approach Preparations

(1) Check aircraft serviceability status.
(2) Check airport equipment status.
(3) Check crew qualification status.
(4) Determine runway minima approved for SALC operations.
(5) Wind conditions/runway length within limitations.
(6) Holding time - calculate available holding time.
(7) Diversion - check weather at alternate. Program route to alternate.
(8) Discuss contingencies in event of aircraft/ground equipment degradation.
(9) Flaps:
- Flap 50 is recommended.

(10) Autobrakes:
- Minimum setting to be used is MED.

(11) BARO/RA setting:
- BAROSET control knob set to CAT I minimum.
- RA control knob set to applicable DH. For Fail Operational, set DH to 0.

NOTE: If VAPP is 140KIAS or less, and DH is selected to 50 feet or less, the aural warning "MINIMUMS" is inhibited. If it is required to set DH to less than50 feet, manually select a VAPP sufficiently high to keep airspeed above140 knots during flare.

Low Visibility Operations (Continued)

Go-Around

Execute go-around when:
- ALIGN does not annunciate below 130 feet RA.
- FLARE does not annunciate below 50 feet RA.
- Any condition in the Approach Requirements is not fulfilled.
- Any condition exceeds the stable approach parameters specified in Flight Operations Manual.

AFTER SHUTDOWN
Recording of Autoland
Flight and Maintenance Log: Record autoland as required.
Update Autoland Expiry Date (present date plus 28 days).

Required Aircraft Equipment

Equipment	Fail Passive	Fail Operational
	Monitored	
AUTOPILOT	ONE	TWO
AUTOTHROTTLE	ONE	ONE
FCC	ONE	TWO
HYD SYSTEM	3 & 1 Or 3 &2 (The requirement of Hyd 3 is not monitored by FMA)	THREE
IDG	TWO	TWO
IRU	TWO	THREE
ILS SYSTEMS	TWO	TWO
LSAS CHANNELS	TWO, provided the remaining channels are on the same FCC	FOUR
Radio Altimeter	TWO	TWO
Yaw Damper	TWO, provdied the remaining channels are on the same FCC	FOUR
	Not Monitored	
ANTI-SKID SYSTEM	ONE	ONE
AUTO PITCH TRIM	ONE	ONE
CADC	ONE, Dual Display	TWO
DEU	TWO	TWO
DU	FOUR	FIVE
ENGINES	TWO	THREE
FMA	ONE (CM2)	ONE(CM2)
MCDU	ONE	ONE
FCP	ONE	ONE

(Continued)

(Continued)

Note:

Monitored: *The FMA status (DUAL LAND/SINGLE LAND/APPR ONLY) will reflect the actual required equipment status.*

Not Monitored: *The FMA status (DUAL LAND/SINGLE LAND/APPR ONLY) will not be affected by the change of required equipment's status or number.*

DE/ANTI-ICING PROCEDURE

Upon completion of the de/anti-icing procedure the ground personnel supervising the process will contact the flight crew and confirm that the process is complete and that the "AIRCRAFT IS CLEAN".
Record details of de/anti-icing procedures in the Flight and Maintenance Log (see FOM Chapter 8 -Adverse Weather).

Standard Cockpit/Ground Communications will be used.

NOTE: Do not shut down APU, if taxying to a remote area to de-ice with engines shut down.

Prior to de/anti-ice spraying,

HYD AUX PUMP #1 (if required) . . . . ON

PARK BRAKE . . . . ON

STAB TRIM . . . . 3º ANU

FLAPS/SLATS . . . . UP/RET

AIR APU Switch . . . . Off

If engine running:

AIR System Controller . . . . MANUAL

BLEED AIR Switches #1, #2, #3 . . . . OFF

1 minute after de/anti-icing is completed, or clear of de-ice area,

AIR System Controller . . . . .Verify Auto

NOTE: If AIR System is being operated in Manual mode due to unserviceable leave AIR System Select Switch in Manual.

AIR APU Switch (if required for engine start) . . . . ON

If required, start all engines using the normal Start Procedure.

In all cases, complete the After Start Procedure and check.

CAUTION: Before moving the flaps/slats to the takeoff position, select SD CONFIG synoptic page and closely monitor flap movement. If the flaps should stop, prior to reaching the selected setting, the flap handle should be placed immediately in the same position as indicated to prevent damage to the flap system. Cause of the flap restriction should be corrected before takeoff.

(CONTINUED)

FLAPS,SLATS . . . . 50/EXT, then set takeoff flaps

NOTE: Following selection of a flap position greater than 28 º; the "AUTOBRAKE FAIL" Level 2 Alert will be displayed for the period that the flap position is greater than 28 º; with the AUTOBRAKE selector set to T.O.

Flight controls . . . . Check

FLAPS,SLATS . . . . As Required

If the taxi route is through slush or standing water in low temperatures, or if precipitation is falling with temperatures below freezing, taxi with flaps up. Taxiing with flaps extended subjects flaps and flap drives to snow and slush accumulations from the main gear wheels. Leading edge devices are also susceptible to slush accumulations. Prudent taxi speeds on contaminated taxi areas provide sufficient protection from contamination of exposed flap/slat surface areas.

[END]

Crosswind Limitations

Runway Condition	Brake Action	Friction Value (u)	Crosswind Limits (kts)	
			T/O	L/D
Runway Condition	Brake Action	Friction Value (u)	Crosswind Limits (kts)	
Dry/Damp			30	30
Wet			24	24
Wet	Good	≥ 0.4	24	24
	Medium to Good	0.39 - 0.36	21	21
Contaminated or Slippery	Medium	0.35 - 0.30	19	19
	Medium to Poor	0.29 - 0.26	17	17
Icy Runway (High risk of hydroplaning)	Poor	0.25 - 0.20	15	15
	Unreliable	< 0.2	--	--

NOTE:

(1) Wind Component includes gust value. The PIC may at his discretion reduce his own crosswind limit.

(2) Reduce landing crosswind limit by 5kt on wet and contaminated runway when using asymmetric reverse thrust..

(3) The wet R/W crosswind limit should be used only if the braking action/friction value is not known. Otherwise refer to the limit for the value given.

(4) Without reported friction value, crosswind limits on runway with:

(i) Standing water/Slush : Takeoff 15 kt; Landing 20 kt.

(ii) Dry Snow: Takeoff 15 kt; Landing 25 kt

(5) When runway width or cleared runway width is less than 45 meters (148 feet), reduce maximum landing crosswind component by 10 knots.

[END]

RUNWAY SURFACE CONDITIONS

No Takeoff and Landing if:

- Slush / Standing water depth > 13mm (> 0.5 inch)
- Dry snow > 102mm (>4 inches) for Takeoff
- Dry snow > 153mm (>6 inches) for Landing
- Braking Coefficient: < 0.20µ

THRUST SETTING:

- FULL THRUST takeoff with contaminated or slippery runway.

Intentionally
Blank

ENGINE START WITH APU INOPERATIVE

If pushback prior to taxi out is not required, start all three engines on stand.

Starting All Three Engines On Stand

Complete Normal Start Procedure.

On SD ELEC synoptic page,

GRs . . . . Verify Close

BTRs. . . . . .Verify Open

EXT PWR Switch . . . . .Off

BTRs. . . . . Verify Close

The ground crew will disconnect the electrical and air supplies on CM1's instruction - "DISCONNECT ELECTRICS", "DISCONNECT AIR".

NOTE: The AIR System Controller will reconfigure the AIR System for normal operation within one minute following the removal of the external air supply.

If pushback prior to taxi out is required, start one engine on stand.

Starting One Engine (normally #3) On Stand

Complete Normal Start Procedure for #3 engine.

On SD ELEC synoptic page,

#3 GR . . . . Verify Close

BTRs. . . . . .Verify Open

EXT PWR Switch . . . . .Off

BTRs. . . . . Verify Close

The ground crew will disconnect the electrical and air supplies on CM1 instruction - "DISCONNECT ELECTRICS", "DISCONNECT AIR".

After pushback, complete Supplemental Procedure - ENG/APU - CROSSBLEED START.

[END]

CROSS BLEED START

NOTE: Cross-bleed starts are not permitted during push-back. The aircraft must be pushed back, the brakes parked and the tug disconnected before initiating a cross-bleed start. If the thrust increase required for a cross-bleed start is above the normal idle range, a clearance is required from both air traffic control and the ground engineer.

Complete BEFORE START procedure.

ECON Switch. . . . . OFF

Push ECON switch and observe "ECON OFF" alert is displayed.

AIR SYSTEM MANUAL

NO

BLEED AIR Switch (Engine Supply Bleed). . . . . .On

BLEED AIR Switches (Engines Being Started). . . . . OFF

Appropriate ISOL VALVE Switch(es). . . . . ON

PACK Switches (ALL). . . . . OFF

ENG START Switch. . . . . Pull

Pull ENG START switch and observe switch light illuminates indicating start valve is open.

"START AIR PRES LO" ALERT DISPLAYED

NO

Advance throttle on supplying engine to maintain a minimum of 25 psi.

(CONTINUED)

CROSS BLEED START (Continued)

Start engine(s) using normal starting procedure.

NOTE: On ground, if throttle on supplying engine was advanced previously, it may be reduced to idle after starter disengagement.

AIR SYSTEM MANUAL

NO

After engine(s) are started,

ISOL VALVE Switches. . . . . Off

BLEED AIR Switches (ALL). . . . . On

PACK Switches (ALL). . . . . On

ECON Switch. . . . . .As Required

Complete AFTER START procedure.

[END]

Intentionally
Blank

RNP

Pre-flight:

MD11 fleet is authorized to operate in the following RNP airspace:

Required RNP	RNP Airspace
RNP5 or BRNAV	ECAC, Middle East, Japan
RNP10	NOPAC, Australian and New Zealand Oceanic FIR, CENPAC, South China Sea and EMARSSH Area (Europe, Middle East, Asia Route Structure South of the Himalayas)

For RNP Departure / Arrival / Approach procedure,

Procedure	RNP Type
P-RNAV Departure / Arrival	RNP1

In-Flight:

It is the crews' responsibility to ensure that the navigation accuracy is maintained. In particular, the following common mistakes must be avoided:

Insertion errors - Coordinates are inserted incorrectly into the system. (Particular care must be taken in case of a new ATC clearance).

De-coupling - If the pilot allows the autopilot to become de-coupled from the equipment which he thinks is providing steering output.

Using faulty equipment - The pilot might continue to use a navigation system which has become inaccurate.

(CONTINUED)

RNP (Continued)

Contingencies Procedure:

Refer to JEPPESEN Vol 1 **AIR TRAFFIC CONTROL** for details if required.

If, as a result of a failure of the RNAV system or degradation of it below RNP 5, an aircraft is unable to either enter the designated airspace or continue operations in accordance with the current air traffic control clearance, a revised clearance shall, whenever possible, be obtained by the pilot.

When a verbal coordination process is being used, the sending air traffic control unit shall include the phrase '**NEGATIVE-RNAV**' at the end of the message. The phrase '**NEGATIVE-RNAV**' shall be also included by the pilot immediately following the aircraft call sign whenever initial contact on an ATC unit frequency is established.

State differences:

Refer to JEPPESEN Vol 1 **AIR TRAFFIC CONTROL** for detail if required

AUSTRALIA, HONG KONG, JAPAN, NEW ZEALAND AND PAC IS, TAIWAN, THAILAND, UNITED STATES and US PAC TERRITORIES.

[END]

RVSM

Before entering RVSM airspace:

The following equipment should be operating normally:
a. Two primary altimetry systems;
b. One automatic altitude-keeping device;
c. One altitude-alerting device.

Within the RVSM airspace:

The pilot must notify ATC whenever the aircraft
a. is no longer RVSM compliant due to equipment failure;
b. experiences loss of redundancy of altimetry systems;
c. encounters turbulence that affects the capability to maintain flight level.

Contingency Procedures:

Refer to JEPPESEN Vol 1 AIR TRAFFIC CONTROL for details if required.
The procedures below extracted from ICAO 4444, have been adopted in the following Flight Information Regions:

- Australian Oceanic FIRs, Anchorage Oceanic FIR, Auckland Oceanic FIR / South Pacific Area, Fukuoka FIR, Kota Kinabalu / Kuala Lumpur FIRs, Oakland Oceanic FIR, Singapore FIR.

(The following FIRs have adopted similar procedures, except that the lateral track separation for in-flight contingencies in oceanic airspace is 25 NM:
Bangkok FIR, Ho Chi Minh FIR, Hanoi FIR, Hong Kong FIR, Manila FIR, Phnom Penh FIR and Taipei FIR).

The radiotelephony distress signal (MAYDAY) or urgency signal (PAN PAN) preferably spoken three times shall be used as appropriate.

For aircraft unable to maintain altitude

a. A revised clearance shall be obtained, whenever possible, prior to initiating any action or,
b. Prior clearance cannot be obtained:
    1. Turn on all aircraft exterior lights and maintain a watch for conflicting traffic both visually and by reference to TCAS
    2. Leave the assigned route or track by initially turning 90 degrees to the right or to the left.

(CONTINUED)

RVSM (Continued)

3. Following the turn, the pilot should; acquire and maintain in either direction a track laterally separated by 15 NM from the assigned route; and once established on the offset track, climb or descend to select a flight level which differs from those normally used by 500 ft (150 m). (enroute diversion across the prevailing traffic flow).

For weather deviation

Rapid response may be obtained by stating "WEATHER DEVIATION REQUIRED".

- If ATC is unable to approve deviation or prior clearance cannot be obtained:
  1. Advise ATC of intentions by the most expeditious means available.
  2. If possible, deviate away from an organized track or route system.
  3. Watch for conflicting traffic both visually and by reference to TCAS. Turn on all aircraft exterior lights.
  4. For deviations less than 10 NM, aircraft should remain at the level assigned by ATC.
  5. For deviations greater than 10 NM, when the aircraft is approximately 10 NM from track, initiate a level change based on the criteria in the table below:

Route center line	Deviations > 10 NM	Level Change track
EAST 000° –179° MAGNETIC	LEFT RIGHT	DESCEND 300 ft (90 m) CLIMB 300 ft (90 m)
WEST 180° – 359° MAGNETIC	LEFT RIGHT	CLIMB 300 ft (90 m) DESCEND 300 ft (90 m)

  6. Establish communication and alert nearby aircraft by broadcasting; at suitable intervals: aircraft identification, flight level, position (including the ATS route designator or the track code, as appropriate) and intentions on the frequency in use and on 121.5 MHz (or, as a back-up, 123.45 MHz).

RVSM (Continued)

7. If contact was not established prior to deviating, continue to attempt to contact ATC to obtain a clearance. If contact was established, continue to keep ATC advised of intentions and obtain essential traffic information.
8. When returning to track be at assigned level, when the aircraft is within approximately 10 NM of centerline.

Wake turbulence avoidance in Oceanic Airspace

Apply SLOP (Strategic Lateral Offset Procedure).

1. Strategic lateral offset shall be established at a distance of 1 NM or 2 NM to the right of centerline relative to the direction of flight.
2. Pilots may contact other aircraft on 123.45 *MHz* to coordinate offsets.
3. Pilots are not required to inform ATC that a strategic lateral offset is being applied.

NOTE: ATC clearance is required for lateral offsets within Canadian and US domestic airspace (DRVSM).

State differences:

Refer to Jeppesen Volume 1 **Air Traffic Control** for detail if required:

CAMBODIA, FIJI ISLANDS, NEW ZEALAND, AND PAC IS and US PAC TERRITORIES.

[END]

RVSM (Continued)

OPERATION IN CHINA RVSM AIRSPACE

When entering or within China airspace.

1. ATC will issue flight level clearance in METER.
2. Both PF and PM check corresponding FEET in China RVSM FLAS (Flight Level Allocation Scheme) and determine closest conventional FL.
3. PF select closest conventional FL in FCP Altitude Window and initiate climb or descend.
4. PF ensure FMA display "HOLD xxxxx" and PFD V/S indicator shows vertical speed 0 (diamond shape) when aircraft level at closest conventional FL.
5. PF select Vertical Speed not more than +500 fpm toward corresponding FEET.
6. PF push Altitude Hold on FCP when approaching corresponding FEET.
7. PF check FMA and FCP Altitude Window display FEET is close to corresponding FEET.
8. PM verifies or edits new Flight Level in FMC INT page.
9. PF engage PROF mode.

Note:

1. *Step 3 and 4 may be omitted if altitude change is less than 500 ft.*
2. *If only one crew is present on flight deck while receive or execute ATC clearance, exercise extra vigilance on this procedure.*

(Continued)

RVSM (Continued)

China ATC will issue the Flight Level clearance in meters. Pilots shall use this table to determine the corresponding flight level in feet.

The aircraft shall be flown using the flight level in feet.

PR OF CHINA (RVSM between FL291 and FL411)

180 ° - 359	360 ° - 179 °	180 ° - 359	360 ° - 179 °
FLIGHT LEVELS In hundreds of feet = in meters	FLIGHT LEVELS In hundreds of feet = in meters	FLIGHT LEVELS In hundreds of feet = in meters	FLIGHT LEVELS In hundreds of feet = in meters
20 600m	**30 0900m**	**236 7200m**	**246 7500m**
39 1200m	**49 1500m**	**256 7800m**	**266 8100m**
59 1800m	**69 2100m**	**276 8400m**	**291 8900m**
79 2400m	**89 2700m**	**301 9200m**	**311 9500m**
98 3000m	**108 3300m**	**321 9800m**	**331 10100m**
118 3600m	**128 3900m**	**341 10400m**	**351 10700m**
138 4200m	**148 4500m**	**361 11000m**	**371 11300m**
157 4800m	**167 5100m**	**381 11600m**	**391 11900m**
177 5400m	**187 5700m**	**401 12200m**	**411 12500m**
197 6000m	**207 6300m**	**430 13100m**	**449 13700m**
217 6600m	**226 6900m**		

"Pilots should expect FL instructions by ATC."

Note: ATC will issue the Flight Level clearance in meters. Pilots shall use the PR of China RVSM FLAS Diagram to determine the corresponding flight level in feet. The aircraft shall be flown using the flight level in FEET. Pilots should be aware that due to the rounding differences, the metric readout of the onboard avionics will not necessarily correspond to the cleared Flight Level in meters however the difference will never be more than 30 meters.

Intentionally
Blank

Quick Reference Miscellaneous

Chapter QR
Section 40

FLIGHT CREW OXYGEN SYSTEM

The following table is the crew oxygen dispatch pressure, reading on the crew oxygen cylinder gauge should meet the requirement.

FLIGHT CREW SYSTEM

(Total PSI = Sys 1 + Sys 2)

Operating Ambient Temperature	Number of Crew				
°C	2	3	4	5	6
-9	664	933	1167	1401	1652
-4	676	951	1189	1427	1684
1	689	969	1211	1454	1715
6	702	987	1233	1480	1746
11	715	1004	1256	1507	1777
16	727	1022	1278	1533	1809
21	740	1040	1300	1560	1840
26	753	1058	1322	1587	1871
31	765	1076	1344	1613	1903
36	778	1093	1367	1640	1934
41	791	1111	1389	1666	1965
46	804	1129	1411	1693	1997

NOTE: For oxygen duration in-flight, refer FCOM Vol. II - Reference Data (RD) Chapter.

HYDRAULIC SYSTEM MANUAL OPERATION

During a flight sequence when the hydraulic system is in MANUAL, the following procedures must be completed. After the appropriate phase of flight, the buff-colored Normal Checklist (when any system is operated in MANUAL) must be used to verify that these procedures have been completed.

NOTE: Minimum acceptable pressure for hydraulic system test is 2400 psi.

Cockpit Preparation - CM1

SD HYD Cue Switch . . . . Push

AUX PUMP 1 . . . . ON

Observe AUX PUMP 1 switch illuminates ON. On SD HYD synoptic, verify AUX PUMP 1 indicates ON and SYS 3 pressure indicates in normal range.

AUX PUMP 2 . . . . ON

Observe AUX PUMP 2 switch illuminates ON. On SD HYD synoptic, verify AUX PUMP 2 indicates ON.

1-3 RMP. . . . . ON

Observe 1-3 RMP switch illuminates ON. Verify HYD SYS 1 PRESS light extinguishes. On SD HYD synoptic, verify 1-3 RMP indicates on, SYS 1 pressure indicates in normal range.

On the EAD, verify “HYD 1 QTY LO” and “HYD 3 QTY LO” alerts are not displayed.

1-3 RMP. . . . . Off

Observe 1-3 RMP switch ON light extinguishes. On SD HYD synoptic, verify 1-3 RMP indicates off, and SYS 1 pressure decreases.

2-3 RMP. . . . . ON

Observe 2-3 RMP switch illuminates ON. Verify HYD SYS 2 PRESS light extinguishes. On SD HYD synoptic, verify 2-3 RMP indicates on, SYS 2 pressure indicates in normal range.

(CONTINUED)

HYDRAULIC SYSTEM MANUAL OPERATION (Continued)

2-3 RMP . . . . .Off

Observe 2-3 RMP switch ON light extinghuishes. On SD HYD synoptic, verify 2-3 RMP indicates off, and SYS 2 pressure decreases.

AUX PUMP 1 . . . . .Off

Observe AUX PUMP 1 switch ON light extinguishes. On SD HYD synoptic, verify system 3 pressure output by AUX PUMP 2 indicates in normal range.

AUX PUMP 2 . . . . .Off

Observe AUX PUMP 2 switch ON light extinguishes.

[END]

After Start - CM2

SD HYD Cue Switch . . . . Push

AUX PUMP 1 and 2 . . . . .Off

RMPs 1-3 and 2-3 . . . . .Off

On SD HYD synoptic, verify AUX PUMPS and RMPs indicate off, and system pressure indicates in the normal range.

L PUMP Systems 1, 2 and 3. . . . . OFF

On HYD panel, observe L PUMP switches illuminate OFF and R PUMP switches lights remain extingulish. On SD HYD synoptic, verify L PUMPs have been commanded off, R PUMPs have been commanded on, and system pressure indicates in the normal range.

L PUMP Systems 1, 2 and 3 . . . . .On

On HYD panel, observe L PUMP switches lights extinguish, and R PUMP switches lights remain extinguish. On SD HYD synoptic, verify L PUMPs have been commanded on, and R PUMPs indicate armed after approximately 20 seconds.

[END]

(CONTINUED

HYDRAULIC SYSTEM MANUAL OPERATION (Continued)

Take-off (prior to takeoff) - CM2

RMPs 1-3 and 2-3 . . . . ON
Observe ON lights are illuminated.

[END]

Take-off (after airborne) - PM

RMPs 1-3 and 2-3 . . . . Off

Observe ON/DISAG lights are extinguished. On SD HYD synoptic, verify RMPs are off.

[END]

Approach - PM

RMPs 1-3 and 2-3 . . . . ON

Observe ON lights illuminate. On SD HYD synoptic, verify RMPs are on.

[END]

Parking - CM2

RMPs 1-3 and 2-3 . . . . Off

Observe ON/DISAG lights are extinguished. On SD HYD synoptic, verify RMPs are off.

[END]

AIR SYSTEM MANUAL OPERATION

During a flight sequence when the air system is in MANUAL, the following procedures must be completed. After the appropriate phase of flight, the buff-colored Normal Checklist (when any system is operated in MANUAL) must be used to verify that these procedures have been completed.

NOTE: This procedure is valid when air is being supplied from the APU. When using external pneumatic air or air-conditioning, refer Supplemental Procedures SP20 AIR - Air Conditioning Using External Pneumatic to Operate Packs/Air Conditioning Using External Airconditioned Air.

Cockpit Preparation - CM1

AIR CONDITIONING FROM APU

NOTE: When using external conditioned air or external pneumatic air, refer Supplemental Procedures SP.20 - Air, Air Conditioning Using External Conditioned Air, or Air Conditioning Using External Pneumatics To Operate Packs, as appropriate.

NOTE: When the air system is in manual, the air system preflight test will be performed by the maintainence according to the MEL..

SD AIR Cue Switch . . . . Push

AIR APU Switch . . . . ON

Observe AIR APU switch illuminates ON.

1-2 and 1-3 ISOL Switches . . . . ON

Observe ISOL DISAG lights illuminate momentarily then ON lights illuminate.

PACK 1, 2 and 3 Switches . . . . .On

Observe FLOW and OFF lights extinguish. On SD AIR synoptic, check system and adjust zone temperatures as required.

[END]

Before Start - CM2

NOTE: For start with an external air source, the AIR APU switch ON light should be extinguished.

If using APU air for engine start:

AIR APU Switch . . . . ON

Verify ON light is illuminated.

Using any air source for engine start:

1-2 and 1-3 ISOL Switches . . . . ON

Verify ON lights are illuminated.

PACKS 1, 2 and 3 Switches . . . . OFF

Observe PACK FLOW lights are extinguished and OFF lights are illuminated.

BLEED AIR 1, 2 and 3 Switches . . . . OFF

Observe OFF lights are illuminated..

[END]

After Start - CM2

1-2 and 1-3 ISOL Switches . . . . Off

Verify ON lights extinguish and DISAG lights are not illuminated.

BLEED AIR 1, 2 and 3 Switches . . . . On

Observe OFF lights are extinguished.

PACKS 1, 2 and 3 Switches . . . . On

Observe PACK FLOW and OFF lights are extinguished.

[END]

Before Take-off - CM2

PACKS 1, 2 and 3 Switches . . . . As Required

Select ON for PACKS ON takeoff.

Select OFF for PACKS OFF takeoff.

BLEED AIR 1, 2 and 3 Switches. . . . . As Required

Select ON for PACKS ON takeoff, or AIR FOIL ANTI-ICE required.

Select OFF for PACKS OFF takeoff (AIR FOIL ANTI-ICE not required).

NOTE: Power should be advanced within 20 seconds of selecting BLEEDS OFF to prevent "AIR SYSTEM OFF" alerts.

[END]

Take-off (after airborne) - PM

BLEED AIR 1, 2 and 3 Switches . . . . .On

Observe OFF lights are extinguished.

PACKS 1, 2 and 3 Switches . . . . .On

Observe OFF lights are extinguished.

[END]

FUEL SYSTEM MANUAL OPERATION

During a flight sequence when the fuel system is in MANUAL, the following procedures must be completed. After the appropriate phase of flight, the buff-colored Normal Checklist with MANUAL System(s) must be used to verify that these procedures have been completed.

Cockpit Preparation - CM1

SD FUEL Cue Switch . . . . Push

1, 2 and 3 Tank PUMP Switches . . . . On

Observe OFF light extinguishes. On SD FUEL synoptic, observe each pump indicates on and no pump indicates low pressure.

1, 2 and 3 Tank PUMP Switches . . . . OFF

Observe OFF light illuminates. On SD FUEL synoptic, observe each pump indicates off.

1 and 3 FILL Switches . . . . Hold On, then Release

Observe ARM light illuminates and remains illuminated while switch is held. On SD FUEL synoptic, observe fill valve spigots are displayed. Release FILL switches.

NOTE: Tank 1 and 3 fill valves will remain armed after switch is released and spigots will remain displayed if tank 2 contains more than 18,144 kilograms.

If FILL switch ARM lights remain illuminated, push tank 1 and 3 FILL switches and observe ARM lights extinguish. On SD FUEL synoptic, observe fill valve spigots are no longer displayed.

1, 2 and 3 XFEED Switches . . . . .ON

Observe ON lights illuminate. DISAG lights will illuminate momentarily as crossfeed valve transitions to open. On SD FUEL synoptic, observe crossfeed valves are open.

1, 2 and 3 XFEED Switches . . . . Off

Observe ON lights extinguish. DISAG lights will illuminate momentarily as crossfeed valve transitions to close. On SD FUEL synoptic, observe crossfeed valves are closed.

(CONTINUED)

FUEL SYSTEM MANUAL OPERATION (Continued)

1, 2 and 3 TRANS Switches . . . . ON

Observe ON lights illuminate. On SD FUEL synoptic, observe tanks 1, 2 and 3 transfer pumps indicate on and no pump indicates low pressure.

1, 2 and 3 TRANS Switches . . . . .Off

Observe ON lights extinguish. On SD FUEL synoptic, observe tanks 1, 2 and 3 transfer pumps indicate off.

If aux tank contains usable fuel:

AUX TANK L and R TRANS Switches . . . . ON

Observe ON lights illuminate. On SD FUEL synoptic, observe each pump indicates on and no pump indicates low pressure for a tank that contains usable fuel, and upper aux fill valve spigot is displayed.

TANK 2 FILL Switch . . . . Push

Observe ARM light illuminates. On SD FUEL synoptic, observe tank 2 fill valve spigot is displayed.

NOTE: Tank 2 fill valve will not remain armed without an auxiliary transfer pump on.

AUX TANK L and R TRANS Switches . . . . .Off

Observe ON lights extinguish and ARM light for tank 2 FILL switch extinguishes when both L and R TRANS switch ON lights are extinguished. On SD FUEL synoptic, observe each pump indicates off and observe tank 2 fill valve spigot is no longer displayed.

If tail tank contains usable fuel:

TAIL TANK TRANS Switch. . . . . ON

Observe ON light illuminates. On SD FUEL synoptic, observe each pump indicates on and no pump indicates low pressure, and upper aux fill valve spigot is displayed.

TAIL TANK ALT PUMP Switch . . . . ON

Observe ON lights illuminate. On SD FUEL synoptic, observe each pump indicates on and no pump indicates low pressure.

(CONTINUED)

FUEL SYSTEM MANUAL OPERATION (Continued)

TAIL TANK TRANS Switch . . . . Off

Observe ON light extinguishes. On SD FUEL synoptic, observe each pump indicates off and upper aux fill valve spigot is no longer displayed.

TAIL TANK ALT PUMP Switch . . . . Off

Observe ON light extinguishes. On SD FUEL synoptic, observe each pump indicates off.

[END]

Before Start - CM2

1, 2 and 3 Tank PUMP Switches . . . . On

Observe OFF and LOW lights are extinguished.

[END]

Takeoff (after airborne) - PM

AUX TANK L and R TRANS Switches . . . . As Required

Select ON if usable fuel contained in aux tank.

TAIL TANK TRANS Switch . . . . As Required

Select ON if usable fuel contained in tail tank.

1, 2 and 3 FILL Switches . . . . As Required

Select ARM when tank 2 quantity more than tank1 or tank 3 quantity.

TANK 2 TRANS Switch . . . . As Required

Select ON when tank 2 quantity more than tank1 or tank 3 quantity.

CAUTION: Do not use ballast fuel.

When “TAIL PUMPS LO” alert is displayed:

TAIL TANK TRANS Switch . . . . Off

When “AUX UPR PUMPS LO” alert is displayed:

AUX TANK L and R TRANS Switches . . . . Off

When tank 2 quantity equals tanks 1 and 3:

TANK 2 TRANS Switch . . . . Off

[END]

Parking - CM2

1, 2 and 3 Tank PUMP Switches . . . . OFF

Observe OFF lights illuminate. If any transfer pump is on, push appropriate TRANS pump switch and observe associated ON lights extinguish. Verify all crossfeeds are closed.

[END]

Actual Landing Distances For Different Runway Conditions (For In-Flight Reference Only)

These tables provided actual (unfactored) landing distance with **Maximum Manual Braking** for in-flight use only.

Dry Runway Estimated Landing Distance (Feet)

No Reverse Thrust, Standard Day, No Wind, Zero Slop (includes air run distances)

FLAP 50/EXT

Pressure Altitude (ft)	**Landing Weight (1000 Kg)**					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	3608	3772	3936	4100	4264	4428
2000 FT (STD=11°C)	3805	3969	4166	4330	4494	4658
4000 FT (STD=7°C)	4002	4166	4363	4560	4724	4920

Pressure Altitude (ft)	**Landing Weight (1000 Kg)**						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	4592	4789	4882	5059	5226	5394	5561
2000 FT (STD=11°C	4855	5052	5108	5276	5472	5659	5846
4000 FT (STD=7°C	5117	5347	5374	5581	5787	5994	6201

CORRECTIONS:

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -10ft per °C
ABOVE standard day : + 23ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -63ft per +1% slope
Downhill : +200ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -27ft per kt
Tailwind: +79ft per kt

(CONTINUED)

Dry Runway Estimated Landing Distance (Feet)

(CONTINUED)

No Reverse Thrust, Standard Day, No Wind, Zero Slop (includes air run distances).

FLAP 35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	170	180	190	200	210
S.L. (STD=15°C)	3871	4035	4232	4396	4592	4756
2000 FT (STD=11°C)	4068	4264	4461	4658	4855	5019
4000 FT (STD=7°C)	4297	4494	4691	4888	5117	5314

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	230	240	250	260	270	280
S.L. (STD=15°C)	4953	5117	5345	5512	5699	5886	6073
2000 FT (STD=11°C	5216	5412	5590	5778	5984	6191	6407
4000 FT (STD=7°C	5544	5740	5886	6092	6319	6565	6811

CORRECTIONS:

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -10ft per °C
ABOVE standard day : + 23ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -63ft per +1% slope
Downhill : +200ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -30ft per kt
Tailwind: +92ft per kt

[END]

Wet Runway Estimated Landing Distances (Feet) (Not Considering Hydroplaning)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances)

50/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	4789	4986	5183	5412	5609	5839
2000 FT (STD=11°C)	5052	5280	5478	5740	5970	6232
4000 FT (STD=7°C)	5347	5576	5839	6068	6331	5693

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	6068	6331	6570	6810	7050	7300	7550
2000 FT (STD=11°C)	6462	6724	6990	7250	7520	7790	8060
4000 FT (STD=7°C)	6856	7151	7410	7690	7970	8250	8540

CORRECTIONS:

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -10ft per °C
ABOVE standard day : + 37ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -125ft per +1% slope
Downhill : +433ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -46ft per kt
Tailwind: +132ft per kt

(CONTINUED)

1 thrust reverser inop	
Landing Weight (1000 Kg)	Landing Distance increase (feet)
160	33
170	92
180	151
190	214
200	273
210	332
220	394
230	453
240	512
250	575
260	634
270	693
280	752

Wet Runway Estimated Landing Distances (Feet) (Not Considering Hydroplaning) (CONTINUED)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distance)

35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	5150	5380	5609	5839	6101	6331
2000 FT (STD=11°C)	5412	5675	5937	6200	6495	6724
4000 FT (STD=7°C)	5740	6036	6331	6593	6888	7184

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	6593	6856	7120	7370	7630	7890	8130
2000 FT (STD=11°C)	7020	7282	7560	7840	8110	8380	8660
4000 FT (STD=7°C)	7479	7774	8060	8350	8660	8950	9250

CORRECTIONS:

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -14ft per °C
ABOVE standard day : + 40ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -132ft per +1% slope
Downhill : +499ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -46ft per kt
Tailwind: +142ft per kt

1 thrust reverser inop	
Landing Weight (1000 Kg)	Landing Distance increase (feet)
160	122
170	194
180	266
190	335
200	407
210	476
220	548
230	621
240	693
250	762
260	834
270	906
280	978

[END]

Wet Runway Estimated Landing Distances (Feet) (Considering Hydroplaning)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distance)

FLAP 50/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	5117	5380	5675	5937	6232	6528
2000 FT (STD=11°C)	5445	5740	6068	6364	6659	7020
4000 FT (STD=7°C)	5839	6134	6495	6823	7151	7544

END]

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	6856	7151	7460	7780	8100	8420	8750
2000 FT (STD=11°C)	7315	7676	8000	8340	8680	9030	9380
4000 FT (STD=7°C)	7872	8266	8610	8980	9350	9730	10100

CORRECTIONS:

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -17ft per °C
ABOVE standard day : + 46ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -142ft per +1% slope
Downhill: +699ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -60ft per kt
Tailwind: +210ft per kt

1 thrust reverser inop	
Landing Weight (1000 Kg)	Landing Distance increase (feet)
160	230
170	306
180	378
190	453
200	525
210	598
220	673
230	745
240	817
250	893
260	965
270	1040
280	1113

(CONTINUED)

Wet Runway Estimated Landing Distances (Feet) (Considering Hydroplaning) (CONTINUED)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distance)

FLAP 35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	170	180	190	200	210
S.L. (STD=15°C)	5576	5872	6167	6495	6823	7118
2000 FT (STD=11°C)	5937	6265	6593	6354	7315	7643
4000 FT (STD=7°C)	6364	6724	7085	7479	7840	8233

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	230	240	250	260	270	280
S.L. (STD=15°C)	7479	7807	8130	7475	8810	9150	9510
2000 FT (STD=11°C)	8036	8364	8700	9070	9430	9800	10175
4000 FT (STD=7°C)	8627	9020	9410	9800	10200	10600	11000

CORRECTIONS

Temperature (From STD -21° C to STD +40° C)
BELOW standard day : -17ft per °C
ABOVE standard day : + 50ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -174ft per +1% slope
Downhill : +837ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -60ft per kt
Tailwind: +236ft per kt

1 thrust reverser inop	
Landing Weight (1000 Kg)	Landing Distance increase (feet)
160	325
170	417
180	512
190	604
200	699
210	791
220	886
230	978
240	1070
250	1165
260	1257
270	1352
280	1444

Contaminated Runway Estimated Landing Distances (Feet)

Slush/Loose Snow/Standing Water Depth = 12.7 mm (0.5 in)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances)

FLAP 50/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	5839	6101	6396	6692	6954	7282
2000 FT (STD=11°C)	6167	6462	6790	7085	7413	7741
4000 FT (STD=7°C)	6528	6856	7216	7544	7872	8266

Pressure Altitude (ft)	Landing Weight (1000 K						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	7577	7938	8260	8590	8940	9280	9630
2000 FT (STD=11°C)	8069	8463	8800	9170	9530	9910	10280
4000 FT (STD=7°C)	8627	9020	9410	9810	10210	10620	11040

CORRECTIONS:

Temperature (From STD -21°C to STD +40 °C)
Below standard day: -17ft per °C
Above standard day: +43ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -240ft per +1% slope
Downhill : +834ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -60ft per kt
Tailwind: +187ft per kt

(CONTINUED)

Contaminated Runway Estimated Landing Distances (Feet)
Slush/Loose Snow/Standing Water Depth = 12.7 mm (0.50 in)

(Continued)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances

FLAP 35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	6331	6659	6954	7282	7643	7971
2000 FT (STD=11°C)	6724	7052	7413	7774	8135	8496
4000 FT (STD=7°C)	7151	7512	7872	8266	8692	9053

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	8299	8660	9010	9380	9720	10090	10450
2000 FT (STD=11°C)	8856	9250	9630	10000	10390	10780	11160
4000 FT (STD=7°C)	9480	9873	10280	10690	11100	11520	11930

CORRECTIONS:

Temperature (From STD -20 °C to STD +40 °C)
Below standard day: -17ft per °C
Above standard day: +46ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -243ft per +1% slope
Downhill : +942ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -66ft per kt
Tailwind: +204ft per kt

[END]

Contaminated Runway Estimated Landing Distances (Feet) Slush/Loose Snow/Standing Water Depth = 25.4 mm (1 in)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances

FLAP 50/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	5314	5576	5839	6068	6364	6659
2000 FT (STD=11°C)	5609	5872	6134	6429	6757	7085
4000 FT (STD=7°C)	5904	6200	6528	6856	7184	7577

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	6954	7282	7600	7930	8250	8590	8930
2000 FT (STD=11°C)	7413	7807	8260	8540	8900	9290	9680
4000 FT (STD=7°C)	7905	8332	8700	9110	9520	9930	10350

CORRECTIONS

Temperature (From STD -20º C to STD +40º C)
BELOW standard day : -17ft per ºC
ABOVE standard day : + 40ft per ºC

Slope (From STD -2% downhill to +1% uphill)
Uphill : -210t per +1% slope
Downhill : +683ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -46ft per kt
Tailwind: +161ft per kt

(CONTINUED)

Contaminated Runway Estimated Landing Distances (Feet) Slush/Loose Snow/Standing Water Depth = 25.4 mm (1 in) (CONTINUED)

Full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances).

FLAP 35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	5806	6068	6364	6692	7020	7348
2000 FT (STD=11°C)	6101	6429	6757	7118	7479	7840
4000 FT (STD=7°C)	6462	6823	7184	7577	7971	8364

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	7708	8036	8400	8780	9130	9490	9860
2000 FT (STD=11°C)	8200	8561	8920	9310	9700	10090	10480
4000 FT (STD=7°C)	8791	9184	9580	10000	10420	10870	11300

CORRECTIONS

Temperature (From STD -20° C to STD +40° C)
BELOW standard day : -17ft per °C
ABOVE standard day : + 43ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -224ft per +1% slope
Downhill : +778ft per -1% slope

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -53ft per kt
Tailwind: +178ft per kt

[END]

Wet Ice Runway Estimated Landing Distances (Feet)

Braking Action 0.25μ, full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances).

FLAP 50/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	**170**	**180**	**190**	**200**	**210**
S.L. (STD=15°C)	9348	9676	10037	10365	10693	11054
2000 FT (STD=11°C)	9906	10267	10660	11021	11349	11743
4000 FT (STD=7°C)	10496	10890	11316	11676	12071	12464

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	**230**	**240**	**250**	**260**	**270**	**280**
S.L. (STD=15°C)	11415	11808	12190	12580	12970	13370	13760
2000 FT (STD=11°C)	12104	12530	12910	13310	13740	14150	14570
4000 FT (STD=7°C)	12858	13350	13730	14180	14610	15070	15520

CORRECTIONS

Temperature (From STD -20°C to STD +40 °C)
Below standard day: -14ft per °C
Above standard day: +56ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -817t per +1% slope
Downhill : Landing on downhill slope is not recommended due to excessive required distance.

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -109ft per kt
Tailwind: +250ft per kt

(CONTINUED)

Wet Ice Runway Estimated Landing Distances (Feet)

(CONTINUED)

Braking Action 0.25μ, full reverse thrust, standard day, no wind, zero slope, 3 engines at max reverse thrust to 80 KIAS, then reverse idle to 60 KIAS, then 3 engines at forward idle to stop (includes air run distances)).

FLAP 35/EXT

Pressure Altitude (ft)	Landing Weight (1000 Kg)					
	160	170	180	190	200	210
S.L. (STD=15°C)	10234	10595	10923	11284	11677	12038
2000 FT (STD=11°C)	10857	11218	11612	12005	12399	12792
4000 FT (STD=7°C)	11513	11907	12333	12760	13186	13612

Pressure Altitude (ft)	Landing Weight (1000 Kg)						
	220	230	240	250	260	270	280
S.L. (STD=15°C)	12399	12792	13198	13600	14000	14410	14810
2000 FT (STD=11°C)	13186	13612	14040	14700	14910	15360	15800
4000 FT (STD=7°C)	14072	14465	14910	15360	15830	16290	16760

CORRECTIONS

Temperature (From STD -20°C to STD +40 °C)
Below standard day: -17ft per °C
Above standard day: +63ft per °C

Slope (From STD -2% downhill to +1% uphill)
Uphill : -998t per +1% slope
Downhill : Landing on downhill slope is not recommended due to excessive required distance.

Wind (From -10 knot tailwind to +20 knot headwind)
Headwind: -115ft per kt
Tailwind: +263ft per kt

[END]

Takeoff Performance Considerations - Wet/Slippery Runways

<table>
<tr><th>Condition</th><th colspan="2">Performance Data and Considerations</th></tr>
<tr><td rowspan="3">WET</td><td>OPSWIN</td><td>RAM</td></tr>
<tr><td>Use: Runway - WET
Flap - Optimum</td><td>Use: WET column.
Procedure as specified in RAM.</td></tr>
<tr><td colspan="2">Maximum crosswind - 24 kts</td></tr>
<tr><td rowspan="2">SLIPPERY</td><td>Use: Runway mu value as reported.</td><td>Use: Procedure as specified in RAM.</td></tr>
<tr><td colspan="2">Maximum crosswind - Depends on actual mu.
Use: Crosswind Limitation Table.
No tailwind</td></tr>
<tr><td rowspan="2">SLUSH
WET SNOW
STANDING WATER</td><td>Use: Contaminated 1/4in or Contaminated 1/2in.</td><td>Use: 1/4in SSW or 1/2in SSW columns.
Procedure as specified in RAM.</td></tr>
<tr><td colspan="2">Maximum crosswind - 15 kts
No tailwind</td></tr>
<tr><td rowspan="2">DRY SNOW
Max. 10.2cm (4in)</td><td>Depth up to 5.08cm (2in), use:
Contaminated 1/4in.
Depth from 5.08cm (2in) to 10.2cm (4in), use:
Contaminated 1/2in.</td><td>Use: 1/4in SSW or 1/2in SSW columns.
Procedure as specified in RAM.</td></tr>
<tr><td colspan="2">Maximum crosswind - 15 kts
No tailwind</td></tr>
<tr><td rowspan="3">ICY</td><td>Use: Runway mu factor as reported.</td><td>Use: reported mu value in ‘Slippery Takeoff Reference’.</td></tr>
<tr><td colspan="2">CAUTION: Takeoff is not permitted when the reported mu value is less than 0.20.</td></tr>
<tr><td colspan="2">Maximum crosswind - 15 kts
No tailwind</td></tr>
</table>

Landing Performance Considerations - Wet/Slippery Runways

Condition	Performance Data and Considerations	
WET	OPSWIN	RAM
	Use: Opswin- Runway WET	Use: QRH - Wet Runway Estimated Landing Distances Not Considering Hydroplaning
	Maximum Crosswind - 24 kts	
SLIPPERY	Use: Landing Performance Module - Check landing distance under MAX autobrake Use the runway reported mu value.	Use: • RAM section 3-3 • GOOD = 0.4 mu + • MED = 0.33 to 0.39 mu • POOR = 0.30 to 0.32 mu
	Maximum crosswind - Depends on actual mu. Use Crosswind Limitation Table No tailwind	

(CONTINUED)

Landing Performance Considerations - Wet/Slippery Runways (Continued)

<table>
<tr><th>Condition</th><th colspan="2">Performance Data and Considerations</th></tr>
<tr><td>CONTAMINATED
SLUSH
WET SNOW
STANDING WATER</td><td>Use:
1/4" CONTAMINATED, or
1/2" CONTAMINATED as required for wet snow, slush or standing water.
For dry snow up to 75 mm (3in) use:
1/4" CONTAMINATED
For dry snow more than 76mm (3in) up to 152 mm (6in) use:
1/2" CONTAMINATED</td><td>For depths of wet snow, slush or standing water up to 6mm (1/4in) and dry snow up to 75mm (3in) use:
Wet Runway Estimated Landing Distances (Feet) Considering Hydroplaning and,
Contaminated Runway Estimated Landing Distances (Feet) Slush/Loose Snow/Standing Water Depth = 12.7mm (0.5 in)
See Instructions on next page</td></tr>
<tr><td rowspan="4">ICY</td><td>OPSWIN</td><td>RAM</td></tr>
<tr><td>Use:
Reported mu value.</td><td>For reported Mu values between 0.25 and 0.29, use:
Wet Ice Runway Estimated Landing Distances Table (QRH QR.40)</td></tr>
<tr><td colspan="2">CAUTION: Landing is not permitted when reported mu value is less than 0.20.</td></tr>
<tr><td colspan="2">Maximum crosswind: Ice not melting - 15 kts.
No tailwind</td></tr>
</table>

Instructions Calculating Estimated Landing Distances for Runways with Contaminated, Slush, Wet Snow, or Standing Water

For depths of wet snow, slush or standing water up to 6mm (1/4in) and dry snow up to 76mm (3in), use charts - Wet Runway Estimated Landing Distances (Feet) Considering Hydroplaning (*QRH-QR.40*), and Contaminated Runway Estimated Landing Distances (Feet) Slush/Loose Snow/Standing Water Depth = 12.7mm (0.5 in) (*QRH -QR.40*)

(CONTINUED)

Landing Performance Considerations - Wet/Slippery Runways (Continued)

From the two values obtained for the particular landing weight, corrected for Temperature, Slope and Wind, calculate the mean; this will be the landing distance required.

For depths of wet snow, slush or standing water, more than 6mm (1/4in), up to 12.7mm (1/2in), and for dry snow, more than 76mm (3in) up to 152mm (6in); use chart - Contaminated Runway Estimated Landing Distances (Feet) Slush/Loose Snow/Standing Water Depth = 12.7mm (0.5 in) . Make corrections for temperature, slope and wind.

OVERWEIGHT LANDINGS

General

When electing to land overweight the maximum landing Flap 50 setting should be used whenever possible. In some situations the approach speed may be higher than the maximum flap limit speed, in which case; the alternate landing Flap 35 setting must be used.

The maximum landing weight and inflight (Landing Flaps) weight should be observed for all normal flight operations. When conditions exist that result in a decision to make an overweight landing, both the maximum landing weight and the inflight (Landing Flaps) weight can be exceeded. The use of Flap 28 to land overweight following a critical emerency is only recommended in the case of an abnormal landing configuration, as dictated by the Emergency or Abnormal Checklist. The use of a landing Flap position that is not Flap 50 or Flap 35 will disable the autospoilers and autobrakes. Coupled with a higher approach speed, the use of restricted landing flap will only compound the problems of an overweight landing.

For estimated landing distance overweight, refer to Estimated Landing Distance Tables in the QRH.

Climb Limited Landing Weight (Tons)
(based on Flap 28 for Go-Around)

Airport Elevation - ft	Temperature - °C				
	-50 to +16	17 to 24	25 to 32	33 to 40	41 to 50
SL - 1000	283.7	281.8	275.5	261.0	235.8
1001 - 2000	276.5	275.7	266.7	250.5	226.3

Subtract 27.0T if approach will be made in icing conditions.

[END]

Normal & Abnormal Configuration Reference Speeds (V_{REF})

LDG WT (1000 KG)	NORM CONF V_{REF}		ABNORMAL CONFIG V_{REF}									
	FLAP/SLAT		FLAP/SLAT									
	35/EXT (1,4)	50/EXT (1)	UP/RET (2)	0/EXT (3)	10/EXT (1)	15/EXT (1)	22/EXT (1)	25/EXT (1)	28/EXT (1,4)	15/RET	25/RET	28/RET (1)
130	131	129	169	137	136	130	129	127	127	158	153	152
140	131	129	175	142	138	135	133	132	131	165	159	158
150	134	132	181	147	143	140	138	137	136	171	165	164
160	138	135	187	152	148	145	142	141	140	176	171	170
170	142	139	193	157	152	149	147	146	145	181	176	175
180	147	143	198	161	156	154	151	150	149	187	181	180
190	151	146	204	166	161	158	155	154	153	192	186	185
200	155	150	210	170	165	162	159	158	157	197	191	189
210	159	153	215	174	169	166	163	162	161	202	195	194
220	162	156	220	178	173	170	167	166	165	207	200	199
230	165	160	225	182	177	173	171	169	169	212	205	204
240	169	163	229	186	181	177	174	173	173	216	209	208
250	173	166	234	190	185	181	178	177	176	221	214	*
260	176	169	239	194	188	185	182	180	179	225	218	*
270	179	172	243	198	192	188	185	183	182	229	*	*
280	183	175	248	201	195	191	189	187	186	234	*	*
290	186	178	252	205	199	195	192	190	189	238	*	*

NOTES:

(1) V_{APP} is the greater of V_{REF} + 5 or V_{REF} + wind additive (see Note (5)).

(2) V_{APP} is V_{REF} + 5. DO NOT ADD WIND

(3) V_{APP} is the greater of V_{REF} + 15 or V_{REF} + wind additive (see Note (5)).

(4) If HYD 2 & 3 failure, V_{APP} is the greater of V_{REF} + 8, or V_{REF} + wind additive (See Note (5)).

(5) Wind additive is 1/2 of the steady state wind greater than 20 knots or full gust, whichever is greater (max 20 knots).

* Exceeds flap placard.

LB1-2-0391

Intentionally
Blank

BRIEFINGS

BRIEFING FOR FLIGHT DECK AUTHORIZED PERSONNEL

- USE OF OXYGEN MASK
- USE OF SHOULDER HARNESS AND SEAT BELT
- EMERGENCY EQUIPMENTS AND EXITS
- EVACUATION PROCEDURES
- STERILE COCKPIT ENVIRONMENT

BRIEFING FOR NON-REVENUE PASSENGERS

The PIC is responsible for ensuring that all non-revenue passengers are briefed on:

- AIRCRAFT AND CABIN SAFETY
- CABIN SECURITY
- OXYGEN MASK OPERATIONS
- EMERGENCY EXIT OPERATIONS
- EVACUATION PROCEDURES
- SAFETY INSTRUCTION CARD

(CONTINUED)

BRIEFING (Continued)

DEPARTURE BRIEFING

- RUNWAY IN USE
- TAKE-OFF FLAP SETTING
- THRUST SETTING
- DEPARTURE CHART/TITLE, DATE AND NUMBER
- DEPARTURE ROUTE, SPEED/ALTITUDE RESTRICTIONS
- FMS SETUP AND NAVIGATION AID SELECTION
- SPECIAL ENGINE OUT PROCEDURE
- ADDITION ITEMS IF APPLICABLE:
  a. Special procedures due to an MEL/CDL item, as appropriate.
  b. Relevant NOTAMs.
  c. Aircraft configuration, if different from normal, handling considerations.
  d. Taxi routing
  e. Icing conditions, use of aircraft systems on the ground and for take-off.
  f. Weather conditions and runway surface conditions, if significant.
  g. Critical conditions affecting the STOP/GO decision (take-off at limiting weight, degraded runway surface friction, crosswind).
  h. Considerations relating to low visibility operations, minimum RVR minimum visual segment.
  i. Noise abatement requirements.
  j. Terrain and obstructions.
  k. Contingency planning-landing at departure airport/departure alternate.
  l. Any other items the pilot considers noteworthy.
  m. Briefing for flight deck authorized personnel.

BRIEFING (Continued)

APPROACH BRIEFING

- CHART/TITLE, DATE AND NUMBER
- ARRIVAL ROUTE, SPEED AND ALTITUDE RESTRICTIONS, STEP DOWN FIXES (IF APPLICABLE), ALTITUDE OVER FAF/MINIMUM ALTITUDE FOR THE FINAL APPROACH SEGMENT
- DH, DA OR MDA AS APPROPRIATE
- FMS SETUP, NAVIGATION AID SELECTION
- APPROACH SPEED/WIND CONDITIONS, LANDING FLAP SETTING
- MISSED APPROACH PROCEDURE AND ALTITUDE
- AUTOBRAKE SETTINGS, USE OF REVERSE THRUST, PLANNED RUNWAY EXIT
- ADDITIONAL ITEMS IF APPLICABLE:

  a. Special procedures due to an MEL/CDL item, as appropriate.

  b. Relevant NOTAMs.

  c. Contingency planning: fuel state, available holding time, routing to the alternate airport, weather conditions at the alternate airport.

  d. Considerations relating to low visibility operations, approach light system runway lights, required visual segment.

  e. Special handling requirements due to unserviceable items, if applicable.

  f. Terrain and obstructions.

  g. Speed and flap setting during descent and approach.

  h. Icing conditions, use of aircraft systems during approach and after landing.

  i. Airport (runway) elevation.

  j. Runway condition, runway length, critical conditions resulting from factors such as high landing weight, degraded runway surface friction, crosswind.

  k. Taxi routing after landing, special communications procedures at a particular airport.

  l. Any other items the pilot considers noteworthy.

Intentionally
Blank

Intentionally
Blank

EMERGENCY LANDING

Couriers………………………………………………………. ALERT

NOTE: Notify the Couriers as to the nature of the emergency, time available for preparation and brace signal to be used.

Two minutes before landing announce "TWO MINUTES BEFORE TOUCHDOWN".

Thirty seconds before landing announce "BRACE, BRACE, BRACE" for landing.

After aircraft has stopped, if required:

EVACUATION

2	OUTFLOW VALVE	.VERIFY OP
1	PARK BRAKE.	ON
1	SPOILER Handle	RET
1	FUEL Switches.	OFF
1	Evacuation Command	ANNOUNCE
	If courier seat occupied "EVAC, EVAC, EVAC" on INT phone	
1	Tower.	.ADVISE
2	ENG FIRE Handles.	DOWN, DISCH
2	APU FIRE Handle.	...PULL, TURN
2	EMER PWR Selector	.OFF
2	BAT Switch	.OFF

Emergency Equipment to be carried out by crew members:

CM1	CM2
Flashlight	Flashlight
Crash Axe	First Aid Kit

[END]